# BESTLINK COLLEGE OF THE PHILIPPINES BULACAN CAMPUS - HOSPITAL MANAGEMENT SYSTEM (PATIENT MANAGEMENT)

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February 2026

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# DEDICATION

This Research project is wholeheartedly dedicated first to our fellow researchers, whose passion, hard work, and collaboration made this study possible. This work is not only a product of individual effort but also of teamwork and shared dedication to the pursuit of knowledge.

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# ABSTRACT

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Healthcare institution continuously strive to improve efficiency and service quality through the use of digital technologies. Most hospital still use a manual processes including Dr. Eduardo V. Roquero Memorial Hospital, which relies on manual, paper-based processes for managing patient records. With the use of manual processes, it is prone to inefficiencies, errors, and data loss, creating operational challenges and hindering the quality of patient care. To address these issues, a centralized digital platform is proposed for creating, storing, managing, and instantly retrieving records, which does the Patient Management

System for automating registration, scheduling, bed tracking and managing patient records. The core features of this system are digitalized patient records, automates appointment scheduling, and bed tracking. It is anticipated that the adoption of PMS will greatly simplify the working processes, decrease the load of administration on the workers of healthcare facilities, decrease the number of mistakes and unnecessary delays, and help access the correct information about patients easily. The system enables the healthcare professionals to spend more time attending to patients directly by automating their routine duties and this eventually improves the overall efficiency and quality of services offered by the hospital.

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# CHAPTER 1

## INTRODUCTION

Healthcare institutions continuously strive to improve efficiency and service quality through the use of digital technologies. Most hospitals still use a manual processes for managing patient records that causes data loss and delays, like Dr. Eduardo V. Roquero Memorial Hospital, because this hospital is one of the hospitals that uses manual recording of the patient information. The development of the web-based Patient Management System aims to streamline and simplify the workflow of healthcare staff by providing quick and easy access to patient records. This system is designed to reduce time-consuming manual processes and minimize the risk of data loss by ensuring secure and centralized data storage. The researchers will ensure to fully address the clients with their problems and concerns and deliver a solution that effectively resolves their operational challenges. The researchers will do their best to build this system and especially, by doing this we learned to build unity and cooperation from other members or groups because while developing this system we earned extra knowledge from each group and members.

A Patient Management System (PMS) helps the client by centralizing all patient information such as medical histories, diagnoses, lab results, and prescriptions into a single, secure platform that can be accessed in real time by authorized staff. This eliminates the delays and errors caused by fragmented records or poor communication between departments. By automating tasks like patient registration, scheduling, discharge processes, and bed tracking, a Patient Management System reduces the administrative burden on healthcare staff, allowing them to focus more on direct patient care. Real-time updates ensure that healthcare professionals can make informed decisions quickly, improving treatment accuracy and patient safety.

In addition to improving record management, a PMS strengthens coordination across hospital units by providing features like automated alerts for critical lab results, and dashboards that track the total reports or analytics, like appointments, inpatient, outpatient, bed occupancy rate. This integration leads to faster workflows, reduced operational costs, and higher patient satisfaction, as care becomes more efficient, transparent, and responsive. Ultimately, a PMS supports hospitals in delivering safe, timely, and high-quality care while building trust between patients and healthcare providers.

## 1.1 Background of the Capstone Project

The Patient Management System is a web-based and a digital platform system created to oversee the daily functions of healthcare institutions by arranging patient data and simplifying administrative duties, that could help the client and users to lessen their burden in recording and storing patient information, storing medical records, bed tracking and appointment scheduling. By streamlining patients records and automating workflow processes, healthcare professionals can access information more efficiently, reduce administrative burden, and ultimately deliver optimal patient care services. It's essentially the digital backbone that helps healthcare professionals to manage all aspects of the patient records, organize appointments, and manage complete digital records digitally. As healthcare professionals the priority must be caring for patients; various paperwork, such as manually recording patient records and scheduling can be a hurdle. A Patient Management System is indicated as an automated system that helps administrative personnel for handling administrative duties, keeping track and can quickly access patient information and records. In this system the client can expect features for creating, managing, storing, and accessing patient records, scheduling appointments and other available features.

The hospital was established on December 30, 1981, and was originally led by Dr. Eduardo V. Roquero who was at that time a physician and served the community as both a rural health doctor and NGO family planning physician in the 1980s, and today is the manager of the then Roquero General Hospital in Sapang Palay, San Jose del Monte Bulacan. The facility was named after Dr. Eduardo V. Roquero Memorial Hospital after his death which answered his medical and civic services to the people and is currently under the leadership of Dr. Isabelita S. Roquero. Dr. Eduardo V. Roquero Memorial Hospital is a healthcare institution that provides a wide range of healthcare services such as emergency care, surgery, obstetrics, pediatrics and internal medicine among other services with a vision of offering affordable and accessible healthcare to residents of the area.

However, the hospital faces operational challenges as it has existing operational challenges, like the hospital currently relies heavily on manual processes such as paper-based patient records, and handwritten appointment logs. These processes often lead to delays, data inaccuracies and difficulty retrieving patient history, which affect the quality of services. To prevent this problem the proposed system could help the client in a situation like this, by designing and implementing a computerized system that digitizes patient information recording and storage, this replaces the manual processing of inputting patient details with a computerized system that will save digitally. Having a system like this could avoid the errors, reducing the workloads, speed up daily tasks and speed up information retrievals. Inefficient in Storing the Patient Records is one of the problems that affects the client, having features in a system that could help the client and users to store records and instantly retrieve records. Switching to a secure and organized digital database or cloud-based system allows quick searching, organized filing, and remote accessibility. Delayed Processing of Patient Records is also one of the problems that affects the client, as when the results are not shared immediately between departments or staff leading to delays and can also result in duplication of records. Having a system that implements a real-time automated system that alerts and updates the staff immediately when results are available.

In order to address these problems, a more effective and reliable solution is the creation of a computerized system called the Patient Management System. This system keeps all the patient data in a digital database that is centrally stored and allows easy access, updates and is secure in handling the data. In this project, the developer used PHP and the Object-Oriented Programming (OOP) to develop a well-structured and maintainable application. PHP OOP enables the system to be made with components that can be reused thus simplifying the management, scaling and adding features in the future. Moreover, it will incorporate with an Artificial Intelligence (AI) in the Beddings and Linens module to assist the hospital in determining the patient length of stay and assist in the proactive allocation of beds and in the process of discharge planning. The system combines the most important patient and clinical variables, such as diagnosis, age, and the level of severity, to provide real-time predictions of predicted inpatient days, which is accurate. Such predictions allow the medical personnel to predict the availability of beds, patient placements, and patients who would be discharged, despite their particular time. Simultaneously, the system aids operational processes by predicting bed turnover and linen demand so that coordination with housekeeping services regarding linen replenishment and cleaning can be done in time. The technique of continuously updating the predictions, as patient conditions change, helps the AI system to reduce delayed discharges, reduce idle time on the bed, improve the flow of patients, and assist in more efficient utilization of the hospital resources without losing the clinical control and decision-making of the healthcare workers.

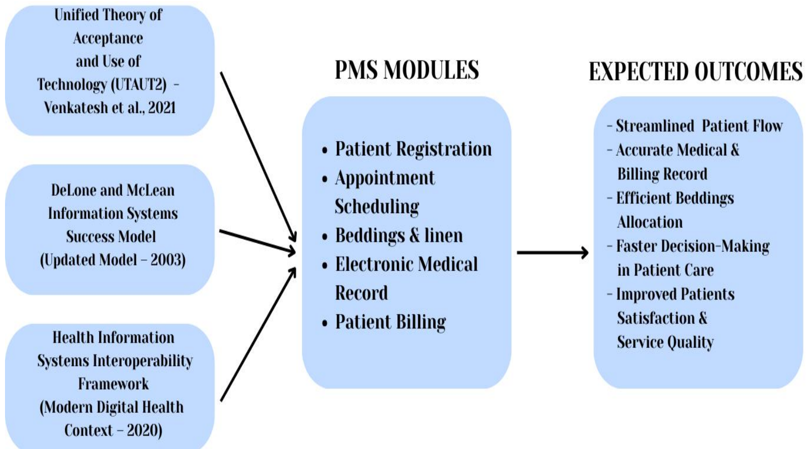


Figure 1 Theoretical Framework

The theoretical framework is providing a structured foundation that guides the Patient Management System (PMS) for understanding a problem, designing a system, and interpreting results by linking the work to existing knowledge and established theories.

**Unified Theory of Acceptance and Use of Technology (UTAUT2) - Venkatesh et al., 2021**

**Focus**: Explains technology acceptance based on performance expectancy, effort expectancy, social influence, and facilitating conditions.

**Application**: The successful adoption of PMS depends on healthcare professionals (doctors, nurses, and admin staff).

**DeLone and McLean Information Systems Success Model (Updated Model – 2003)**

**Focus**: Measures system success through system quality, information quality, service quality, user satisfaction, and net benefits.

**Application**:

- Accurate and timely patient data
- Reliable EMR and billing systems
- Secure access to patient information

**Health Information Systems Interoperability Framework (Modern Digital Health Context – 2020s)**

**Focus**: Ensures seamless data exchange between healthcare systems.

**Application**:

- Integration with laboratory systems, pharmacy systems, and insurance providers.
- Real-time access to patient records across departments.

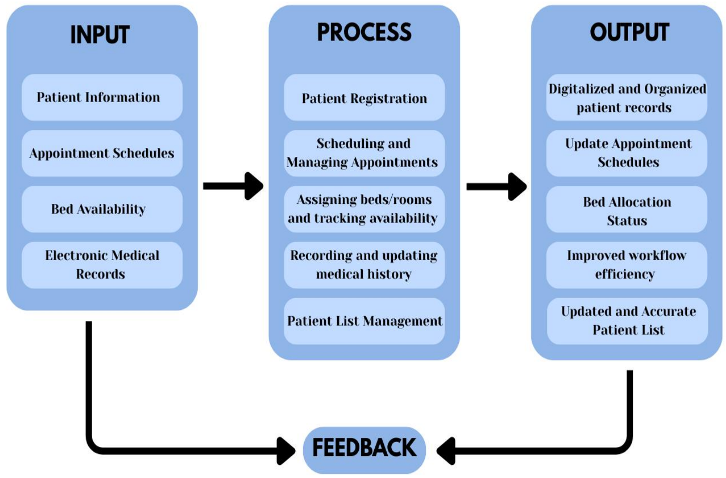


Figure 2 Conceptual Framework

Conceptual Framework of Patient Management System, illustrates how data flows within the system using the Input-Process-Output (IPO) model with a Feedback mechanism to produce effective and efficient healthcare services.

**Input**

All the data that enters the system, all the information required to make the system effective. These figures are the raw material which the system utilizes.

- **Patient Information** – basic information about the patient including full name, age, gender, address, telephone number, and emergency contact, eye color, email address, and previous medical records.
- **Appointment Schedules** - information on the visits of patients including the date, time, the doctor they have been

assigned and the department. This prevents any cases of duplicity in booking and manages time efficiently on both the side of patients and staff.

- **Bed Availability** - live information regarding the current status of a hospital bed such as available, occupied, or under maintenance. This informs the system of the individuals who can be admitted, the ones who can be discharged, and the appropriate ways of utilizing the bed.
- **Electronic Medical Records (EMR)** - a set of medical history, treatment, diagnosis, lab findings and medication of the patient. These records make care continuity and provide clinicians with up-to-date information.

**Process**

Process part presents how the system transforms input data into valuable output by means of automation and connection.

- **Patient Registration** – stores the basic information of the patient in a central database. It prevents the use of paper in writing, eliminates data redundancy, and simplifies the search of the data.
- **Scheduling and Managing Appointment** – allows the system to schedule visits fast. The employees will be able to

make, modify or cancel appointments and maintain the coordination between patients and physicians.

- **Assigned Beds/Rooms and Availability Tracking** – make sure that the patient receives a bed or a room that suits his or her needs and bed availability. The system displays the bed info continuously as individuals come, move, or leave.
- **Recording and Updating Medical History** – make sure that the medical record of the patient is updated as he or she visits, receives treatments, and follow-ups. It transforms medical information to be precise and assists physicians to make decisions.
- **Patient List Management** – This process involves organizing, viewing, and updating all registered and admitted patients in a centralized list. It allows healthcare staff to easily search and access patient information, reducing delays and improving coordination among departments.

**Output**

Output component is what the system produces after processing the input information.

- **Digitalized and Organized Patient Records** – safe, neat, and convenient database that enhances the process of data finding and makes the list of data less missing or incomplete.

- **Update Appointment Schedules** – provides patients and doctors with up to date schedule information, which reduces waiting time and absence of visits.
- **Bed Allocation Status** - displays bed occupancy and vacant bed in real time enabling the hospital managers and staff to utilize resources efficiently.
- **Improved Workflow Efficiency** – achieves the reduced number of manual operations, reduced number of errors, and increased rate of hospital work, which enhances the quality of services and patient satisfaction.
- **Updated and Accurate Patient List** – This output provides a complete and updated list of registered and admitted patients. It supports efficient monitoring, reporting, and decision-making by healthcare staff.

**Feedback**

The feedback loop allows the system to continue to become better. The outputs such as updated medical records, appointment schedules and bed status are verified and validated. Any mistakes, alterations or updates of the system are returned to the input section ensuring that the data remains correct, the system remains stable and the process continues to become better and better.

## 1.2 Context and Scope

The Patient Management System will be developed to work in the healthcare setting, in particular, hospitals like Dr. Eduardo V. Roquero Memorial Hospital, which still use a manual processes. These processes are digitized and streamlined by the PMS which offers a web based platform in which authorized personnel are able to make patient recordings, access and update patient information in a secure manner. The following list is included in this system project:

- **Patient Registration -** can add new patients and also to gain access to the system. It is to collect and save important information about the patient for the first time they visit to the hospital. This involves collecting and recording essential patient information, such as demographics (name, date of birth, address, contact details), and emergency contacts.
- **Patient List -** all patient information is stored and managed. It shows a list of all the patients who have been registered in the system and admit in hospital, making it easy for all healthcare staff to view, update, and access patient details whenever needed.
- **Electronic Medical Records (EMR) -** save, store and manage the medical information of each patient, such as medical history,

diagnoses, prescriptions, and etc. It allows authorized staff to easily view and update a patient’s health records. EMRs improve accessibility, enhance care coordination among healthcare providers, and reduce errors associated with paper records.

- **Appointment Scheduling -** organize and manage patient appointments with doctors. The authorized staff can view and set a new booking appointments for check-ups or follow-ups.
- **Beddings and Linens -** Allowed staff to monitor bed availability, assign patients to beds, track linen usage, and manage linen inventory to ensure clean and adequate supplies for patient care.

### 1.3 Problem and Statement

These challenges do not only cause stress for healthcare staff but also result in frustration for patients who experience delays and lack of easy access to their medical information.

- **Writing Manually of the Patient Information using an Index Card -** During a patient’s check-up at the hospital, the receptionist or the attending nurse records the patient's information manually using an index card as part of the initial assessment process. In the manual recording process, the attending nurse often puts in incomplete information and does not write some of the essential details like the complete address. It should include the full details such as block, street, region, and city. Should also consistently record the exact date and time the patient arrived at the hospital. These details are essential for maintaining accurate and reliable patient records, and if left incomplete can lead to delays and negatively affect overall service quality.

- **Inefficient in Storing the Patient Records -** Currently, patient records are stored manually by compiling them in envelopes along with other patients' documents. When a patient returns for another check-up, the nurse must search through these envelopes to retrieve their previous records. This process is time consuming and burdensome, especially during emergency situations. Since the hospital uses index cards to record patient information, and due to the high volume of incoming patients, the storage has become overwhelming and disorganized, making it increasingly difficult to locate specific records. Moreover, the hospital maintains patient records for up to 10 years, which adds to the growing bulk of files and further complicates record retrieval. This traditional method of record-keeping is inefficient and poses risks to timely and accurate healthcare delivery.

- **Delayed Processing of Patient Records Results -** The lack of sharing of vital patient records and the resulting delays in doctors receiving the results to make decision-making about treatment. For example, if test results from the laboratory are not reviewed, or made available in a timely manner after it is generated or collected, doctors may miss critical findings, causing treatment delays. Similarly, poor communication and failure to review the results immediately between the departments, and wards can result in duplicated work, incomplete records, or missed patient transfers. These gaps not only slow down hospital operations but also increase the risk of medical errors, reduce patient satisfaction, and undermine trust in the healthcare system. Delays in the preparation and delivery of patient records can hinder patient care because delayed or incomplete information sharing can lead to treatment errors, missed interventions, and operational inefficiencies. When departments and staff are not aligned whether due to late lab results, poor handovers, or unclear responsibilities critical decisions may be based on outdated or missing data.
- **Manual and Delayed Verification of Bed Preparation -** The hospital is using manual check of every bed to know whether it

has changed the linens and it is ready to be used by the next patient. This is a manual system that is time-consuming, unequal, and subject to negligence particularly when there is a high workload or patients. The fact that employees have to go to time wasting activities of physically checking the rooms to confirm bed availability creates unnecessary time wastage in ensuring rooms are ready to receive new patients.

## 1.4 Objectives and Goals

### Objectives

The objectives specify the concrete and measurable actions that the system intends to accomplish. These objectives translate the system's goals into specific functions and features that address identified problems in patient management. They guide in guiding the system construction process as they demonstrate precisely what the system has to do to make patient data more accurate, efficient, secure, and easily accessible.

- **Digitize Patient Information Records -** By replacing the manual process through index card method with a computerized system, that all patient details will be entered, typed and saved digitally into secure forms, ensuring that important fields are completed and consistent. This method is effective and can prevent data loss from being manually recorded from the patient information. The information will be stored in a secured database, which can be searched instantly using a search function to find patient information in seconds. Instead of keeping physical index card, all patient details are securely stored in a computerized database. Using the system, the users can reduce workloads, errors, prevent loss or damage of records, speed up daily tasks, and speed up information retrievals. And with a fully functional Patient Registration service that allows new patients to be added, updated, and managed securely.

- **Organized Digital Storage System -** This replaces the traditional physical storage with a computerized Patient Management System, that stores patient information in a centralized, searchable database. This system will allow authorized users to instantly retrieve and update records, reduce storage space needs and prevent the loss or damage of documents. This will help the users to search the information instantly, so they don’t need to search in a bulk of records, could speed up the retrieval of records and make it easier to share patient records between departments or facilities.

- **Automate a reliable real-time system -** This system has features like real-time automated result updates and alerts to immediately notify staff when lab results or critical data are available. This system can have features like real-time data entry, automated verification of information, mandatory fields to prevent missing data, and instant updates. It can also include tracking and notification tools to alert staff when records are pending or delayed.
- **Implement a features that tracks bed status -** With an effective tracking and computerized system, authorized personnel can immediately view the status of each bed availability without having to physically check it at any given time to ensure they can assign a bed easily, guard against the occurrence of double bookings and the arrangement of patients in a more effective manner.

**Goals**

The goals describe the overall purpose and long-term outcomes that the system aims to achieve. These goals focus on improving the general performance of healthcare operations by enhancing the management of patient information, reducing inefficiencies, and supporting better patient care.

- **Streamline registration, scheduling, and records** - The manual process is replaced with automated and computerzied data entry. It allows staff to easily input, search, and update patient information, making hospital operations faster and more reliable.
- **Easy viewing and retrieval  of records -** Allows authorized staff to quickly access patient files anytime, ensures accuracy and transparency in healthcare data and that all information is available when needed.
- **Ensures accuracy and security of records -** A digital system include required fields, validation checks, and auto-saving features, ensuring all  critical information is recorded properly even during busy or rushed moments, and maintaining complete and accurate medical records.
- **Improved the overall user experience -** Automation speeds up record processing and result generation. Staff can update patient data in real time. This makes the workflow smoother and reduces waiting times, as the system provides online services, giving more convenient and engagement in their healthcare.
- **Ensure Bed Readiness -** Prepared quickly for faster preparation and properly by coordinating linen availability with real-time bed status updates. The system will help the staff to

know exactly when a bed needs fresh linens, when linens have already been replaced, and which beds are ready for patient use.

### 1.5 Significance and Relevance

The Patient Management System plays a vital role in modernizing hospital operations, especially in storing and managing patient records, appointments scheduling, medical information and viewing patient bills. Knowing the significance and relevance of the study, can help the developers to develop a system that fits to the clients and user’s needs.

The importance of the study is to address the problems in the hospital that are still using the manual procedures to record and manage the patients information like in Dr. Eduardo V. Roquero Memorial Hospital. By developing a Patient Management System, it aims to help healthcare professionals to transition from paper-based processes to a more organized, and digitalized system. Manual recording and storing using index cards and envelopes often results in incomplete information, delays in locating patient history, and increased workload for staff, especially during emergencies. Having a Patient Management System could reduce these problems and improve overall healthcare delivery for the client and the users.

Since the healthcare institutions still lack automated systems, causing delays, inefficiencies, and errors. In emergency situations, quickly finding a patient’s medical history is critical to saving lives, therefore healthcare professionals need centralized and organized information to make informed decisions. The relevance of the study is the ability to solve real-word challenges faced by the hospitals that still use manual and outdated methods for handling patient records. This system is highly relevant to healthcare facilities that experience high patient volume and must deal with the burden of managing large amounts of paper records. The development of a Patient Management System, supports the healthcare sector’s growing need for digital transformation. It also makes sure that the critical patient records can be accessed, updates, and secured efficiently, by the authorized users especially during emergencies or follow-up consultations.

- For the patients as end-users is that the system enhances the everyday experiences of patients, by providing a faster registration or admission reducing long waiting times through organized queuing. Also allow patients to receive faster, more accurate, and minimizing waiting time through an organized appointment scheduling and giving accurate records of personal and visit information.

- For the receptionist and admin as users, the system helps them to simplify their tasks, by automating data entry, scheduling, record retrieval, and allowing for easy monitoring of patient flows, appointments, and records. Reducing the time spent on manual filing and searching for documents and improving reporting and tracking of daily operations.
- For the client Dr. Eduardo V. Roquero Memorial Hospital, the system will support the goals of the hospital by enhancing operational efficiency, cost-effectiveness, preventing data loss with secure digital storage, back-up and providing real-time analytics and reports for better decision-making.
- For the future researchers, they can use the system or this study as a guide for their study in the future.

## 1.6 Structure of the Document

| METHOD | DESCRIPTION |
| --- | --- |
| Introduction | This part is the discussion about the capstone project, the background information of the client, stating what their problem to address, introduction of the functions of the modules, outlining of the objectives and goals, identifying what and who would benefit, and giving the summary of the document structure. |
| Literature Review | In this part, the developers review a relevant and existing literatures, including an overview of agile scrum methodology, concepts of enterprise architecture, defining microservices architecture, discussion about development operation (DevOps) and continuous integration/continuous delivery (CI/CD), reviewing the previous studies and related research, and explain how the information systems are integrated in enterprise environments. |
| Methodology | In the third part are the outlines of the methodology that used in the project, focusing in |

| | |
|---|---|
| | the agile scrum principles, roles -such as Scrum Master, Programmer, System Analyst, Business Specialist, Document Specialist, and Product Owner, Sprint Cycles, Scrum Artifacts, explaining how each module of the system works independently but is connected to others, how will manage the codes and updates, and highlighting the unique features of the system. |
| TOGAF | Is a framework essentially a structured approach to designing, planning, implementing, and governing an organization’s architecture. |
| Reference | The full detail of a source that is used in the research. It appears at the end of the thesis and allows readers to find and verify the original source and give credit to the original author. |
| Appendices | Contains supplementary materials that support the project. These can include survey questionnaires, interview guides, raw data, technical documents, and additional tables related to the study. |

Table 1 Structure of the Document

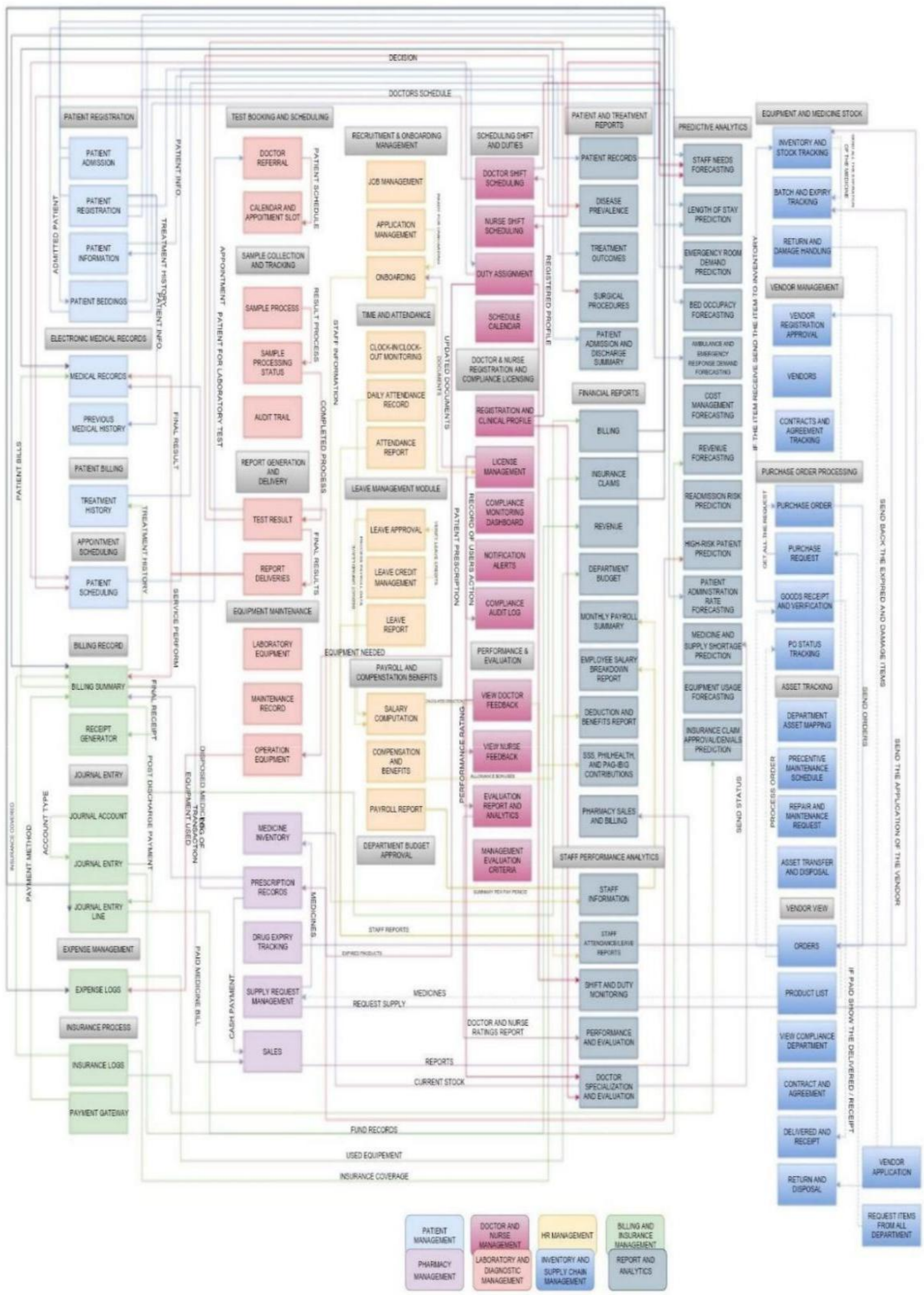

Figure 3 BPA Level 1

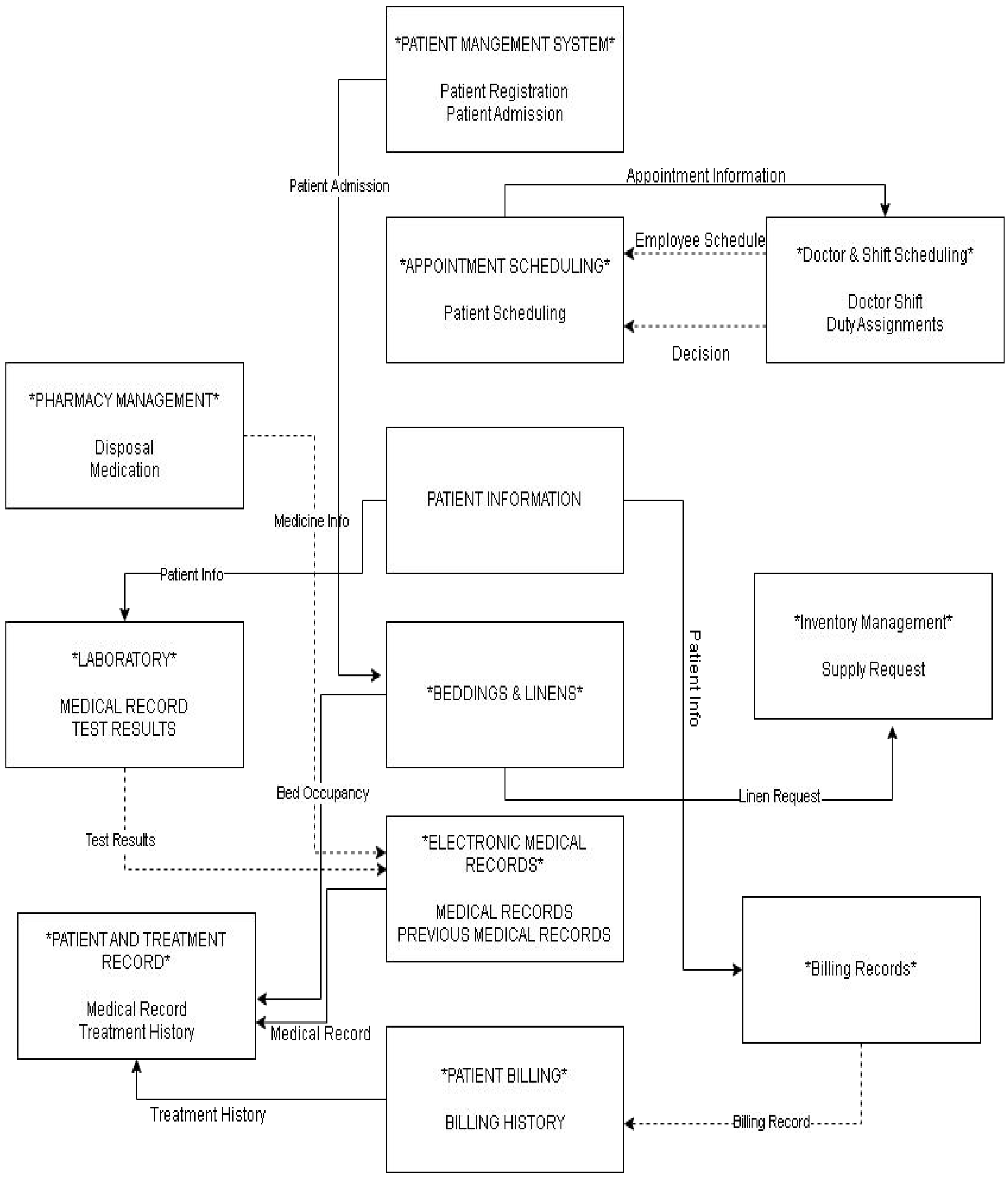


Figure 4 BPA Level 2

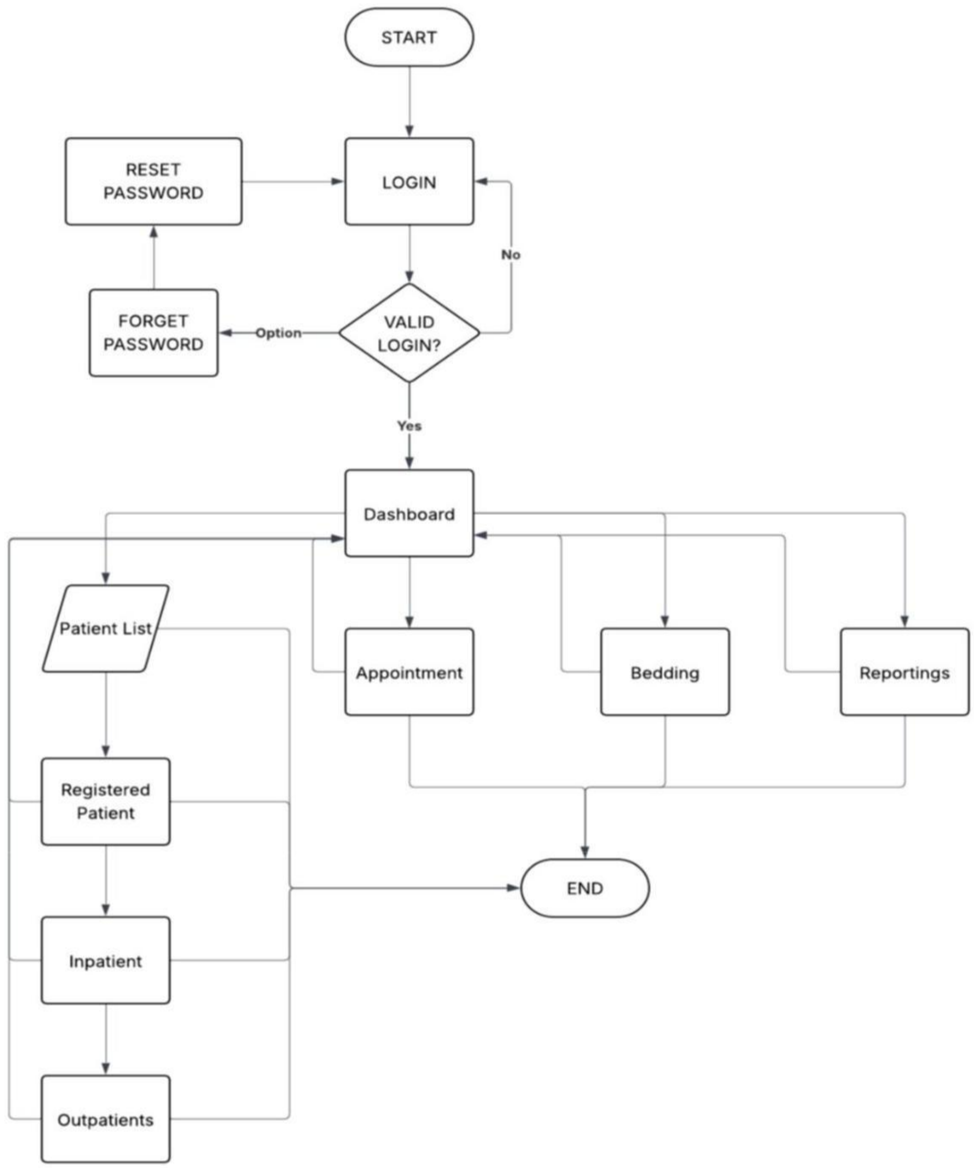


Figure 5 Flowchart

# CHAPTER 2

## RELATED STUDIES AND LITERATURE REVIEW

The development of a Patient Management System (PMS) is anchored in the broader body of knowledge on Health Information Systems (HIS), Electronic Medical Records (EMR), and Hospital Management Systems (HMS). Prior studies have emphasized that traditional manual processes—such as recording patient information on index cards—often result in inefficiencies, incomplete data, and risks of record loss (Alotaibi & Federico, 2017). Digitized systems address these challenges by ensuring data accuracy, accessibility, and security.

Research in health informatics has demonstrated that EMR systems improve patient care outcomes, reduce medical errors, and enhance operational efficiency (Menachemi & Collum, 2011). Similarly, studies on cloud-based hospital management systems highlight their scalability and flexibility in managing large volumes of patient data, which is crucial for modern healthcare institutions (Sultan, 2014).

Furthermore, the adoption of agile methodologies such as Scrum in healthcare software projects has shown success in delivering reliable systems in iterative cycles, ensuring stakeholder collaboration and adaptability to change (Pikkarainen et al., 2012). This aligns with the need for continuous improvement and rapid deployment in healthcare environments.

## 2.1 Agile Scrum Methodology Overview

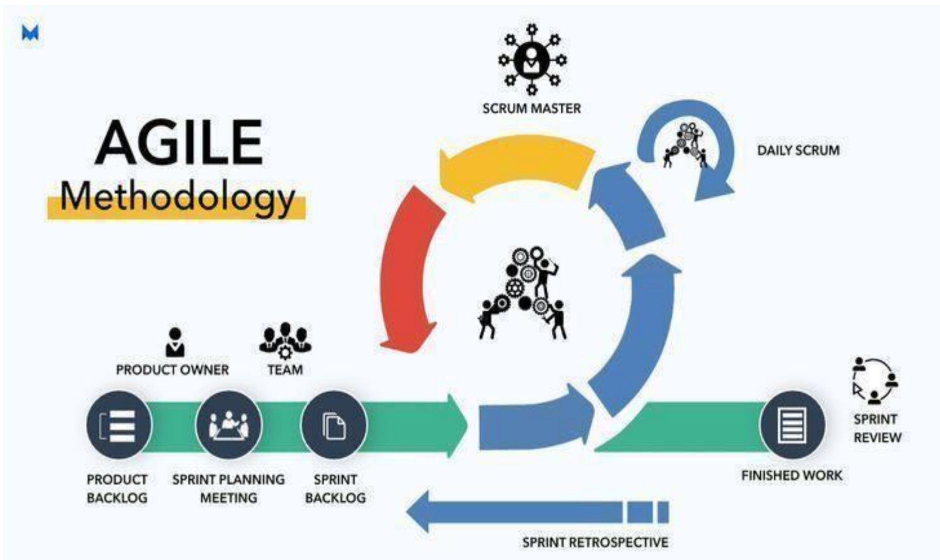


Figure 6 Agile Scrum Methodology Overview

Agile Scrum is a framework for applying the Agile methodology's principles to the management and completion of complicated software projects. It breaks down software development into brief, time-boxed cycles called sprints, which typically last two to four weeks. During these sprints, small, functional software components are designed, developed, tested, and delivered. Scrum places a strong emphasis on teamwork, flexibility, and ongoing feedback. It involves important roles like the Development Team (who create the product), the Scrum Master (who facilitates the process), and the Product Owner (who oversees requirements).

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Scrum is widely used because it allows teams to handle the uncertainty and changing requirements that are common in software projects. Instead of waiting until the end to deliver a complete system (like in the traditional Waterfall model), Scrum delivers working features early and often, giving stakeholders the opportunity to test and provide feedback during development. This reduces risks, improves product quality, and ensures the final software actually meets user needs. It also promotes team collaboration, transparency, and accountability, while maintaining flexibility to adjust priorities in every sprint. In short, Scrum help software projects become more efficient, adaptive, and user focused.

The use of Agile Scrum in developing the PMS provides significant benefits to the project team and to the overall system development. Through its iterative and increment approach. Scrum allows the team to divide the project into manageable sprints and to deliver the project step-by-step, this reduces the risks of delays and overlooked requirements, while making it easier to adapt to changes requested by the client. Scrum also promotes close collaboration among the team members such as the system analyst, programmer, document specialist and business specialist by assigning clear responsibilities to minimize time. It also improves the quality of the system, as the client can review, test and give feedback during every sprint so if error occurs could find it earlier.

## 2.2 Enterprise Architecture Concepts

Enterprise Architecture (EA) for a Patient Management System can also be defined as a comprehensive approach to designing and managing the system's structure, ensuring that its business operations, data management, applications, and technologies are properly aligned to meet healthcare goals. It provides a high-level blueprint that guides how patient information is collected, processed, and shared across different modules, enabling better decision-making, improved service delivery, transaction, and adaptability to future healthcare needs.

To implement Enterprise Architecture (EA) concepts in building a Patient Management System (PMS), the process begins by defining the scope, identifying stakeholders, and establishing guiding principles such as compliance with health data privacy regulations. The architecture development cycle aids in directing design from conception to execution by utilizing frameworks such as TOGAF. Data domains such as patients, beds, appointments, and results are defined by the information architecture. The logical system components such as registration, scheduling, patient administration, billing history and reporting dashboards are then specified by the application architecture, to ensure different limits and modular services. Governance plays a central role, with architecture review boards and data councils ensuring adherence to standards, while privacy-by-design and role-based access controls safeguard patient information. Reduced patient wait times, better resource utilization, accurate billing, and higher-quality care are just a few of the observable advantages that result from the PMS becoming scalable, interoperable, and compliant by aligning business needs, information flows, applications, and technology under EA principles.

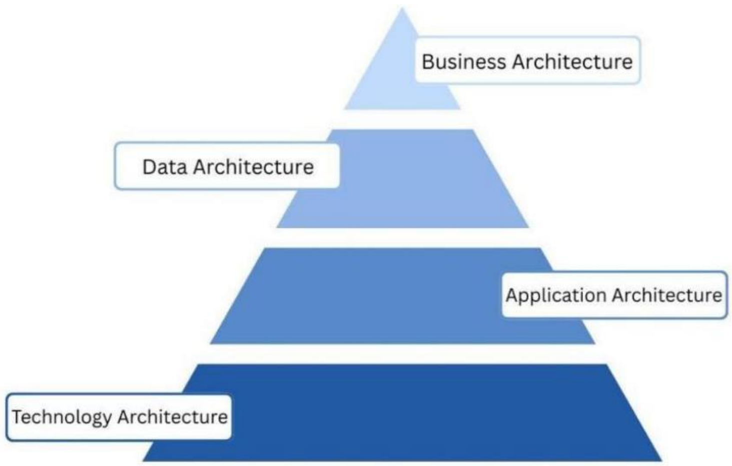


Figure 7 Enterprise Architecture

Building further, the application of Enterprise Architecture in a Patient Management System also emphasizes interoperability, governance, and long-term adaptability. EA promotes a service-oriented, modular architecture in which elements such as clinical documentation, billing, and scheduling can change on their own without affecting the system as a whole. By encouraging continuous improvement, this strategy enables the hospital to react swiftly to modifications in laws, patient demands, or technological advancements in healthcare. The architecture also guarantees that sensitive patient data is safeguarded through audit trails, role-based access control, encryption, and consent management by integrating security and privacy by design. In the end, applying EA principles ensures that a PMS is sustainable, effective, and prepared for the future by guiding its technical development and coordinating it with the hospital's strategic goals.

These architectural layers work together to provide a clear blueprint for the system's design and implementation. By applying these EA concepts, healthcare organizations can develop a PMS that not only meets current needs but is also well-structured, scalable and adaptable to future changes in technology and healthcare regulations. It also helps the hospital to provide better, more efficient care by making sure that every system works well with each other. This holistic approach ensures that the PMS is not only functional on its own but can also be integrated with other hospital systems, and a strategic asset that enhances patient care and operational efficiency.

## 2.3 Microservices Architecture

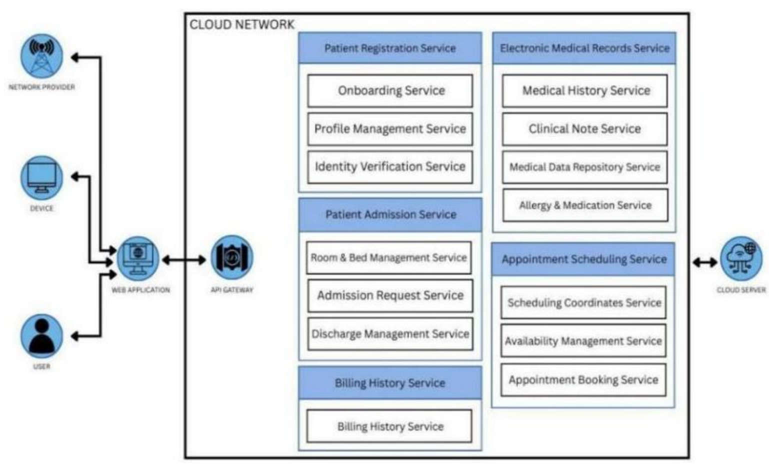


Figure 8 Microservices Architecture

Microservices architecture is a way of designing software systems where the application is broken down into small, independent services. Each service runs on its own, has its own database (or shared if needed), and communicates with others via APIs. Unlike monolithic systems (where everything is bundled together), microservices allow flexibility, faster updates, and scalability. Patient registration, Patient Admission, Appointments Scheduling, Electronic Medical Records, and Billing History, are just a few of the complex duties that the patient management system in a hospital setting oversees. A single changes could have an impact on the entire system if it is constructed as a single monolithic system. As microservices allow each module to function independently, hospitals can update or fix a single service without having to stop the PMS as a whole.

To ensure effectiveness and adaptability, a patient management system's design should adhere to modern software concepts like scalability, independent services, and API-driven architecture. Core modules like patient registration, patient admission, appointment scheduling, billing history in patient portal, and medical records can operate independently while still interacting with one another by utilizing independent services, which are comparable to microservices. Because of this division, the system is simpler to maintain, and updates or fixes can be applied to a single module without affecting the application as a whole. Another important idea is scalability, since hospitals may see increases in staff, data, and patient volume over time. By supporting more users, storing more medical data, and continuing to function as demand increases, a scalable PMS can manage growing workloads. Lastly, the PMS's API-driven design enables smooth integration with other hospital information systems, including HRIS for staff management, CRM for patient communication, and ERP for finance. Together these principles assure the Patient Management System is dependable, flexible, and future proof for the changing needs of healthcare organizations. APIs also facilitate interoperability with third-party healthcare services, including laboratory systems, insurance systems, and pharmacy systems.

The Patient Management System benefits greatly from principles such as modularity, scalability, and continuous deployment. Through modularity, it can be easily maintained, updated, and debugged without affecting the entire application. Examples of these components include patient registration, patient admission, appointments, billing history, and medical records. Additionally, because of this separation, hospitals can implement new features like laboratory integration or pharmacy without interfering with their current operations. Such as the appointment module can be scaled independently during peak seasons without putting unnecessary strain on other modules like appointment scheduling. Scalability assures that the system can grow with the hospital's needs, handling more patients, larger medical databases, and increased staff activity while maintaining performance. All of these advantages make the Patient Management System adaptable, dependable, and able to change with the changing demands of modern healthcare organizations. Continuous deployment makes it possible for updates and bug fixes to be automatically tested and deployed to the live environment, making sure that physicians, nurses, and staff always have access to the most recent developments without long waits or system failures.

Despite its advantages, a Patient Management System also faces several challenges, particularly when built using modern architectures like microservices. One major challenge is data consistency, since patient information may be spread across different modules such as medical records, billing, and pharmacy; ensuring all services are synchronized and always up to date can be difficult. Another challenge is system complexity, as dividing the PMS into multiple independent modules makes development, monitoring, and troubleshooting more complicated compared to a simple monolithic design. Finally, there is a strong DevOps dependency, because continuous integration, testing, deployment, and monitoring require advanced DevOps pipelines, automated tools, and skilled personnel. Without these practices, the system may experience downtime, slow updates, or integration failures. These challenges must be carefully managed to ensure that the Patient Management System remains reliable and effective for hospital operations.

## 2.4 DevOps and CI/CD

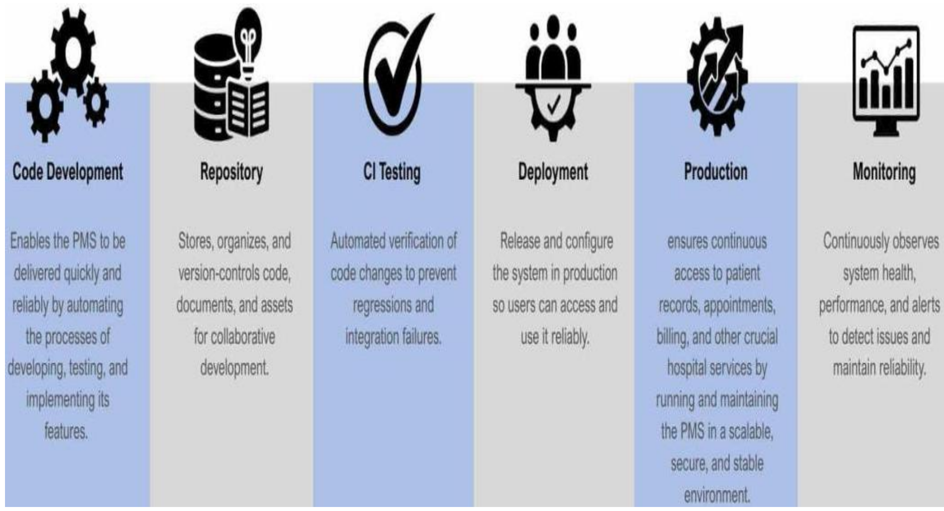


Figure 9 DevOps and CI/CD

1. **Code Development**

   **What happen:** Developers work with doctors, nurses, and administrative staff to create or improve parts of the Patient Management System (PMS), like patient registration, electronic medical records, scheduling appointments, and beddings.

   **Purpose:** To make sure the PMS really helps hospitals with things like speeding up patient registration, admissions, making records more accurate, and retrieving information easier.

**Key activity:** Writing and updating code based on how things work in the hospital, feedback from staff, and changing hospital rules.

2. **Repository**

   **What happen:** A central repository (like GitHub, GitLab, or Bitbucket) stores and keeps track of all PMS source code, updates, and technical documentation.

   **Purpose:** to make it possible for the development team to collaborate effectively, keep an accurate change history, and quickly roll back earlier iterations when necessary.

   **Key activity:** Clear documentation management within the repository, including branching, merging, and committing.

3. **CI Testing**

   **What happen:** Automated pipelines test PMS modules (such as registration forms, patient records, appointment scheduling, and beddings) whenever developers submit code to the repository.

   **Purpose:** to avoid bugs or integration issues that might compromise hospital workflows or the accuracy of patient data.

   **Key activity:** setting up automated regression, integration, and unit tests for each of the PMS's essential features.

4. **Deployment**

**What happen:** Updated PMS builds are made available to staging and production servers after passing testing, allowing hospital staff to use new features without interruption.

**Purpose:** To provide enhancements, like improved billing screens or quicker patient lookups, consistently and with the least amount of downtime possible.

**Key activity:** executing deployment scripts, setting up environments, and packaging builds for PMS servers.

5. **Production**

**What happen**: Serving nurse, administrators, and patients, the live patient management system operates in a reliable and secure setting.

**Purpose:** to ensure that pharmacy orders, consultation notes, and patient registration are always correct and available.

**Key activity:** Maintaining database performance, applying security patches, and keeping an eye on uptime even during busy hospital hours.

6. **Monitoring**

**What happen:** Monitoring dashboards, analytics, overview, and resource utilization allow for constant observation of the PMS.

**Purpose:** to keep staff and patients reliable and promptly identify problems.

**Key activity:** adjusting resources, checking logs, and setting up alerts to keep the system reliable and quick.

## 2.5 Relevant Studies and Research

**Local**

**“Online Scheduling System for Doctors and Patients in a Hospital”**

**Author:** Marion Rylan Q De Guzman, Jonn Louis N Ordoñez, Ronald O Somocierra, Gloren S Fuentes

**Year:** 2021

The Outpatient Department (OPD) of a hospital is where none emergency patients with issues visit for consultations without the need for being admitted. It consists of various doctors who specialize in different areas, as well as specific consulting rooms and machinery. The focus of this research is to create an online appointment scheduling system for the OPD. The system will provide ease and organization in booking appointments. Patients will be able to view the availability of doctors and book their appointments according to it, thus saving them time from waiting long at the hospital. Doctors, too, can view their day's set of appointments, enabling them to have a better grip on their schedules. (https://ieomsociety.org/proceedings/2021monterrey/277.pdf)

**“Dental Clinic Management System”**

**Author:** Rose Jean P. Bayot, Rona Jane L. Desoyo, and Gina A. Mangyao

**Year:** 2023

A Dental Clinic Management System (DCMS) is a comprehensive software solution designed to streamline and enhance the efficiency of dental clinic operations. This capstone project involves the development of a robust and user-friendly system that integrates various functionalities to facilitate the smooth functioning of a dental clinic. (https://www.scribd.com/document/638765050/Dental-ClinicManagement-System-Capstone-Project)

**“Implementing Electronic Health Records”**

**Author:** Anton Elepaño, Carol Stephanie Tan-Limp, Regine Ynez De Mesa, Mia Rey, Josephine Sanchez, Leonila Dans, Antonio Miguel Dans

**Year:** 2025

This implementation study used an explanatory mixed methods design. Two EHR systems were deployed: an Open Medical Records System (OpenMRS)-based platform in 2016, and a Microsoft-based system in 2021. Both systems integrated clinical documentation, pharmacy, laboratory, and reporting modules. Implementation strategies included training workshops and materials, iterative user feedback loops, and infrastructure cofinancing with local governments. Surveys were administered yearly to all end users.
(https://pmc.ncbi.nlm.nih.gov/articles/PMC12283060/)

**Foreign**

**“Electronic Patient Management”**

**Author:** Enebeli Margaret Ifelunwa

**Year:** 2021

This project title is written to help hospitals especially Lagos State University Teaching Hospital (LASUTH), Lagos state in the areas they encounter problems in keeping their attendance scheme for patient and the solution given to tackle problem such as transforming the existing manual attendance scheme for patients system in which the existing problems involved at the time was laziness of the Doctors to work, misplacement of files, excessive loitering around of patient for their files and loitering of paper in the office. This software reports on our pilot evaluation of an Electronic Patient Management System and their Doctors. The aim is to improve the quality of care to patients and the information about them, as indicated by an improvement in the effectiveness and efficiency of care and in an increase in patient satisfaction. (https://www.scribd.com/document/517153335/Patient-Managementsystem )

**“Developing A Computer-Based Record Management System”**

**Author:** Tomas U. Ganiron Jr.

**Year:** May 2023

This paper is to develop a computerized program for patient medical and dental record management system. The system integrates clinical, scheduling, electronic medical record, and data consolidation/reporting components that enable clinics to provide patients with quality care in a timely and cost-effective manner. As a result, the patient record management system offers a number of benefits to the user and can capture data, store, view, add and delete the records entered the data can also be posted information to the database. the notion of user made it easy for end users to save their files without a help of a flash drives.

(https://www.researchgate.net/publication/370607649_Developing_A_Computer-Based_Record_Management_System)

**“Patient Records Management System in Magomeni Hospital”**

**Author:** Rogers Felix

**Year:** July 2023

The Patients Records Management System has been designed to help so many Health Centers in Tanzania that need to computerize their operations to improve services delivered to their customers and increase the productivity of the entire organization. This system seeks to provide hospital employees with a new state-of-the-art, simple-to-use software since every basic concept, method and procedure has been explained fully in a precise clear language that is understood by many, and that is systematic and procedural. The project design work was done using a well-researched methodology according to the nature of the problem stated and the requirements or needs which are intended to be met, which avails the user with important illustrative information and content.

(https://www.researchgate.net/publication/372131512_Patient_Records_Management_System_in_Magomeni_Hospital)

## 2.6 Integration of Information Systems in Enterprise Environment

Various systems are used in hospitals because they have different departments to handle various operations. Making these systems stand alone may result in mistakes, slowness, and delays. Integration in relation to Information System (IS) implies a connection between modules and systems, such that the information flows freely between departments. The combination of the PMS with the rest of the systems will ensure that information flows automatically further streamlining the processes, increasing their accuracy, and making them faster and smoother. A healthcare organization has a collection of Information Systems (IS) such as Doctors and Nurse Management, Billing and Insurance Management, Pharmacy Management, and Laboratory and Diagnostic Management, including Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Human Resource Information System, that are integrated in a single workflow, which is the Patient Management System (PMS). Some of the basic hospital activities that are handled by the PMS are patient registration, appointments, bills, and medical records. This data is then shared with other systems by use of integration.

As an example, ERP and PMS collaborate to handle finances of the hospitals, medication stocks, and purchases of medical supplies. CRM is incorporated to track follow-ups and interactions with patients, and to improve the patient experience through a certain communication. HRIS controls scheduling, payment, and distribution of the workload of staff based on the needs of a ward and appointments of patients. This integration leads to better decision making, and results in higher efficiency, patient satisfaction, and hospital performance.

# CHAPTER 3

## METHODOLOGY


Figure 10 SDLC Methodology

1. **Analysis Phase -** In this phase, the researchers gather and analyze all relevant information from the intended users and stakeholders of the system, such as hospital staff, administrators, and other end users. Different requirement-gathering techniques are used, including interviews, observations, and document reviews, to fully understand the existing problems, workflows, and system limitations.
2. **Design Phase -** The Design phase transforms the requirements in the analysis phase into a step-by-step plan of the system. In this

stage, the researchers will decide on how the system will be constructed and how all the parts will interact with one another. Their role is to design the overall architecture such as database structure, software modules, data flow, and the connection of the parts. Additionally, this phase includes the design of the user interface to ensure the system is user-friendly and easy to navigate for hospital staff and patient.

3. **Implementation Phase -** The Implementation phase is where the actual system development takes place. At this phase, the programmer begin writing the program code based on the approved design from the previous phase. Appropriate programming languages, frameworks, and development tools are used to develop the system's features and functionalities. Each module of the system is coded, integrated, and configured to ensure proper operation. The database is created and connected to the system, and interfaces are implemented according to the design.

4. **Testing Phase -** The Testing phase involves a thorough evaluation of the developed system to identify and correct errors, bugs, and performance issues. During this phase, the researchers review the system's code and test all functionalities to ensure that the system works as intended and meets the specified

requirements. Each feature is checked to confirm that inputs produce the correct outputs and that the system performs reliably under normal operating conditions.

5. **Deployment Phase -** Deployment phase is the stage at which the complete and test system is availed to the users. During this stage, the system is mounted and put into use in the hospital where it can be used by the staff in actual operations.

6. **Maintenance Phase -** Maintenance phase is concerned with the continuous maintenance and enhancement after the launch of the system. The researchers or system administrators monitor the performance of the system, correct the issues reported and make frequent updates on the system to ensure the system is working well. Feedback from users is collected to identify possible enhancements or new features that can improve system usability and performance.

### 3.1 Agile Scrum Methodology in the Project


Figure 11 Agile Scrum Methodology in the Project

The project of the Patient Management System will be led by Scrum, which will offer a formal and at the same time a flexible model of managing the development. The initial step of the project is to create a product backlog that lists all the required features such as patient registration, appointment booking, beddings, and electronic medical records. The Scrum team then pick and develop a given backlog items into a working software in sprint, which are short development cycles averaging between two and four weeks. The review meeting is held at the end of every sprint to demonstrate the progress, such as an appointment module, and receive feedback about the work done by the stakeholders, hospital administrators, and physicians. Development team remains on track through daily stand-up meetings that help them to spot and fix problems within a short period of time. To achieve continuous improvement within the project, the team considers what was good and what can be done better in a retrospective that is conducted after every sprint. To reduce risk, adapt to changing hospital requirements, and ensure that the developed system is reliable, usable, and conforming to the healthcare needs, the patient management system may be developed in the form of small and delivery-specific increments with the help of Scrum.

The Patient Management System is developed using Scrum and not any other project management approach due to its high flexibility in meeting the needs of the healthcare organization which are complex and dynamic. In contrast to the traditional models, like the Waterfall model, which presupposes the strict adherence of phases, Scrum enables the project to be developed in iterative sprints, which means that it is easier to release small functional modules, such as patient registration or appointment booking, at the initial stages of the process. This will make testing and feedback by hospital staff fast and enable the team to respond to the requirements change before the project is finalized. Since Scrum is based on very specific sprint cycles, reviews, and retrospectives, which focuses on continuous flow of tasks without strict timeboxing. This guarantees monotony and transparency. Scrum also promotes high collaborations and stakeholder engagement and relentless improvement that is vital in creating a dependable Patient Management System where accuracy, usability, and responsiveness are paramount.

### 3.2 Roles and Responsibilities

| ROLES | NAME AND CONTACT INFORMATION | RESPONSIBILITIES |
|---|---|---|
| Scrum Master | Princess Anne Norbie A. Sikat princessannsikat@gmail.com | Scrum Master is the team member who acts as the facilitation and coordinator of the group. Serve as a bridge between stakeholders and the team. |
| Programmer | Jhon Robert B. Delmigues jhonrobertdelmigues@gmail.com | A programmer is a professional who writes tests and maintain codes that tells a computer how to perform specific tasks. Who takes the conceptual ideas for a system, how it should look, and how it should |

| | | |
|---|---|---|
| | | behave and make them a reality? |
| System Analyst | Evelyn B. Suarez evelyn37467@gmail.com | A System Analyst is a professional who studies, analyzes, and designs information systems to solve problems and improve processes within an organization or project. Work closely with programmers to ensure correct implementation. |
| Business Specialist | Mario L. Balangatan mariobalangatan2@gmail .com | A Business Specialist is the member of the team who focuses on the business side of the system being developed. Who begin by gathering and analyzing stakeholder and business requirements to fully understand the needs and |

| | | |
|---|---|---|
| | | objectives of the project. |
| Document Specialist | Jonel Cristian C. Balute jonelcristianbalute@gmail.com | A Document Specialist is the responsible for organizing, preparing, and managing all project related documents. A key part of their job is to ensure that all documentation is consistent in its formatting and style, maintaining a professional and unified appearance across all materials. |

Table 2 Roles and Responsibilities

### 3.3 Sprint Cycles

| SPRINT | FEATURES | TASKS/DEVELOPED | DURATION |
|---|---|---|---|
| Sprint 1 | Dashboard | • Create UI Design<br>• Layout for the analytics and the overview. | 15 days |
| Sprint 2 | Patient Registration | • Create UI Design Registration<br>• Design patients table | 17 days |

| | | | |
|---|---|---|---|
| | | • Implementation validation<br>• Add Patient list function for inpatient and outpatient<br>• Working patient registration and search<br>Patient data stored in database | |
| Sprint 3 | Appointment Scheduling | • Design appointment table<br>• UI for scheduling appointment<br>• Functional appointment booking system<br>• View of scheduled appointment in list type<br>Filter by doctor, patient and date | 19 days |
| Sprint 4 | Beddings and Linens | • Design a bed list<br>• UI for beddings and linens<br>• Enable linen request button<br>• Display assigned patient | 13 Days |

| | | | |
|---|---|---|---|
| | | and available bed | |
| Sprint 5 | Patient Portal | • IU Dashboard<br>• Layout for view billing, medical history and appointment history. | 14 Days |

Table 3 Sprint Planning

**Daily Standups**

| TEAM MEMBER | ACCOMPLISHMENTS | TASKS | IMPEDENTS/ CHALLENGES |
|---|---|---|---|
| Scrum Master | Facilitated team progress | Support ongoing tasks | None |
| Programmer | Completed UI Dashboard | Start working on other features | Waiting for Feedback |
| System Analyst | Defined data requirements | Work on design of modules | Need access to a specific resource |
| Business Specialist | Provided Business insights | Review curriculum module progress | None |
| Document Specialist | Updated documentation | Organize and maintain documentation | None |

Table 4 Daily Standup

## 3.4 Scrum Artifacts

### Product Backlog

| USER STORY NO. | USER STORIES | USER STORIES PRIORITIES | REVISED PRIORITY | STATUS |
|---|---|---|---|---|
| 1. | As a patient, I cannot access my account due to an incorrect password or a forgotten password. | 1 | - | Done |
| 2. | As a patient, I want to see where room or bed I assign. | 1 | - | Done |
| 3. | As a patient, I couldn't view or see my bills and history. | 1 | - | Done |
| 4. | As an admin, the patient records do not display the full medical history or latest update. | 1 | - | Done |
| 5. | As an admin, I am having difficulty viewing all the patient's medical history. | 1 | - | Done |

| | | | | |
|---|---|---|---|---|
| 6. | As an admin, I don't want other stuff to see or edit sensitive sections. | 1 | - | Done |
| 7. | As an admin, I want to update patient status or if they discharge. | 1 | - | Done |
| 8. | As an admin, I want to see available beds so that I can assign them to new patients. | 1 | - | Done |
| 9. | As an admin, the results that uploaded are not matched with current consultation. | 1 | - | Done |
| 10. | As an Admin, I can't view the patient medical history. | 1 | - | Done |

Table 5 Product Backlog

**Sprint Backlog**

| FEATURES | NAME & ROLE | DURATION |
|---|---|---|
| Login | Jhon Robert Delmigues<br>Programmer | 6 Days |
| Dashboard | | 15 Days |
| Patient Registration | | 17 Days |
| Appointment Scheduling | | 19 Days |
| Beddings and Linens | | 13 Days |
| Patient Portal | | 14 Days |

Table 6 Sprint Backlog

## 3.5 Integration Approach for Information System

The PMS integration approaches the way how the internal modules interact with each other to provide a unified hospital workflow. Each of these modules is connected through a shared database and internal APIs to ensure that information flows smoothly across departments. The workflow starts with the Patient Registration module, where essential information is stored. Once registered, the patient is automatically added to the Patient List module as inpatient, which serves as the centralized directory of all the admitted or discharged patients. From this list, staff can access records to manage Appointment Scheduling, where doctor's availability is displayed in real time. During consultations, relevant details are stored in the Medical History module, creating a longitudinal record of the patient's past diagnoses, treatments, and medications. Finally, all services availed by the patient, are consolidated in the Billing Summary, ensuring accurate financial records and transparent charging. By integrating these modules, the PMS eliminates data silos, prevents duplication of information, and ensures that all departments access a single source truth.

### 3.6 Microservices

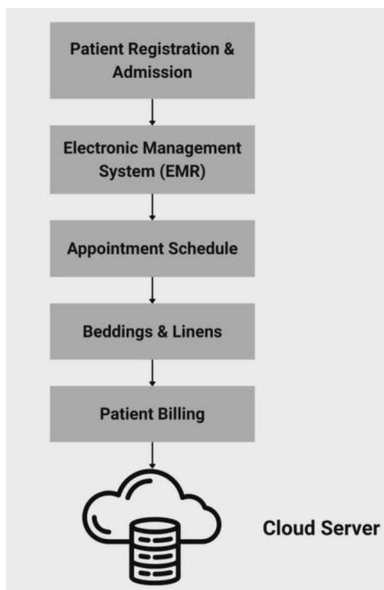


Figure 12 Microservices

- **Patient Registration**

  This module is in charge of gathering and storing patient information, like their name, address, phone number, previous medical history, and insurance information. It makes sure that each patient has a unique profile in the system so that they can be easily found and referred to later.

- **Appointment Scheduling**

  The scheduling of patient appointments with doctors or specialists is handled by this module. It helps avoid scheduling conflicts for doctors and prevent double booking.

- **Beddings & Linens**

  The availability and assignment of hospital beds are tracked and managed by this module. It makes it easier to assign rooms or wards to patients effectively by keeping track of occupied, available, and under maintenance. It guarantees appropriate use of hospital resources and helps cut down on admissions delays.

- **Electronic Medical Records**

  The medical history of a patient, including diagnoses, prescriptions, test results, allergies, and treatment plans, is available digitally in this module. Better decision-making and care continuity are ensured by enabling authorized healthcare professionals to swiftly access current and accurate patient data.

- **Patient Billing** (Patient Portal)

  The patient bills module allows patient to view billing records and statements such as consultation fees, medications, and other hospital expenses. It keeps track of hospital stays, prescription drugs, lab work, consultation fees, and other services. It provides transparency in hospital charges but does not allow editing and processing of payments.

### 3.7 Introduction to TOGAF and the four Architectural Domains

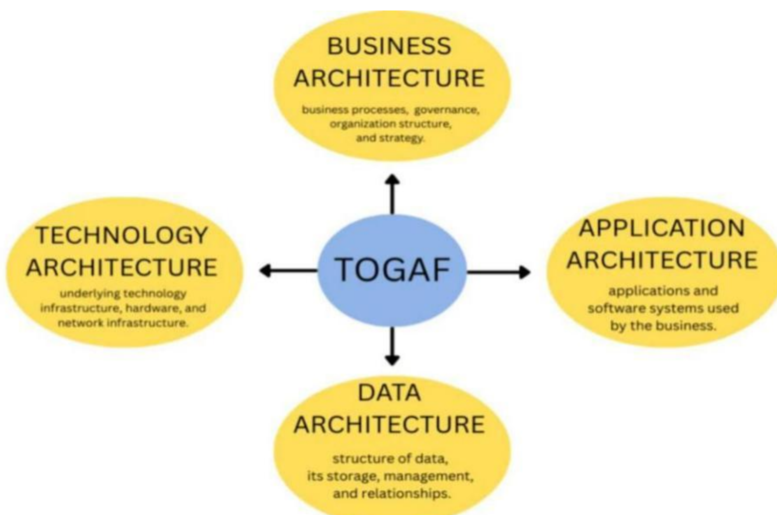


Figure 13 TOGAF Framework

The Open Group Architecture Framework (TOGAF) is one of the widely used framework in Enterprise Architecture (EA), it guides organizations in aligning its goals with technology solutions, and provides a structured method, called the Architecture Development

Method (ADM), which guides the planning, design, implementation of IT solutions. It offers a methodology to align Information Technology (IT) with business goals, ensuring that systems are not only technically sound but also support the organization's long-term vision and strategy (The Open Group, 2022).

In Patient Management System (PMS), TOGAF serves as a guiding framework that ensures the proposed system is well-integrated, scalable, and adaptable for future needs. By applying TOGAF, the PMS can be architected in a way that reduces redundancy and ensures that all integrated systems can work and talk together. TOGAF has four architectural domains that provide an invaluable structured approach for guiding the design integration of the microservices-based system within an enterprise context.

**Business Architecture**

The Business Architecture, this defines the business strategy, and key business processes. For PMS, it is centered on automating and optimizing a specific operational process (registration, bed assignment, discharge, appointment, and medical history). It defines the administrative processes, such as patient registration, electronic medical records, appointment scheduling, and patient billing, that the PMS support.

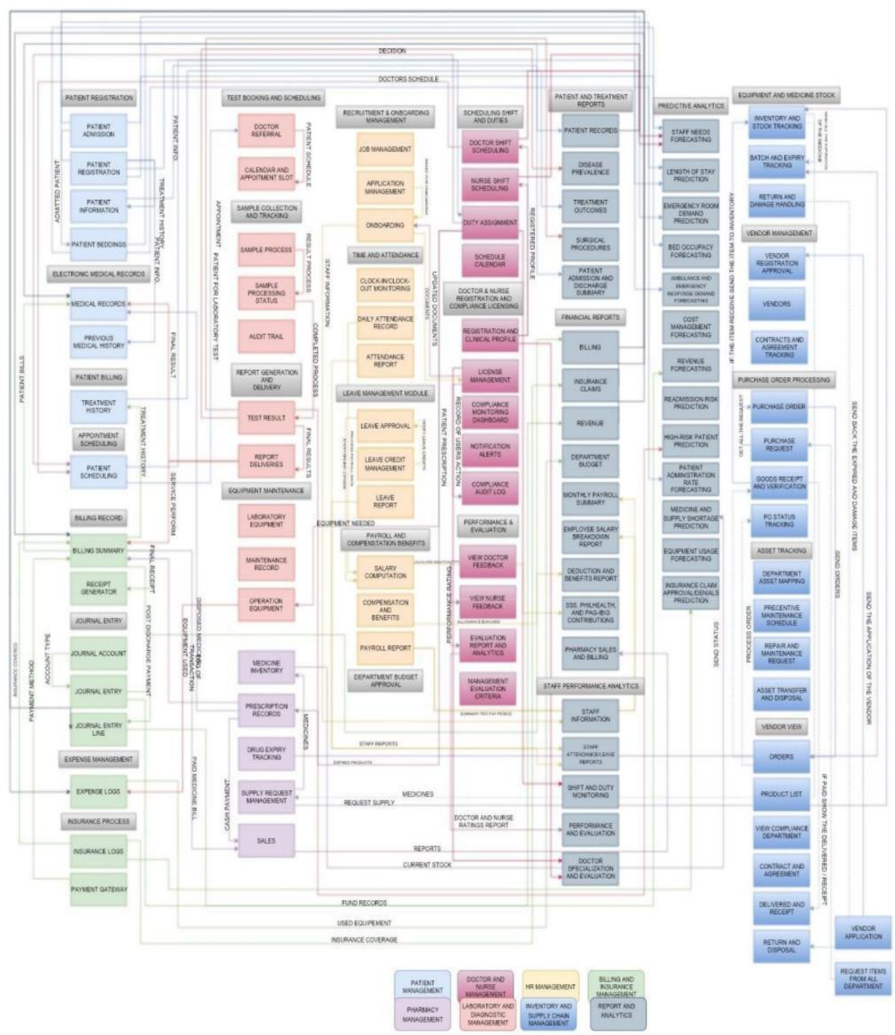

Figure 14 Business Architecture

**Data Architecture**

The Data Architecture, this focuses on how data is stored, and managed. For PMS this involves designing the patient database, implementing encryption for sensitive information, and enabling secure data sharing across hospital departments. It outlines how patient data, including medical history, lab results, and personal information, is structured, stored, and managed, ensuring data integrity and security. This ensures that all of the information is consistent, secure, and accessible, when the other department needs it.

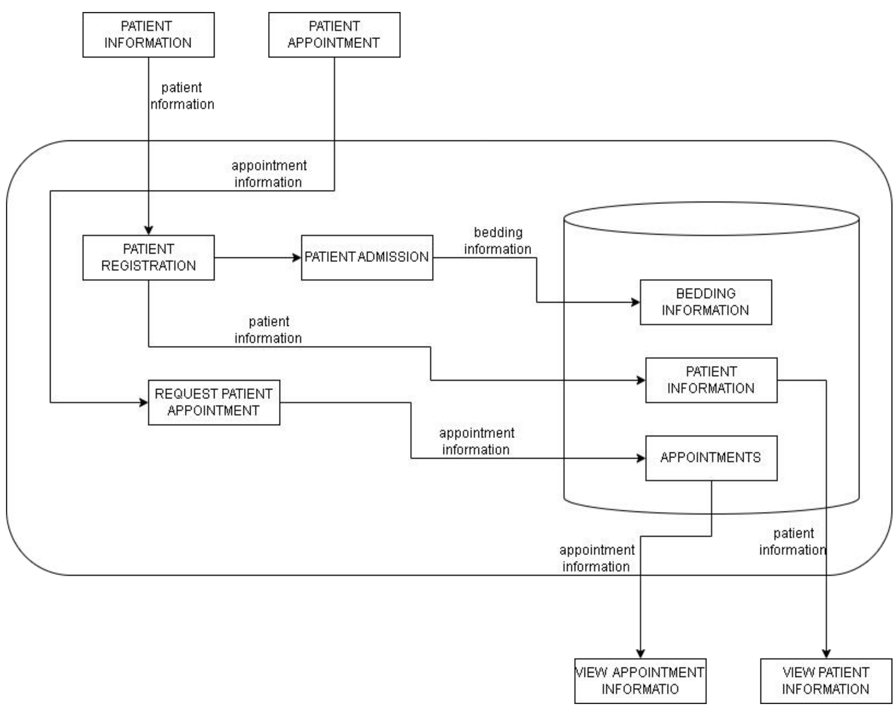


Figure 15 Data Architecture

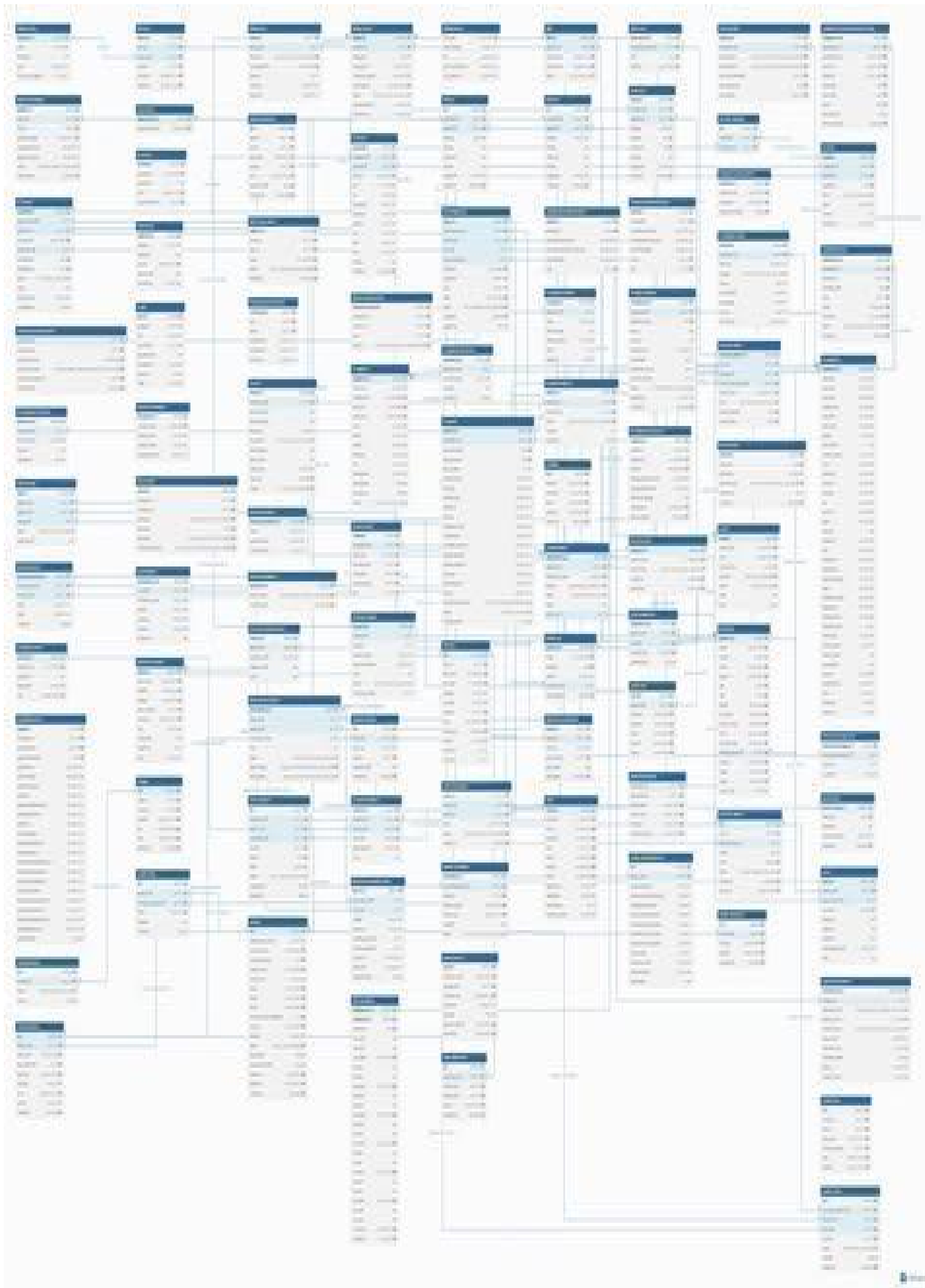

Figure 16 Database Schema

**Application Architecture**

The Application Architecture specifies the modules within the system and with different system components the hospital uses and how they will talk and interact with each other. The PMS manages patient registration, scheduling, records and beddings, while also integrating with other systems. Like the PMS is integrated with Doctors and Nurse Management for appointment scheduling, Laboratory for managing test results, with Pharmacy for prescription information, and with Billing for viewing the billing history. Where this application communicates using APIs to share patient data automatically without manual data entry.

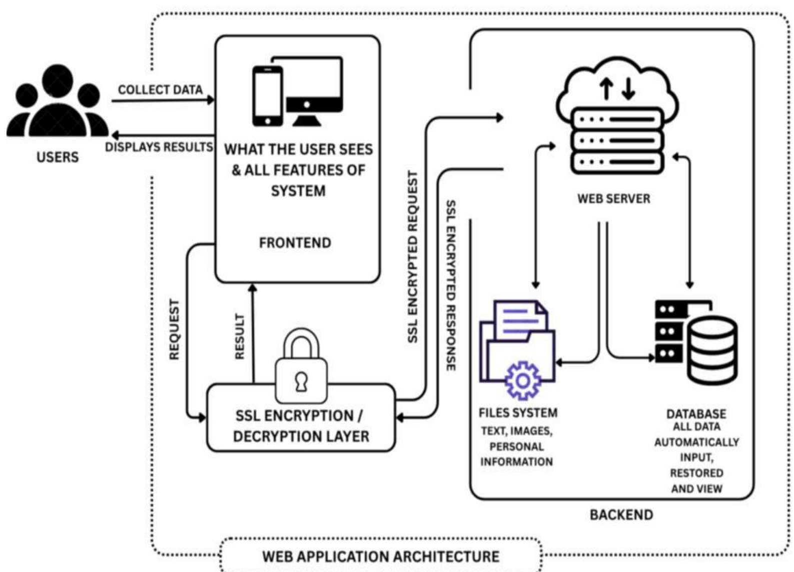


Figure 17 Application Architecture

**Technology Architecture**

The Technology Architecture defines the underlying hardware, software infrastructure and technologies, including servers, databases, and network protocols, required to deploy and support the system. In PMS this includes the use of APIs for integration with external systems, CI/CD pipelines for continuous deployment, and security protocols such as firewalls and role-based access controls. By mapping the PMS design and implementation against these four TOGAF domains, the system ensures a holistic approach to enterprise system development. And with this approach it significantly increases the system's relevance, sustainability, and potential for successful integration into a real enterprise environment. It ensures the technology is reliable, scalable, secure, and fast enough to handle the workload.

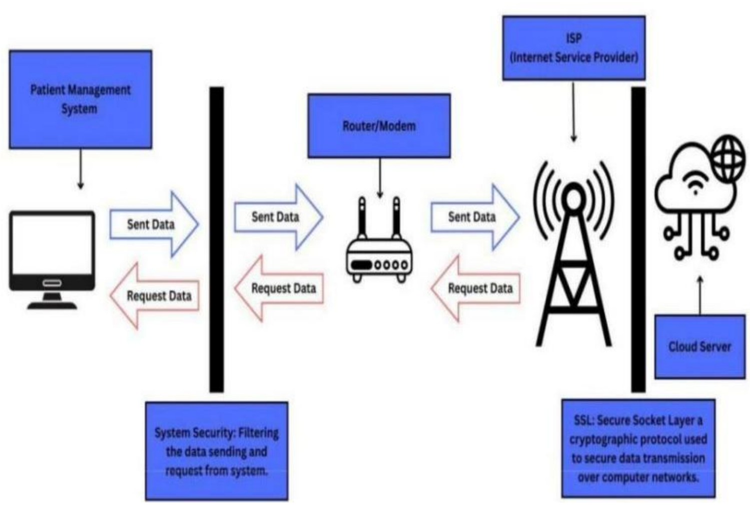


Figure 18 Technology Architecture

# CHAPTER 4

## SYSTEM DEVELOPMENT AND IMPLEMENTATION

### 4.1 Requirements Analysis

Requirements Analysis stage is of significant importance in the development of the Patient Management System (PMS) as it determines the basis in which the system should perform and how it will do it in the hospital settings. In this stage, the research team collected specific knowledge of a broad scope of the stakeholders, such as the hospital administrators, reception staff, nurse and patient. The interviews, direct observation, and close examination of pre-existing processes allowed the team to outline the following issues in the current manual or semi-digital system, the delay in accessing patient records, duplication of data, and the inability to trace patient data. Main aim of this phase is to ensure clarity of both the functional and the non-functional requirements so that the completed system may be efficiently applicable in the hospital running processes and enhance service delivery and patient care.

The functional requirements list the characteristics and functionalities that the system should offer. PMS should enable users with authenticated access to add new patients, update personal data, view previous medical history and manage records at ease. The scheduling of appointments should flow smoothly so that the authorized staff is free to make a schedule, reschedule, or cancel appointments. Also, the bed and linens module enables personnel to view the available beds, move the patients between rooms or beds, and request linens. In order to ensure security and proper access, the system should have functions to manage users, such as role-based access control and secure functions to logout/log in the system.

Non-functional requirements are requirements that describe the performance of the system in the real life scenario. Security is the most important; the data of patients should be secured with the help of the strong authentication, encrypted storage and role-based access control. The system is to record the idle users automatically to avoid unauthorized access. Performance is also essential the system should be able to respond fast to user requests and within a system there should be a number of users on it without delays. Reliability needs frequent backup of databases, as well as the low downtime and the efficient error-handling mechanisms. It must also be usable; the interface should be user-friendly and should not require an extended period of training. Lastly, scalability should be taken into consideration to make sure that the system can handle the number of patients and can also support the possible future additions of other healthcare modules.

### 4.1.1 Stakeholder Identification

Stakeholder identification in Patient Management Systems (PMS) is an essential process that entails recognizing all individuals and groups with a vested interest in the effective implementation and functioning of the system. Internally, this encompasses healthcare personnel such as nurses, and administrative staff who utilize the system on a daily basis to oversee patient records, appointments, and clinical workflows. The management of the hospital or clinic plays a pivotal role in endorsing the system, distributing resources, and ensuring its alignment with organizational objectives and healthcare standards.

External stakeholders may encompass software vendors or system developers responsible for designing, customizing, and maintaining the patient management system. Government and regulatory agencies also play a vital role as stakeholders, as they set forth healthcare regulations, data privacy laws, and compliance requirements that the system must adhere to. Moreover, patients are essential stakeholders since the system has a direct impact on the accuracy of their medical records, the efficiency of service delivery, and the overall patient experience. Recognizing and involving all pertinent stakeholders guarantees that the patient management system operates efficiently, remains compliant, is user-friendly, and has the potential to enhance healthcare delivery and organizational performance.

**Internal Stakeholders**

- Patients: Primary beneficiaries; their medical records, care coordination, and service experience are directly impacted.
- Nurses: Use the system for clinical documentation, diagnosis, and treatment planning.
- Administrative Staff: Manage patient registration, scheduling, and record maintenance.
- IT Department: Implements, maintains, and secures the system while providing technical support.

**External Stakeholders**

- Software Vendors/System Developers: Design, supply, and maintain the patient management system.
- Government and Regulatory Agencies: Set standards for healthcare compliance, data protection, and patient privacy.
- Insurance Companies/Providers – Use system data for, claims processing, and verification of medical services.

- Third-Party IT Service Providers – Offer system hosting, cloud services, cybersecurity, and technical support.
- Accrediting Bodies – Ensure the system supports healthcare quality standards and certification requirements.

### 4.1.2 Requirements Gathering Techniques

The process of requirements gathering for a Patient Management System (PMS) entails the identification of functional, technical, and user requirements essential for the development of an efficient, secure, and user-friendly system aimed at managing patient information and healthcare processes. A variety of techniques may be employed to collect these requirements, including:

- **Surveys and Questionnaires**

  These are distributed to administrative staff to collect structured feedback on system usability, required features, workflow challenges, and expectations for digital patient management.

- **Observation**

  Direct observation of the daily operations within a hospital encompassing patient admission, consultation, discharge, facilitates the identification of inefficiencies, data entry problems, and opportunities for enhancing accuracy and service delivery through automation via the PMS.

- **Focus Groups**

  Focus group discussions with representatives from different hospital departments encourage open dialogue about system requirements, user experience issues, and suggestions for improving patient registration, scheduling, and record management.

- **Interviews**

  One-on-one or group interviews conducted with healthcare professionals, and hospital administrators yield comprehensive insights into existing system limitations, data management requirements, regulatory issues, and the functionalities that are sought after in the system.

- **Pilot Testing and Feedback Collection**

  Implementing the system in a limited environment or department allows users to provide real-world feedback, helping identify additional requirements or necessary improvements before full deployment.

### 4.1.3 User Stories and Use Cases

| STORIES NUMBER | STORIES | STORIES PRIORITIES | REVISED PRIORITY | STATUS |
|---|---|---|---|---|
| Task No.01 | As a Hospital Admin, I want to manage patient records in a centralized system, so that patient information is accurate, updated, and easily accessible by authorized staff. | 4 | – | Done |
| Task No.02 | As a Patient, I want to register and book appointments online, so that I can schedule consultations conveniently without visiting the hospital in person. | 4 | – | Done |
| Task | As a Doctor, I want to | 5 | – | Done |

| | | | | |
|---|---|---|---|---|
| No.03 | view my patients' medical history and test results, so that I can make informed clinical decisions during consultations. | | | |
| Task No.04 | As a Nurse, I want to update patient vital signs and treatment notes in real time, so that care teams are informed of the patient's current condition. | 4 | – | Done |
| Task No.05 | As an Admin, I want patient treatment data to automatically generate financial reports, so that hospital accounting is precise and up to date. | 4 | – | Done |

Table 7 User Stories and Use Cases

**Use Case 1**: Patient Registration

**Precondition**: Hospital Admin has access to the hospital website or registration system.

**User Actions:**

- Hospital Admin creates patient an account and fills in personal, contact, and insurance information.
- Hospital Admin selects a department, doctor, or specialist for consultation.
- Hospital Admin chooses the preferred date and time for the appointment.
- Reception staff can view scheduled appointments and adjust if needed.

**Postcondition:**

Patient is successfully registered and has a confirmed appointment, reducing administrative workload and wait times.

**Use Case 2**: Managing Patient Medical Records

**Precondition:** Patient is registered in the system.

**User Actions:**

- Doctors and nurses access the patient’s profile to view medical history, past diagnoses, allergies, and medications.
- Lab results, imaging reports, and prescriptions are uploaded to the system.

- System maintains version history of updates for audit purposes.
- Authorized staff can search for patient records using filters like name, or patient ID.
- System ensures data privacy and restricts access based on user roles.

**Postcondition:**

Patient medical records are accurate, up-to-date, and securely accessible to authorized personnel.

**Use Case 3:** Tracking Patient Treatment and Care

**Precondition:** Patient is admitted, undergoing treatment, or scheduled for ongoing care.

**User Actions**:

- Nurses record vital signs, symptoms, and daily observations.
- System tracks medications, dosages, procedures, and follow-up schedules.
- Patients can view treatment schedules, medication reminders, and upcoming procedures through a portal.
- Admin generate reports for patients needing extra attention.

**Postcondition**:

Patient care progress is monitored, documented accurately, and communicated efficiently to all authorized staff.

### 4.1.4 Functional Requirements for Integration

| FEATURE | REQUIREMENTS | FUNCTIONALITY |
|---|---|---|
| Data Integration | The system must integrate data from multiple healthcare sources, including patient records, laboratory results, and imaging reports, | Data Gathering: A unified repository for storing and accessing patient data from various systems. |
| User Authentication | Single sign-on (SSO) for healthcare staff. | Integration with hospital authentication system for secure access. |
| Appointment & Scheduling Management | **Appointment Booking:** Allow scheduling of patient consultations with doctors. **Calendar Integration:** View doctor availability and manage time slots. | Efficient scheduling and reduction of missed appointments. |
| Treatment & Care | **Care Plans:** Record and track treatment plans for | Support doctors and nurses in patient |

| | | |
|---|---|---|
| Management | patients.<br>**Medication Management:** Track prescriptions, doses, and refill schedules.<br>**Procedure Management:** Document surgeries, therapies, or diagnostic procedures. | treatment tracking and care coordination. |
| Patient Engagement & Accessibility | **Patient Portal:** Allow patients to view medical records, lab results, and care plans. | Improve patient experience, engagement, and accessibility to care. |
| Compliance & Security | **Regulatory Compliance:** Track mandatory trainings, patient privacy, and safety protocols.<br>**Audit Trails:** Maintain records of patient data access and updates. | Meet healthcare regulations and ensure patient data security. |
| User | **Patient Profiles:** Store | Centralized |

| | | |
|---|---|---|
| Management | patient demographics, medical history, allergies, and current medications.<br>**Role-Based Access:** Define roles (doctor, nurse, patient, admin) with specific permissions.<br>**Self-Service Portal:** Allow patients to view appointments, test results, billing history, bed assign and treatment plans. | management of user accounts and permissions. |

Table 8 Functional Requirements for Integration

## 4.2 Business Process Architecture

### 4.2.1 Identification of Business Processes

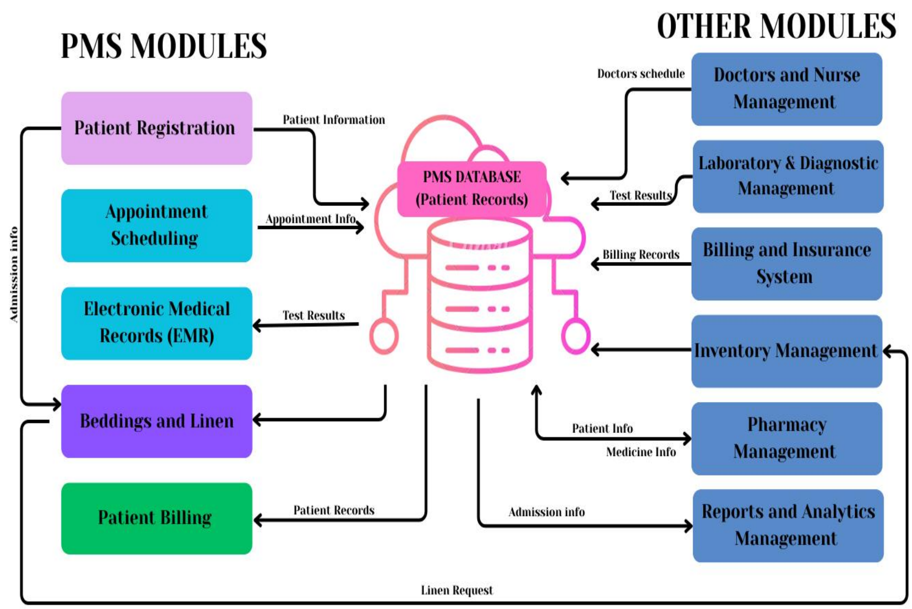


Figure 19 Identification of Business Processes

## 4.2.2 Business Process Diagrams

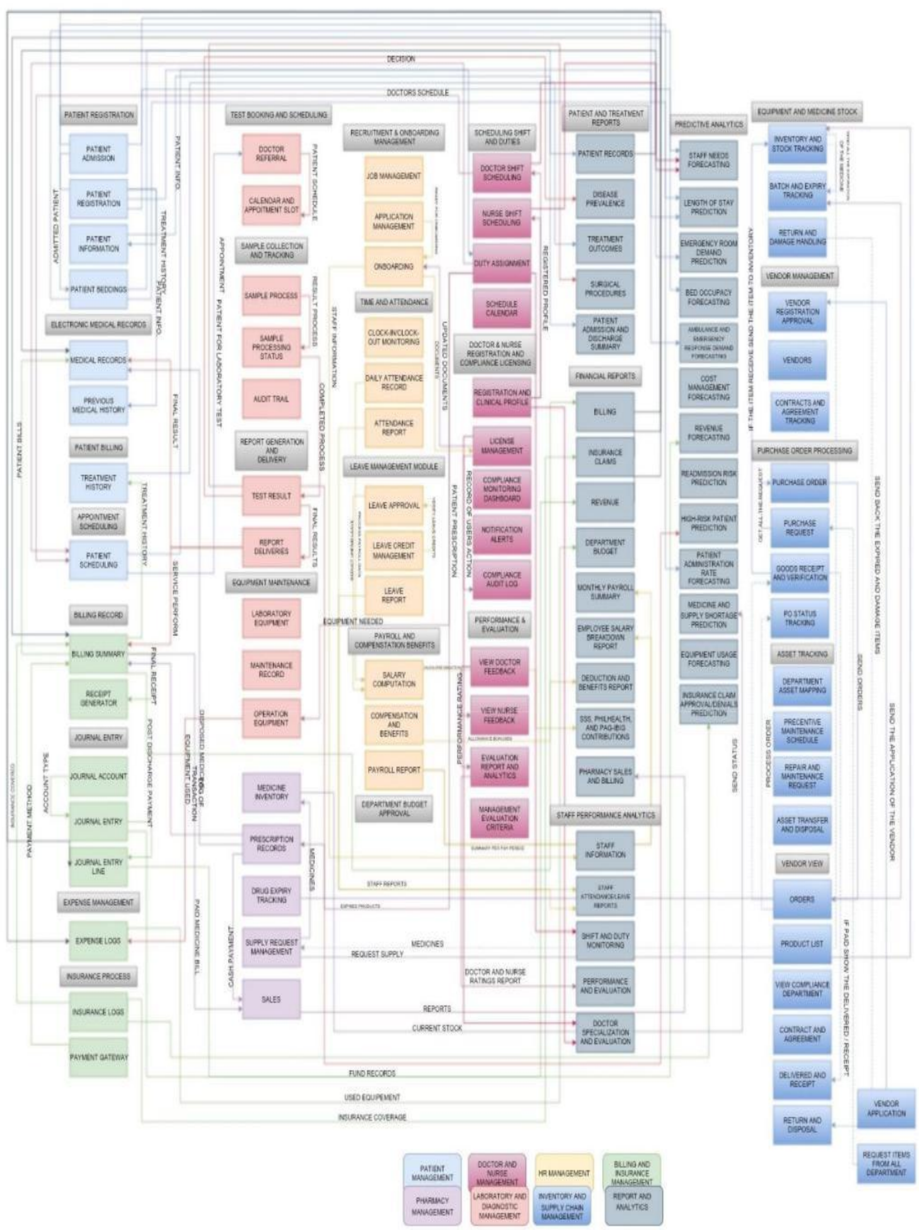

Figure 20 BPA Level 1

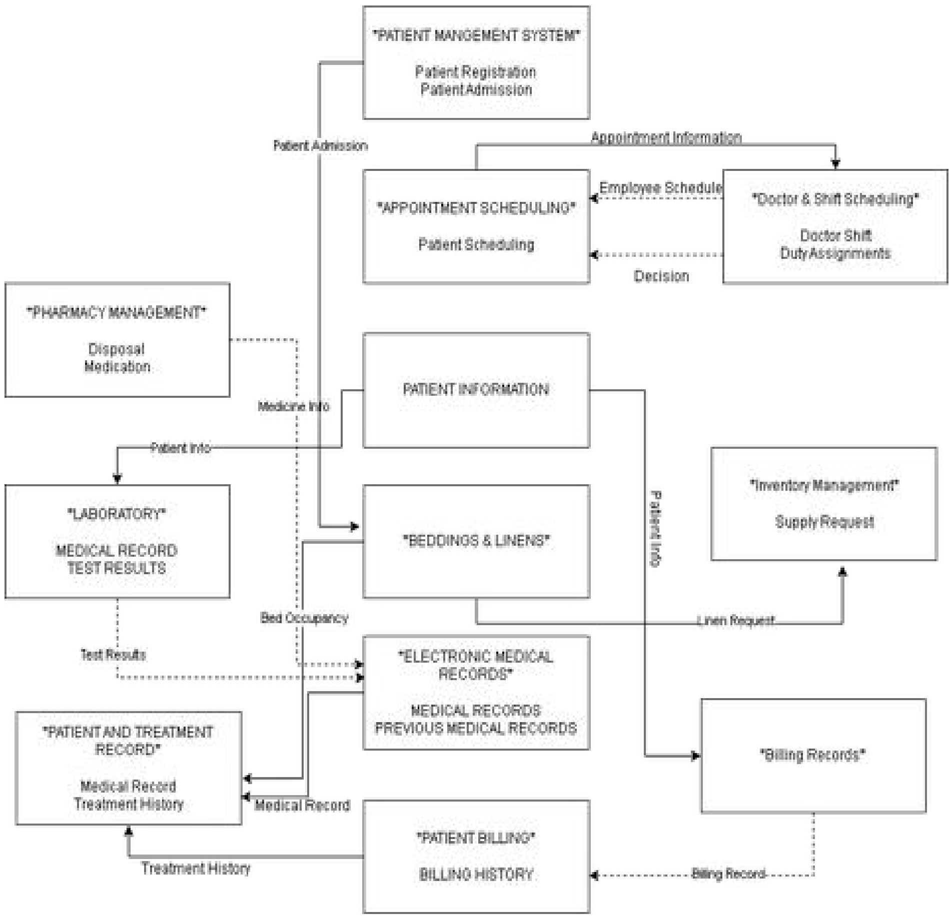


Figure 21 BPA Level 2

### 4.2.3 Alignment of Integrated System with Business Processes

Doctor and Nurse System

Billing and Insurance System

Pharmacy System

Laboratory and Diagnostic System

Inventory and Supply Chain System

Report and Analytics System

1. Doctor and Nurse System

   Contains: Doctor Schedule

2. Billing and Insurance System

   Contains: Billing Records

3. Pharmacy System

   Contains: Medicine Details

4. Laboratory and Diagnostic System

   Contains: Test Results

5. Inventory and Supply Chain System

   Contains: Linens Request

6. Report and Analytics System

   Contains: Patient Information and Bedding Information

### 4.2.4 Business Process Improvements

Developing a computerized Patient Management System resulted in massive enhancement of business processes of the healthcare facility through automation and digitalization of processes which were once manual. By streamlining the patients records on a user-friendly interface, which replacing the manual and paper-based processes with automated and centralized functions. By digitalizing the data entry made registration of patients quicker, more precise, saved time and eradicated duplication of records. A centralized and secure database enhanced the management of medical records where authorized personnel were able to gain access to information about the patient within a short period of time. Electronics records improved consultation and clinical documentation and made them more standardized and legible, which facilitated improves clinical decision-making. Additionally, the integration of all modules reduced redundancy, minimized errors, and improved coordination among healthcare staff. As a result, the system enhanced operational efficiency, supported inform decision-making, and delivered better quality and more reliable healthcare services to patient.

## 4.3 Application Architecture

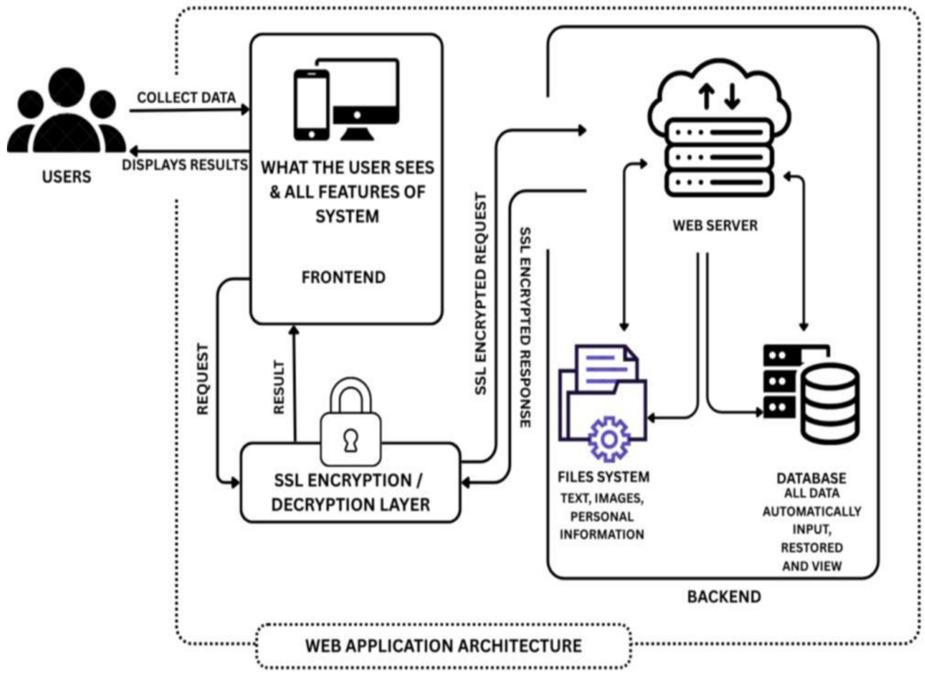


Figure 22 Application Architecture

### 4.3.1 Components of Application Architecture

| COMPONENTS | DESCRIPTION | FUNCTIONALITY |
|---|---|---|
| User Access Component | This component is designed to control how different users enter the system and interact with its features. It defines user roles such as doctors, nurses, | Through this component, users can log in securely, manage sessions, and access system features based on their assigned roles, preventing |

| | | |
|---|---|---|
| | staff, and patients to ensure proper system access. | unauthorized use of sensitive information. |
| Patient Information Processing | At the core of system operations, this component manages the logic behind patient-related activities. It ensures that all actions follow hospital rules and workflows. | Using this component, the system processes patient registration, updates medical records, schedules appointments, assigns beds, and maintains accurate patient status across all departments. |
| Medical Data Storage | Serving as the main data repository, this component stores all medical and administrative information in an organized manner. It supports structured | With this functionality, the system securely saves and retrieves patient profiles, medical histories, laboratory results, billing information, and appointment records |

| | | |
|---|---|---|
| | and reliable data management. | when needed. |
| System Connectivity Component | To enable smooth communication, this component links the patient management system with external and internal services. It supports data exchange across platforms. | As a result, the system can integrate with laboratory systems, pharmacy services, insurance providers, and payment platforms to ensure complete and coordinated patient care. |
| Data Protection and Privacy | Focused on security, this component safeguards sensitive patient and hospital data from threats and misuse. It follows data privacy standards and best practices. | By applying authentication, authorization, encryption, and access control, this component ensures that medical information remains confidential and protected from |

| | | |
|---|---|---|
| | | unauthorized access. |
| System Environment Support | Behind the scenes, this component provides the technical resources required to run the application smoothly. It includes both hardware and hosting services. | Through servers, cloud services, networks, and system configurations, this component ensures the system remains stable, responsive, and available at all times. |
| Performance Tracking and Records | To maintain system quality, this component observes and records system activities. It helps administrators understand system behavior. | By collecting logs, monitoring usage, tracking errors, and analyzing performance data, this component supports system maintenance and continuous improvement. |

Table 9 Components of Application Architecture

### 4.3.2 Application Architecture Diagram

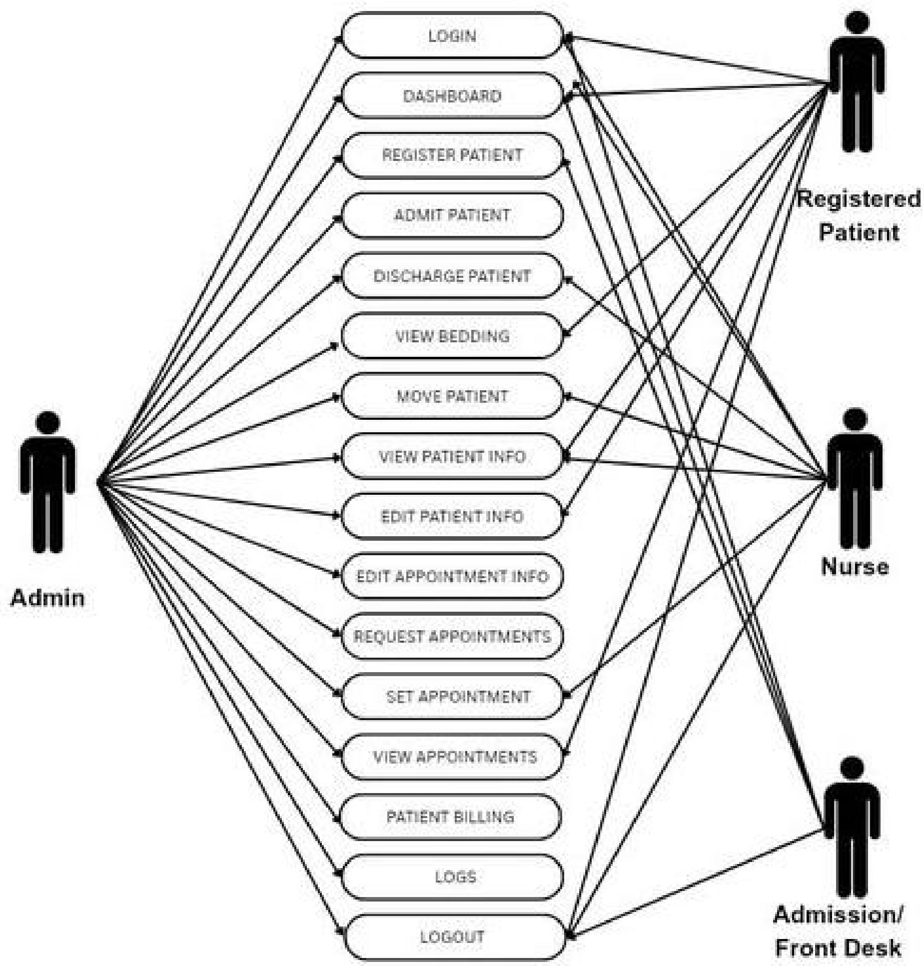


Figure 23 Application Architecture Diagram

### 4.3.3 Integration of Software Modules

Integrating software modules within a Patient Management System allows functions such as patient registration, appointment scheduling, bedding and linen management, EMR, and patient billing to operate harmoniously. Data provided during registration is utilized throughout all modules, facilitating effective care delivery, appropriate resource management, and precise billing while minimizing errors and manual labor

- **Integration Strategy**

  A well-defined integration strategy guarantees seamless communication and data exchange between all modules of the Patient Management System. It establishes data flow and communication protocols to ensure efficient and trustworthy system operations.

- **Database Integration**

  It enables all modules to retrieve and modify patient information from a unified database. This guarantees consistency, accuracy, and minimized duplication of data throughout the system.

- **Integration of User Interface**

  User interface integration offers a cohesive and intuitive design, enabling users to move seamlessly between system modules. A unified interface enhances user experience and efficiency.

- **Evaluation and Quality Control**

  Testing and quality assurance confirm that all modules function properly together. Integration testing guarantees correct data flow, safety, and system efficiency.

- **Deployment and Monitoring**

  Following deployment, the system undergoes monitoring to guarantee consistent performance and detect problems promptly. Ongoing oversight assists in preserving system reliability and effectiveness.

- **Feedback and Iteration**

  User input is utilized to consistently enhance the system. Consistent updates and improvements guarantee the Patient Management System adjusts to user requirements and stays efficient.

### 4.3.4 Communication and Interaction Patterns

- **Staff - System Interaction**

  Front-desk staff interact with the system to create patient profiles and assign unique patient identifiers, which are stored in the centralized database and shared across departments.

- **System - Mediated Interaction**

  Communicate scheduling request through the system, which processes availability and updates schedules, reducing wait times and administrative workloads.

- **Inter - Module Interaction**

  Module interacts with patient and clinical data to support efficient bed allocation based on patient condition and department.

- **Patient - System Interaction**

  Through a secure portal to access their medical records, test results, prescriptions, and appointment schedules through a patient portal.

## 4.4 Data Architecture

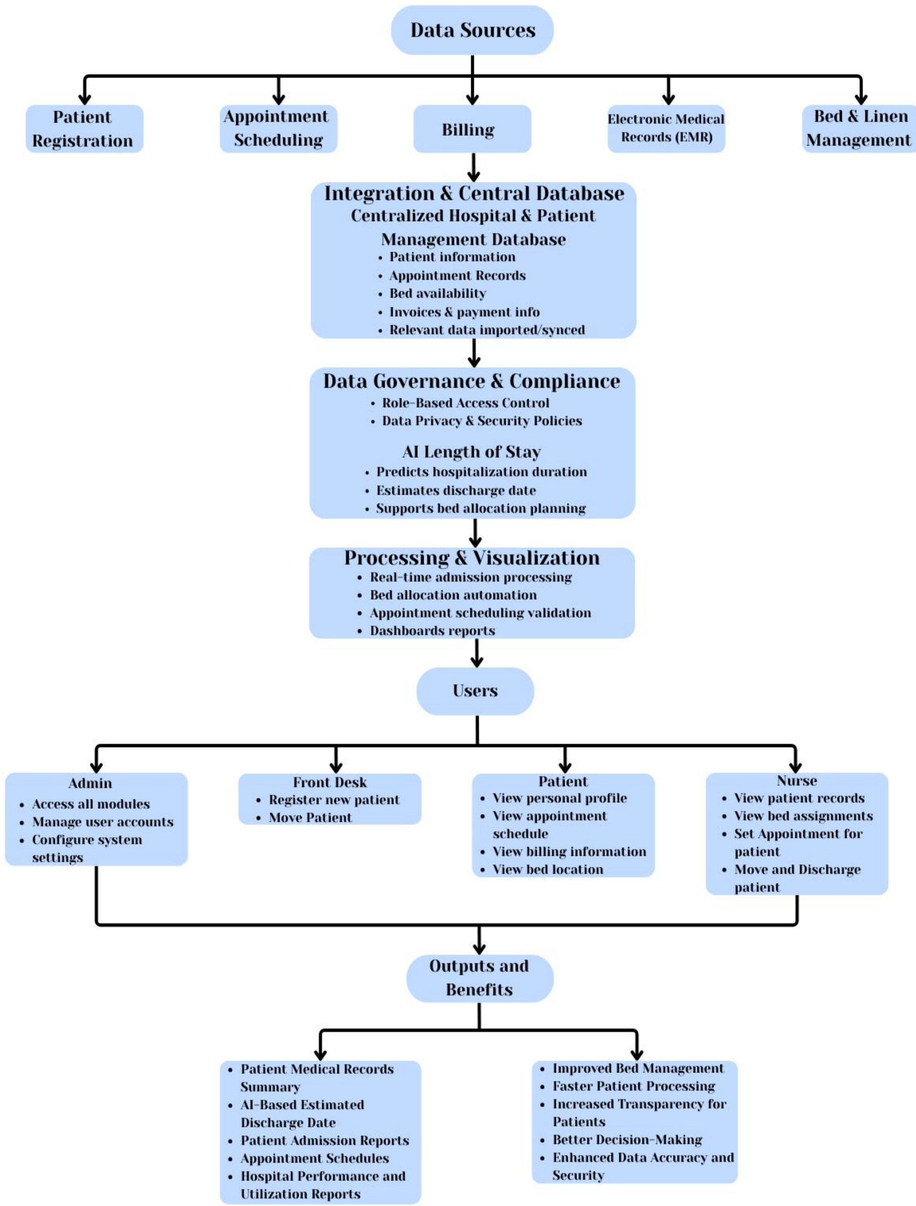


Figure 24 Data Architecture

### 4.4.1 Data Sources and Types

A Patient Management System (PMS) depends on a variety of data sources and data kinds. Both internal and external data sources are possible, and the data types might be unstructured, semi-structured, or organized. To support effective healthcare operations, enhance patient care, and guarantee accurate record-keeping.

**1. Internal Data Sources**

These originate within the healthcare facility and are generated through daily operations.

- **Patient Registration System**

  Collects demographic details such as patient name, age, gender, address, contact information, emergency contacts and insurance information.

- **Electronic Medical Records (EMR)**

  Stores patient medical histories, diagnoses, treatment plans, allergies, immunizations.

**2. External Data Sources**

These data sources come from outside the healthcare organization.

- **Insurance Providers**

  Supplies insurance eligibility data, coverage details, and claim approvals or rejections.

- **Government Health Agencies**

  Provides public health reporting requirements, regulatory compliance data, and standardized health codes.

- **Referral Hospitals and Clinics**

  Exchanges patient referral information and medical summaries.

**3 Data Types**

**1. Structured Data**

Highly organized data stored in predefined formats.

- Patient demographic information
- Appointment schedules
- Medication lists
- Bed availability

**2. Semi-Structured Data**

Data with some organizational structure but flexible formats.

- Lab result files
- System-generated logs

### 3. Unstructured Data

Free-form data that does not follow a fixed structure.

- Doctor and nurse clinical notes
- Scanned documents
- Medical images (X-ray, MRI)

## 4.4.2 Data Flow Diagrams


Figure 25 Data Flow Diagram

### 4.4.3 Data Storage and Managements

For Patient Management Systems (PMS), data management and storage are essential to enhancing the effectiveness, continuity, and quality of healthcare service. The main repository for patient-related data, including personal information, medical histories, diagnoses, treatment plans, test results, prescriptions, and billing records, is a centralized Patient Management System or Electronic Health Record (EHR) platform. Healthcare workers can make informed clinical decisions and coordinate care across departments thanks to this centralized system, which guarantees that they have real-time access to accurate and current patient data.

Many healthcare institutions enhance their PMS with cloud-based storage solutions, enabling authorized medical staff to securely access patient records anytime and anywhere, particularly in multi-branch hospitals or telemedicine environments. For organizations that require greater control over sensitive health data, on-premises databases may be implemented to store patient information within the hospital’s infrastructure, reducing exposure to external risks while maintaining system reliability and performance.

Effective data governance is crucial to patient management systems. These guidelines assist healthcare practitioners in adhering to rules like HIPAA and other legislation pertaining to the protection of patient data. Regular data audits further improve data quality by identifying errors, duplications, or outdated patient information.

Data security is a critical aspect of PMS data management due to the highly sensitive nature of medical information. Strong security measures such as encryption, multi-factor authentication, and role-based access control are implemented to ensure that only authorized personnel can access specific patient data. Additionally, comprehensive data backup and disaster recovery plans are established to protect patient records from data loss caused by system failures, cyberattacks, or natural disasters, ensuring continuity of care even during emergencies.

Healthcare organizations can obtain important insights into patient outcomes, success of treatment, resource use, and population health trends by utilizing Data analytics. Administrator can find gaps in care and enhance service delivery by analyzing indicators like hospital readmission rates, treatment success rates, and patient wait times. Better planning, better patient experiences, and higher-quality healthcare are all supported by this data-driven strategy.

### 4.4.4 Data Synchronization Across Systems

Data synchronization across systems is a critical component of the Patient Management System (PMS), ensuring that patient information remains accurate, consistent, and up to date across all hospital departments. In a healthcare environment where multiple units such as registration, nursing stations, laboratory, billing, and administration rely on shared patient data, synchronized information flow is essential for efficient operations and quality patient care.

A centralized database architecture is used in the proposed Patient Management System for Dr. Eduardo V. Roquero Memorial Hospital to accomplish data synchronization. Data is accessed and updated from a single, cohesive database by all modules, including Patient Registration, Electronic Medical Records (EMR), Appointment Scheduling, and Bed and Linen Management. This method guarantees that any changes made in one module are instantly mirrored throughout the entire system and removes data silos.

For example, when a patient is registered or admitted, their demographic and admission details are instantly available to other modules such as appointment scheduling and bed management. Similarly, updates made by doctors in the EMR such as diagnoses, prescriptions, or treatment notes are synchronized in real time, allowing nurses, and administrators to access the latest patient information without delays. This real-time synchronization minimizes errors caused by outdated or duplicate records and improves coordination between departments.

The PMS also supports role-based access control, ensuring that synchronized data is only accessible to authorized users. While all departments share the same data source, access permissions prevent unauthorized viewing or modification of sensitive patient information. This maintains data integrity while supporting seamless information sharing across systems.

By eliminating unnecessary data entry, expediting patient processing, and facilitating prompt clinical decision-making, data synchronization across systems greatly enhances hospital operations. The PMS improves operational efficiency and ultimately leads to safer and better patient care by guaranteeing that all departments function with consistent and real-time patient data.

## 4.5 Technology Architecture

### 4.5.1 Technology Stack and Infrastructure

| TECHNOLOGIES AND FRAMEWORKS | DESCRIPTION |
|---|---|
| XAMPP | XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends. The name XAMPP stands for Cross-Platform (X), Apache (A),MySQL (M), PHP (P), and Perl (P). It includes Apache HTTP Server, MariaDB (a MySQL fork), PHP, and Perl interpreters, making it a convenient way to set up a local server environment for web development and testing on your computer. |
| | Visual Studio Code (VS Code) is a free source-code editor developed by Microsoft for Windows, Linux, and macOS. It supports various programming languages and comes with features like syntax highlighting, intelligent code |

| | |
|---|---|
| | completion, debugging capabilities, version control integration (Git), and an extensive ecosystem of extensions to enhance functionality. It's widely used by developers for web development, application development, and other programming tasks due to its lightweight design, versatility, and powerful features. |
| | MySQL is an open-source relational database management system (RDBMS) that uses SQL (Structured Query Language) for managing and manipulating data. It is widely used for building web applications and is known for its reliability, scalability, and performance. MySQL is developed, distributed, and supported by Oracle Corporation. MySQL is often utilized in conjunction with popular web development technologies like PHP, Python, and JavaScript. It offers features such as transactions, stored procedures, |

| | |
|---|---|
| | triggers, and views, making it suitable for a wide range of applications from small-scale projects to large enterprise systems. |
| | Github is an online platform designed for hosting, managing, and sharing code through the Git version control system. It enables developers to upload their projects to the web, monitor modifications, collaborate with teammates, and effectively handle various of their code. |
| | PHP is a popular open-source server-side scripting language tailored for web development. It enables the creation of dynamic and interactive websites by handling data on the server, linking to databases, and producing web page content prior to its presentation to users. |

Table 10 Technology Stack and Infrastructure

### 4.5.2 Network Topology and Configuration

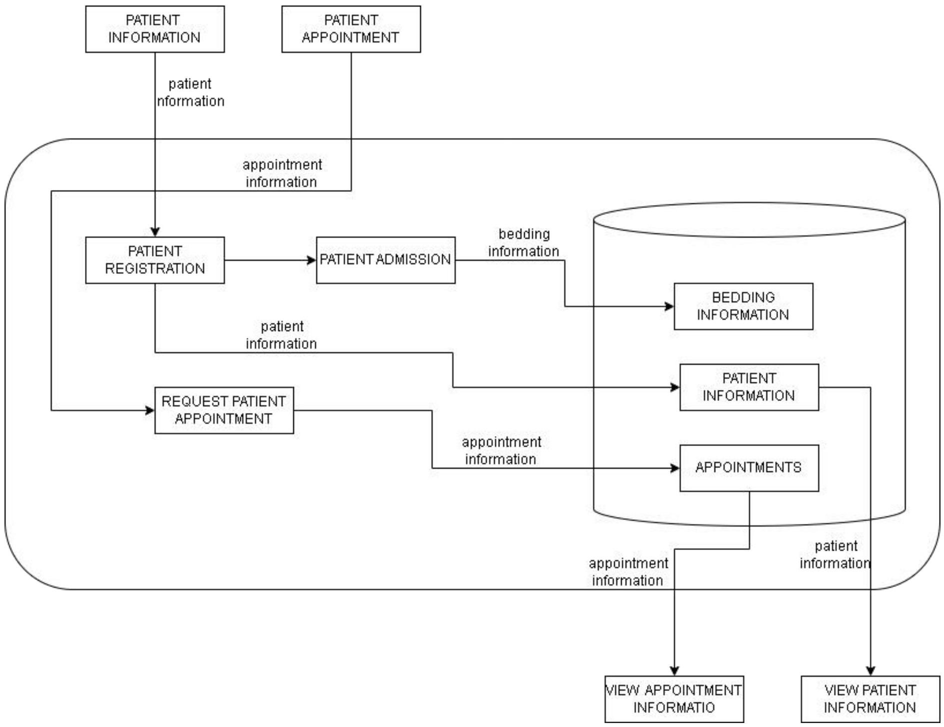


Figure 26 Network Topology and Configuration

### 4.5.3 Software Technologies

| SOFTWARE TECHNOLOGIES | DESCRIPTION |
|---|---|
| JS Query | JavaScript is a programming language commonly used for web development to create interactive and dynamic elements on web pages. It runs on the client side, 89 meaning it's executed by the user's browser rather than the web server. JavaScript can |

| | |
|---|---|
| | be used for tasks like form validation, DOM manipulation, event handling, animations, and AJAX requests. JQuery, on the other hand, is a JavaScript library that simplifies and enhances JavaScript programming. It provides a set of pre-written functions and methods that make it easier to perform common tasks in web development. |
| AJAX | AJAX, or Asynchronous JavaScript and XML, is a web development technique that enables web pages to fetch and display data from a server without needing to refresh the entire page. This asynchronous nature allows for a more 90 seamless and interactive user experience by updating parts of the webpage dynamically. JavaScript is used to initiate AJAX requests, sending data to the server in the background, and processing the response without interrupting the user's browsing experience. While AJAX initially utilized XML for data exchange, JSON has |

| | |
|---|---|
| | become the preferred format due to its simplicity and compatibility with JavaScript. |
| Bootstrap | Bootstrap is a popular front-end framework used for designing and developing responsive and mobile first websites. It provides predesigned templates, components, and JavaScript plugins, making it easier to create a consistent and visually appealing web interface |
| CSS | a language used to style the layout and appearance of HTML elements on a web page. It controls things like colors, fonts, spacing, and overall visual presentation. |
| HTML | HTML (Hyper Text Markup Language) is the standard markup language for creating web pages and web applications. It provides the structure and content of a web page by using tags to define elements like headings, paragraphs, links, images, and more. |

Table 11 Software Technologies

### 4.5.4 Scalability and Performance Considerations

The Patient Management System's success and efficiency are dependent on its scalability and performance. Scalability is required in hospitals to accommodate an increasing number of patients, medical information, and everyday transactions. As the number of patients and system users grows, the system must be capable of handling increased workloads without compromising performance or dependability. Performance optimization is also required to give timely and responsive access to patient information with minimal delays or system downtime.

Techniques such as efficient database administration, caching, and smart resource allocation contribute to system speed, particularly during peak hours. Furthermore, by employing cloud-based infrastructure, the system can increase resources as needed while keeping operational expenses under control. Continuous monitoring and system optimization ensure that the Patient Management System stays reliable, efficient, and capable of sustaining high-quality patient care and healthcare operations.

## 4.6 Development Process

### 4.6.1 Agile Scrum Roles and Responsibilities

| MEMBERS | ROLES | RESPONSIBILITIES |
|---|---|---|
| Scrum Master | A Scrum Master who ensures scrum principles and practices are followed. They help remove obstacles and ensure smooth collaboration within the team. | Facilitates Scrum ceremonies, removes team impediments, coaches Agile practices, and ensures continuous team improvement. |
| System Analyst | A System Analyst is analyzing current systems and processes to identify inefficiencies or areas for improvement. | Converts business requirements into technical specifications, designs system workflows, and supports developers during implementation. |
| Programme | A Programmer develops | Writes, tests, debugs, |

| | | |
|---|---|---|
| r | software by designing, testing and maintaining code. They implement system features based on defined requirements. | and maintains code, implements system features, and collaborates with the development team. |
| Document Specialist | A Document Specialist creates and maintains project and technical documentation. They ensure information is clear, accurate, and properly organized. | Prepares, updates, and maintains project documentation, ensures accuracy and consistency, and manages document versions. |
| Business Specialist | A Business Specialist identifies business needs and translates them into system requirements. They ensure solutions align with organizational goals. | Gathers and analyzes business requirements, communicates with stakeholders, and validates that solutions meet business needs. |

Table 12 Agile Scrum Roles and Responsibilities

### 4.6.2 Sprint Planning and Backlog Management

| PRODUCT BACKLOG ITEM | PRIORITY | REQUIREMENTS |
|---|---|---|
| Patient Registration | High | Capture patient personal details, and contact info |
| Update Patient Information | High | Edit and maintain accurate patient records. |
| Bed Status Monitoring | Medium | Display occupied, available, and maintenance beds. |
| Electronic Medical Records (EMR) | High | Store patient diagnosis, prescriptions, and treatment history. |

Table 13 Sprint Planning and Backlog Management

### 4.6.3 Sprint Execution and Deliverables

| TASK NO. | TASK DESCRIPTION | USER STORY | STATUS |
|---|---|---|---|
| 1 | Patient Registration Form Implementation | As a receptionist, I aim to enroll new patients by collecting their personal and contact information to | In progress |

| | | | |
|---|---|---|---|
| | | ensure their profiles are saved in the system | |
| 2 | Appointment Scheduling | As a patient, I want to schedule, change, or cancel appointments to manage my visits easily. | In progress |
| 3 | Beddings & Linen Tracking Module | As a nurse, I aim to monitor bed availability and linen status to ensure patients are assigned accurately and hygiene is upheld. | In progress |
| 4 | EMR Access & Update | As a doctor or clinician, I want to access and update a patient's electronic medical record. This helps me provide accurate and current care based on the patient's medical history. | In progress |

Table 14 Sprint Execution and Deliverables

### 4.6.4 Challenges Faced in the Development Process

| CHALLENGES | DESCRIPTION |
|---|---|
| Data accuracy and duplication | Ensuring accurate patient data entry and preventing duplicate records, which may result in treatment or billing mistakes. |
| Schedule conflicts and overlaps | Preventing double-booking of doctors, rooms, or equipment; necessitates immediate updates and complex scheduling logic. |
| Inventory management and real-time tracking | Effectively overseeing bed availability, linens, and medical supplies; difficulties involve prompt updates and preventing shortages. |
| Data security, privacy, and compliance | EMR systems handle sensitive health data. |
| Integration with insurance and payment systems | Billing needs to interact with various insurers, verify claims, handle payments, and create invoices without errors |
| Scalability and performance | The system needs to manage a high number of patients and extensive data |

| | |
|---|---|
| | without experiencing delays or failures. |
| User-friendliness and usability | The system needs to be user-friendly for doctors, nurses, and administrative personnel to minimize mistakes and training duration |

Table 15 Challenges Faced in the Development Process

## 4.7 Implementation

### 4.7.1 Technical Implementation Details

| FRAMEWORK | LANGUAGE | UI COMPONENT |
|---|---|---|
| Bootstrap | JavaScript<br>HTML<br>CSS | Font Awesome |
| Back End | | |
| FRAMEWORK | LANGUAGE | SERVER |
| Xampp | PHP | MySQL |
| Database | | |
| DBMS | TOOL | |
| MySQL | SQLyog | |
| Development Tools | | |
| VERSION CONTROL | CODE EDITOR | |
| GitHub | Visual Studio Code | |

Table 16 Technical Implementation Details

### 4.7.2 Tools and Technologies Used

The tools and technologies used in the Patient Management System play an important role in ensuring efficient healthcare operations, reliable data management, and effective system development. These tools support patient data storage, system functionality, reporting, communication, and informed decision-making within healthcare facilities.

| TOOLS/ TECHNOLOGY | DESCRIPTION |
|---|---|
| MySQL | serves as the main database of the Patient Management System. It is responsible for storing and managing patient personal information, medical histories, appointment schedules, and billing records. The use of MySQL ensures data accuracy, fast retrieval of records, and supports the generation of reports needed for healthcare monitoring and decision-making. |
| XAMPP | is used as a local development and testing environment for the Patient Management System. It provides essential components such as Apache, MySQL, and PHP, which |

| | |
|---|---|
| | allow developers to test system features, workflows, and database connections before deployment. This helps ensure system stability, security, and proper functionality. |
| Microsoft Word | is utilized for creating and formatting patient-related documents such as medical reports, discharge summaries, and billing statements. Information generated from the system can be exported to Word for documentation, printing, and official record-keeping purposes within the healthcare facility. |
| Visual Studio Code | used as the primary code editor for developing, modifying, and maintaining the Patient Management System. It supports multiple programming languages, including HTML, CSS, PHP, and JavaScript, and provides tools for debugging and code optimization, allowing efficient system development and customization. |

| | |
|---|---|
| Chrome | Google Chrome serves as the web browser used to access the Patient Management System. It provides a secure and user-friendly interface for doctors, nurses, administrators, and staff to manage patient records, appointments, and reports through a web-based platform. |

Table 17 Tools and Technologies Used

### 4.7.3 Code Integration and Interoperability

**RESTful APIs**

Work on the design of comprehensive RESTful APIs that promote secure and optimal communication between the Patient Management System (PMS) and third-party health software. The design of these APIs will cover the major functionalities such as:

- Accessing and updating patient demographic information
- Managing Appointment schedules
- Handling Electronic Health Records (EHRs)
- Viewing financial/billing records

- Adequate security features will be incorporated for the safety and reliability of the system.

**HL7/FHIR Standards**

Apply HL7 and FHIR standards to achieve robust interoperability and ensure a smooth, standardized flow of healthcare information from and within these systems. Through interoperability, the PMS will be able to share and receive patient information from:

- Hospital institutions
- Laboratories
- Pharmacies
- Other authorized healthcare organizations

**Webhooks**

Implement event-based webhooks. Webhooks will facilitate immediate notifications for certain occurrences such:

- Availability of lab results
- Updates in the patient's file
- The status of an appointment

Such an approach will enable various systems like the doctor's portal, the lab system, and the mobile healthcare applications to be updated immediately. The need for constant checks will be eliminated.

### 4.7.4 Integration Testing and Debugging

| TEST CASE NO. | FEATURE/ COMPONENT | TEST DESCRIPTION |
|---|---|---|
| TC-01 | Login | to ensure proper integration with the authentication system, user database, and access control. This testing includes valid and invalid login attempts, password verification, and session handling. The system verifies that users such as administrators, nurses, staff and patient can log in successfully using correct credentials. It also ensures that unauthorized users are denied access and redirected appropriately. Role-based access was tested to confirm that each user can only access the modules assigned to their role. |
| TC-02 | Dashboard | tested to confirm that it correctly displays information from different system modules. This includes patient counts, appointment schedules, and bed availability. The test |

| | | |
|---|---|---|
| | | ensures that data shown on the dashboard is accurate, updated in real time, and properly integrated with patient records and appointments. Different dashboard views were validated to ensure they are customized based on user roles, such as admin, medical staff, or support staff. |
| TC-03 | Patient Registration | verify its integration with the patient database and electronic medical records. This includes testing the creation of new patient profiles, updating personal information, and validating required fields. The system ensures that patient data is stored securely and can be retrieved accurately. Duplicate record handling and data consistency across modules were also tested. |
| TC-04 | Appointment Scheduling | to ensure smooth integration with patient records, doctor availability, and the dashboard. Test cases include booking |

| | | |
|---|---|---|
| | | new appointments, rescheduling, and canceling existing appointments. The system verifies that appointment details are updated correctly and reflected across all related modules. |
| TC-05 | Electronic Medical Records | was tested to ensure proper integration with patient registration, consultation records, and medical reports. This includes testing the viewing, and retrieval of medical history, diagnoses, and treatment notes. The system ensures data accuracy, confidentiality, and access control, allowing only authorized medical personnel to view or modify sensitive medical information. |
| TC-06 | Beddings | tested to confirm accurate tracking of bed availability and patient movement within the facility. Integration testing ensures that when a patient is admitted, transferred, or discharged, the bed status is updated automatically. The module was tested for |

| | | |
|---|---|---|
| | | real-time updates and synchronization with patient records and the dashboard to prevent overbooking or incorrect bed assignments. |
| TC-07 | Patient Portal | to ensure patients can securely access their personal medical information. This includes viewing appointment history, medical records, billing history, bed assign and treatment summaries. The system ensures that patients can only view their own records and not access other patients' information. |

Table 18 Integration Testing and Debugging

## 4.8 Testing and Quality Assurance

### 4.8.1 Testing Strategies and Methodologies

The testing process for the Patient Management System (PMS) begins with unit testing, where individual components such as patient registration, appointment scheduling, electronic medical records, and bed tracking modules are tested independently using appropriate testing frameworks. This stage allows developers to identify and resolve errors early in the development cycle, ensuring that each function performs as intended before integration.

After unit testing, integration testing is conducted to verify that different modules of the PMS work together seamlessly. For example, this testing ensures that patient registration data correctly flows into the electronic medical records, appointment schedules are properly linked to doctors, and bed assignments update in real time. Functional testing simulates actual hospital workflows in order to verify that the PMS satisfies its requirements. Patient registration, appointment scheduling, medical history recording, and bed assignment, are all replicated by automated testing tools. This guarantees that vital hospital functions operate accurately and consistently in practical situations.

User Acceptance Testing (UAT) involves nurses, administrative staff, and hospital personnel who use the PMS in actual working conditions. Their feedback is essential in evaluating system usability, workflow efficiency, and overall satisfaction. This stage helps identify gaps between user expectations and system performance, ensuring that the PMS supports daily hospital operations effectively.

Performance testing assesses how the PMS responds to various workloads, such as peak patient registration times or concurrent departmental access. This testing guarantees that the system will continue to be responsive, stable, and able to process large amounts of patient data without experiencing any delays or crashes. Given the sensitive nature of medical information, security testing is a critical component of the PMS testing strategy. This process identifies vulnerabilities related to unauthorized access, data breaches, and system misuse. Security testing ensures compliance with healthcare data protection standards and safeguards patient records through authentication, access control, and data encryption mechanisms.

To make that the PMS offers a dependable and consistent user experience across various browsers and used inside the hospital, cross-browser and device testing is also carried out. Every time a system update or improvement is made, regression testing is also carried out to make sure that the new modifications don't interfere with already-existing features.

By adopting Agile testing practices, along with Test-Driven Development (TDD) , the PMS development team maintains close collaboration with stakeholders and continuously aligns system functionality with hospital requirements. Through these comprehensive testing strategies, the Patient Management System delivers a secure, reliable, and efficient platform that enhances hospital operations, improves patient care, and increases user satisfaction.

### 4.8.2 Test Cases and Test Data

**Log In**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-LOGIN-01 | Login with valid username and password | Username: robert@patient Password: ********* | User is successfully logged in and redirected to the dashboard | - | Passed |
| TC-LOGIN-02 | Login with invalid username or password | Username: robert@patient Password: ********* | Error message displayed: Invalid username or passwrord | - | Passed |
| TC- | Login | Usernam | Validation | - | Pas |

| | | | | | |
|---|---|---|---|---|---|
| LOGIN-03 | with empty username and password | e: (blank) Password: (blank) | message: Username and Password are required | | sed |

Table 19 Test Cases and Test Data (Log In)

**Patient Registration**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-PR-01 | Register new patient with complete information | Name: Juan Dela Cruz Age: 45 Address: Block 3, Brgy. San Jose, Bulacan Contact: 09123456789 | Patient record is saved successfully and visible in Patient List | As Expected | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-PR-02 | Emergency quick-entry registration | Name: Pedro Reyes Condition: Chest Pain | System saves partial data and flags record for completion | As Expected | Pass |
| TC-PR-03 | Register patient with missing required fields | Name: Maria Santos Age: (blank) | System displays validation error and prevents saving | As Expected | Pass |

Table 20 Test Cases and Test Data (Patient Registration)

**Patient List**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-PL-01 | Search patient by name | “Juan Dela Cruz” | Correct patient record is displayed | As Expected | Pass |
| TC-PL- | View | Patient | Displays | As | Pas |

| | | | | | |
|---|---|---|---|---|---|
| 02 | complete patient profile | ID: “1” | demographics and medical history | Expected | s |
| TC-PL-03 | Retrieve patient record during peak hours | Multiple concurrent searches | Records load without delay | As Expected | Pass |

Table 21 Test Cases and Test Data (Patient List)

**Electronic Medical Record (EMR)**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-EMR-01 | Add new medical record | Diagnosis: Hypertension Prescription: Amlodipine | Record saved and linked to patient | As Expected | Pass |
| TC-EMR-02 | Update existing medical | Updated diagnosis | Changes reflected immediately | As Expected | Pass |

| | | | | | |
|---|---|---|---|---|---|
| | record | | | | |
| TC-EMR-03 | Unauthorized EMR access | Role: Front Desk | Access denied | As Expected | Pass |

Table 22 Test Cases and Test Data (Electronic Medical Record)

**Appointment Scheduling**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-AS-01 | Schedule appointment for patient | Patient: Juan Dela Cruz Doctor: Dr. Reyes Date: 10/15/2025 | Appointment saved successfully | As Expected | Pass |
| TC-AS-02 | View doctor schedule | Doctor: Dr. Reyes | Displays correct appointment list | As Expected | Pass |

Table 23 Test Cases and Test Data (Appointment Scheduling)

**Bed Tracking and Linen Management**

| TEST CASE ID | TEST SCENARIO | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|
| TC-BT-01 | Assign bed to admitted patient | Bed No: B101 | Bed status changes to “Occupied” | As Expected | Pass |
| TC-BT-02 | Discharge patient | Patient ID: 7 | Bed marked as “Under Maintenance” | As Expected | Pass |

Table 24 Test Cases and Test Data (Bed Tracking)

## 4.8.3 Test Results and Bug Reports

**Test Results Summary**

| TEST CASE ID | DESCRIPTION | STATUS | COMMENTS |
|---|---|---|---|
| TC-01 | User Login | Passed | Login process works correctly with valid credentials |
| TC-02 | Patient Registration | Passed | Patient information is saved and retrieved successfully |
| TC-03 | Appointment Scheduling | Passed | Appointments are scheduled and displayed properly |

| | | | |
|---|---|---|---|
| TC-04 | View Patient Records | Passed | Patient records load and update accurately |
| TC-05 | Generate Medical Reports | Passed | Medical reports are generated and stored correctly |
| TC-06 | Bed Availability Tracking | Passed | Bed status updates correctly in real time |

Table 25 Test Result Summary

**Bug Reports**

| BUG ID | TEST CASE ID | DESCRIPTION | SEVERITY | ASSIGNED TO | COMMENTS | STATUS |
|---|---|---|---|---|---|---|
| BUG-O1 | TC-01 | Login authentication verification | LOW | Development Team | Login logic reviewed and confirmed working correctly | DONE |
| BUG-02 | TC-02 | Patient registration data | LOW | Development Team | Input validation checked | DONE |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | validation | | | and functioning properly | |
| BUG -03 | TC-03 | Appointment scheduling process | LOW | Development Team | Scheduling flow tested and synced correctly with database | DONE |
| BUG -04 | TC-04 | Patient record viewing and updating | LOW | Development Team | Data retrieval and updates verified across the system | DONE |
| BUG -05 | TC-05 | Medical report generation | LOW | Development Team | Report creation and storage tested successfully | DONE |
| BUG | TC-06 | Patient | LOW | Develo | Bed status | DO |

| | | | | | | |
|---|---|---|---|---|---|---|
| -06 | | portal access and data visibility | | pment Team | updates validated and reflected in real time | NE |

Table 26 Bug Reports

### 4.8.4 Quality Assurance Measures

Quality assurance measures are essential to ensure that the Patient Management System is effective, reliable, and aligned with the operational goals of healthcare institutions. This step helps identify the specific functional and performance needs of the system, ensuring that it supports accurate patient data management and efficient healthcare workflows. Once system requirements are defined, the next phase focuses on system design and development. Best practices in software development are applied to create a well-structured, secure, and user-friendly system. Pilot testing is conducted in a controlled environment to evaluate system functionality, data accuracy, and usability.

Continuous testing and monitoring are vital throughout the system's operation. Functional testing, integration testing, and performance testing are performed regularly to ensure that all modules such as patient records, appointment scheduling, and billing in patient portal work correctly and consistently. Regular system reviews help identify errors, performance issues, or security risks that may affect system reliability.

User feedback also plays a crucial role in quality assurance. Healthcare staff are encouraged to provide feedback on system usability, accuracy, and efficiency. This feedback supports continuous improvement by guiding system updates and enhancements based on real-world usage and evolving healthcare needs.

Technology further supports quality assurance by enabling secure data storage, access control, and audit trails. These measures ensure data accuracy, confidentiality, and compliance with healthcare data protection standards. By implementing these comprehensive quality assurance measures, the Patient Management System maintains high standards of performance, reliability, and data integrity, ultimately supporting improved patient care and efficient healthcare operations.

## 4.9 Results and Evaluation

### 4.9.1 Project Outcomes and Deliverables

**Project Outcomes**

- **Improved Patient Record Management –** Patient information is stored digitally in one secure system, making records easy to access, update, and manage while reducing paperwork and errors.
- **Streamlined Hospital Processes –** Automated features such as registration, appointment scheduling, billing, and notifications reduce manual work and improve overall efficiency.
- **Faster and More Accurate Patient Care** – Medical staff can quickly view patient history, medical records, and test results, helping them make better and faster medical decisions.
- **Improved Appointment Scheduling** – The system allows easy booking, of appointments, reducing waiting time and avoiding scheduling conflicts.
- **Better Bed Management and Availability Tracking** – The system tracks bed availability and patient movement,

ensuring proper bed allocation and improving hospital resource management.

- **Enhanced Patient Engagement and Access** – Patients can use the patient portal to view their medical history, appointments, billing history, bed assign, and treatment records, increasing transparency and trust.
- **Secure and Confidential Data Handling** – Role-based access ensures that only authorized users can view or update sensitive patient information, protecting data privacy.
- **Accurate Billing and Payment Trackin**g – The billing system ensures correct charges, proper payment recording, and clear transaction status, reducing billing errors.
- **Data-Driven Decision Making** – System reports and dashboards provide insights into patient flow, resource usage, and hospital performance to support better management decisions.

| DELIVERABLES | DESCRIPTION |
|---|---|
| Login | Secure authentication system for administrators, nurses, and patients to access the patient management system based on their roles. |
| Dashboard | Centralized dashboard displaying key information such as patient statistics, analytics, overview appointment schedules, and bed availability. |
| Patient Registration | Enables efficient registration of new patients by capturing personal details, contact information, and basic medical data. |
| Appointment Scheduling | Allows patients and staff to book, update, reschedule, or cancel appointments with doctors. |
| Electronic Medical Records | Digital storage and management of patient medical histories, diagnoses, prescriptions, lab results, and treatment plans. |
| Beddings Management | Monitors bed availability, assigns beds to patients, and manages patient movement between wards or rooms. |

Table 27 Project Outcomes and Deliverables

### 4.9.2 Alignment with Project Objectives

- **Efficient and Precise Patient Registration**

  This aligns with the aim of getting accurate and timely capturing of patient information, identification, and service delivery, and reliable data for decision-making across boards.

- **Optimized Appointment Scheduling and Patient Flow**

  This supports the goal of optimizing patient flow, enabling timely scheduling, reducing waiting times, and enhancing resource utilization.

- **Effective Management of Bedding and Linen Resources**

  This aligns with the objective for the real-time tracking of bed availability and consumption of linen to ensure efficient patient allocation and hygiene standards.

- **Electronic Medical Records Centralized for Quality Care Electronic Supports**

  The goal of providing centralized, accurate, and accessible patient clinical data, contributing to quality care and enhanced clinical decision-making.

- **Accurate and transparent Patient Billing**

  This aligns with objectives by ensuring correct, clear, timely, and integrated billing of charges from all services to enhance financial efficiency and minimize billing errors.

### 4.9.3 Stakeholder and User Feedback

To make sure that the Patient Management System (PMS) consistently enhances and successfully supports hospital operations, patient care, and administrative efficiency, it is crucial to gather and act upon stakeholder and user input. The system is kept in line with the hospital's operational objectives, healthcare standards, and users' actual demands thanks to this iterative feedback-driven methodology. The PMS can be improved to increase usability, accuracy, and dependability by incorporating feedback from many stakeholders.

- **Patients**

  Patients provide valuable feedback regarding the ease of registration, waiting time reduction, and accuracy of personal and medical information stored in the system. Their insights help identify issues related to appointment scheduling, clarity of medical records, and overall experience during hospital visits.

- **Nurses and Medical Staff**

  During consultations or crises, nurses and accompanying medical personnel evaluate how well the system updates electronic medical records and retrieves patient histories. Their comments highlight how well the system works in high-stress scenarios, such emergency admissions, and point out areas where quick-entry tools, alarms, and data accuracy need to be improved.

- **Receptionists and Administrative Staff**

  Feedback on the effectiveness of patient registration, record retrieval, and appointment scheduling is given by receptionists and administrative staff. Their observations serve in locating data entry issues, validation mistakes, and system navigation problems. This input encourages enhancements to search capabilities, report generation, and workflow automation, which lessen errors and administrative workload.

- **Hospital Administrators**

  Hospital administrators focus on how well the PMS aligns with organizational objectives, such as operational efficiency, data security, compliance, and resource utilization. They evaluate system reports related to patient volume, bed occupancy, and service performance. Their feedback ensures that the PMS delivers measurable value, supports decision-making, and contributes to improved hospital management and planning.

### 4.9.4 Lessons Learned

The design, development, and implementation of the Patient Management System (PMS) provided valuable lessons that can guide future enhancements and similar healthcare information system projects. Key takeaways include the importance of user-centered design, accuracy and completeness of patient data, seamless system integration, continuous stakeholder feedback, and scalability to accommodate growing hospital demands.

Throughout the development and deployment of the Patient Management System, several important lessons emerged. User experience is critical, as the system is used daily by healthcare staff who often work under time pressure. An intuitive interface, clear navigation, and fast system response are essential to ensure high adoption and effective use. A complex or slow system can hinder workflows, increase errors, and negatively affect patient care, particularly during emergency situations.

The significance of data validation and accuracy is another important lesson. Mandatory fields, validation standards, and real-time updates are crucial to preventing inaccurate or incomplete information since patient records directly influence clinical choices. In order to strike a balance between speed and data integrity, features like quick-entry modes for emergency cases and follow-up reminders for record completion became essential.

Continuous feedback from stakeholders, including nurses, administrative staff, and patients, was essential in refining system functionality. Regular feedback sessions helped identify usability issues, missing features, and workflow inefficiencies early in the development cycle. This iterative improvement process ensured that the system evolved in response to real operational needs rather than assumptions.

Finally, ongoing support, maintenance, and monitoring are necessary to ensure long-term system reliability. Regular updates, security patches, and performance monitoring help prevent downtime and protect sensitive patient data. Measuring system impact such as reduced waiting times, faster record retrieval, and improved staff efficiency also provides evidence that the PMS contributes to improved healthcare delivery and operational efficiency.

These lessons emphasize the importance of a reliable, secure, scalable, and user-focused approach in developing patient management systems. Applying these insights will help ensure that future iterations of the PMS continue to enhance patient care, support healthcare professionals, and align with the hospital’s operational goals.

# CHAPTER 5

## SUMMARY, CONCLUSION, AND RECOMMENDATIONS

### 5.1 Key Takeaways and Summary

Patient Management Systems are a vital component of modern healthcare, helping medical institutions deliver efficient, accurate, and patient-centered services. By using technology, focusing on ease of use, and relying on accurate data, healthcare providers can improve patient care, streamline hospital operations, and support better decision-making. An effective patient management system ensures that patient information is well-organized, accessible, and secure, enabling healthcare professionals to respond quickly to patient needs and improve overall service quality.

A Patient Management System is a structured approach to managing patient information, appointments, medical records, and bed tracking. It plays an important role in improving healthcare delivery, enhancing patient satisfaction, and supporting the smooth operation of healthcare facilities.

- **Improved Healthcare Efficiency**

  Patient management systems help hospitals operate more efficiently by reducing paperwork, minimizing errors, and automating routine tasks.

- **Better Patient Care and Safety**

  Accurate and up-to-date electronic medical records allow healthcare providers to make informed decisions, leading to improved patient care and reduced medical risks.

- **Patient Engagement and Satisfaction**

  Patient portals allow patients to access their medical history, view appointments, billing history and stay informed, increasing trust and satisfaction.

- **Technology Integration**

  The use of digital tools such as electronic medical records (EMR), dashboards, and notification systems improves coordination among healthcare staff and ensures timely care delivery.

- **Resource and Bed Management**

  Bedding management systems help track bed availability and patient movement, ensuring optimal use of hospital resources and reducing delays in patient admission or transfer.

- **Data Accuracy and Reporting**

  Reliable data collection and reporting support better planning, monitoring, and evaluation of hospital performance and patient outcomes.

- **Scalability and Adaptability**

  Patient management systems can be expanded and adapted to meet the growing needs of healthcare institutions and changing medical requirements.

- **Secure Information Management**

  Strong security measures ensure patient data confidentiality and compliance with healthcare regulations, building trust between patients and healthcare providers.

## 5.2 Project Achievements and Contributions

The development and implementation of the Patient Management System (PMS) for Dr. Eduardo V. Roquero Memorial Hospital achieved several significant milestones and contributions that addressed the institution's operational challenges and supported its transition from manual to digital healthcare processes.

- **Successful Digitization of Patient Records**

  One of the major achievements of the project was the successful conversion of manual, paper-based patient records into a centralized digital system. The PMS enabled secure electronic storage of patient demographics, medical histories, diagnoses, and visit records, reducing the risks of data loss, misplacement, and duplication associated with index cards and paper files. This contribution significantly improved record accuracy, consistency, and accessibility for authorized hospital staff.

- **Enhanced Operational Effectiveness in Hospital Processes**

  The system optimized essential hospital operations including patient registration, appointment scheduling, management of electronic medical records (EMR), and bed tracking. Through the automation of these workflows, the PMS alleviated administrative burdens, decreased delays in accessing patient information, and enhanced interdepartmental coordination. Consequently, healthcare staff could dedicate more attention to patient care instead of manual record-keeping.

- **Execution of Real-Time and Structured Data Management**

  The PMS implemented a centralized database that facilitates real-time updates and immediate access to patient information. Elements like structured data entry, required fields, and validation checks have significantly decreased the occurrence of incomplete records, particularly in emergency or high-demand scenarios. This advancement has improved the dependability of medical data utilized in diagnostic and treatment choices.

- **Improved Appointment Scheduling and Bed Management**

  The project contributed to better hospital resource utilization by implementing an appointment scheduling module and a bed and linens tracking feature. These modules allowed staff to monitor doctor availability, manage patient bookings efficiently, and verify bed readiness without manual inspection. This reduced patient waiting time and improved admission and discharge processes.

- **Enhanced Data Protection and Access Management**

  Role-based access control was established to guarantee that only authorized individuals such as administrators and medical personnel were able to access or alter sensitive patient information. This implementation support data privacy, integrity, and adherence to healthcare data protection regulations.

- **Scholarly and Practical Contribution**

  In addition to its technical execution, the PMS represented a significant academic contribution by illustrating the application of information systems, enterprise architecture, and agile methodologies within practical healthcare environments. This project offers a reference model for upcoming researchers and developers focused on hospital or patient management systems, especially in organizations moving from manual to digital processes.

## 5.3 Future Works and Enhancements

### Future Trends in Patient Management Systems

The Patient Management System (PMS) landscape is evolving rapidly, driven by advancements in artificial intelligence, automation, data analytics, and patient-centric innovations. As healthcare providers strive to enhance efficiency, improve patient experiences, and streamline operations, several key trends are shaping the future of PMS.

a. **AI-Powered Predictive Analytics and Automation Artificial Intelligence (AI) -** is transforming PMS by enabling predictive analytics, automation, and decision support. With AI, healthcare providers can:

Predict patient health risks based on historical data.

- Automate appointment scheduling, billing, and insurance claims processing.
- Use chatbots and virtual assistants for 24/7 patient support.
- Leverage AI-driven clinical decision support systems (CDSS) to recommend treatment plans.
- AI-driven PMS will help reduce administrative burden, enhance efficiency, and improve clinical outcomes.

**b. Telehealth Integration for Remote Patient Monitoring -** The rise of telemedicine and remote patient monitoring (RPM) is making PMS more accessible and versatile.

Future-ready PMS solutions will integrate with:

- Wearable devices to track vitals like heart rate, blood pressure, and oxygen levels.
- Smart home healthcare tools for monitoring chronic conditions.
- Video consultation platforms to connect doctors and patients seamlessly.
- This integration will bridge the gap between in-person and virtual healthcare, improving accessibility and continuity of care.

**c. Interoperability and Unified Health Data Exchange -** A major challenge in healthcare is fragmented data across multiple systems. Future PMS solutions will prioritize interoperability, ensuring seamless data exchange between:

Electronic Health Records (EHRs)

- Laboratory Information Systems (LIS)
- Pharmacy and Imaging Systems
- Health Information Exchange (HIE) Networks

- Standards like FHIR (Fast Healthcare Interoperability Resources) and HL7 will drive this transformation, allowing real-time data sharing across healthcare networks.

d. **Cloud-Based and Mobile-First PMS Solutions -** The shift toward cloud computing and mobile-friendly PMS platforms will continue to grow, offering:

- Anytime, anywhere access for healthcare professionals and patients.
- Lower infrastructure costs compared to traditional on-premises solutions.
- Enhanced data security and automated backups.
- A mobile-first PMS will enable doctors to access patient records, update charts, and communicate with patients on the go, improving workflow efficiency.

e. **Blockchain for Secure and Transparent Healthcare Data -** Blockchain technology is gaining traction in PMS for secure, transparent, and immutable health record management. With blockchain, PMS can:

- Enhance data security and prevent unauthorized access.
- Facilitate secure patient consent management.
- Enable decentralized health data sharing without intermediaries.

- As data privacy regulations like HIPAA and GDPR tighten, blockchain-driven PMS will play a crucial role in ensuring compliance and patient data protection.

f. **Personalized and Patient-Centric Care -** The future of PMS is shifting towards personalized healthcare, where systems:

- Analyze patient behavior and preferences to offer customized treatment plans.
- Provide personalized health recommendations based on AI-driven insights.
- Use patient engagement tools like self-service portals, chatbots, and digital wellness programs.
- By empowering patients with more control over their health data, PMS will enhance patient satisfaction and treatment adherence.

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# Appendices A

## Detailed Technical Documentation

### 1. System Architecture

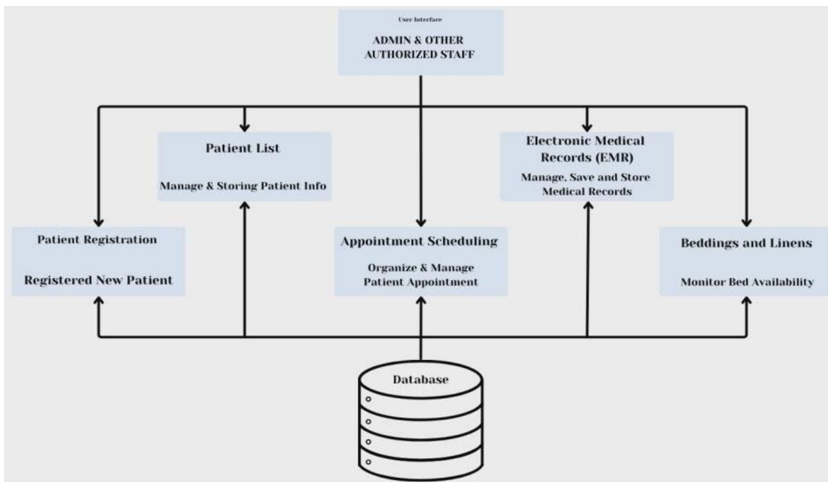


Figure 27 System Architecture

**Overview Of The Overall System Architecture**

The Patient Management System (PMS) is designed using a multi-tier, web-based system architecture that ensures scalability, security, maintainability, and efficient integration of hospital operations.

**1. Presentation Layer**

The Presentation Layer acts as the user interface for the Patient Management System. Authorized users, including administrators, nurses, and front desk, access it via standard web browsers. The interface is designed to be user-friendly, responsive, and role-based, ensuring that each user only accesses functions relevant to their responsibilities.

- Patient registration forms
- Appointment scheduling interfaces
- Bed monitoring dashboards
- Electronic Medical Record (EMR) views
- Patient list and reporting screens

**2. Application Layer**

The Application Layer encompasses the essential logic and features of the Patient Management System. It handles user requests, implements business rules, and manages communication between the Presentation Layer and the Data Layer. This layer is implemented using PHP with Object-Oriented Programming (OOP), allowing modular design, code reusability, and easier system maintenance. Each module operates independently but remains fully integrated within the system architecture, supporting scalability and future enhancements

- Patient Registration Module
- Appointment Scheduling Module
- Bed and Linen Management Module
- User Authentication and Authorization

### 3. Data Layer

The Data Layer is responsible for secure storage, retrieval, and management of patient and hospital data. It uses a centralized relational database that stores. This centralized method removes data duplication, guarantees data uniformity, and enables immediate access throughout hospital departments. Data validation, backup, and access control systems are enacted to ensure data integrity, confidentiality, and adherence to healthcare standards.

- Patient demographic information
- Medical histories and EMRs
- Appointment records
- Bed availability and linen status
- System logs and audit trails

### 4. Infrastructure and Technology Layer

The Infrastructure Layer provides the technical foundation that supports the PMS. The system can be implemented in either an on-site or cloud-based setting, providing adaptability according to hospital needs. Protective strategies like firewalls, encrypted messages, and role-specific access safeguard sensitive patient information

- Web server hosting the application
- Database server for data storage
- Network components (firewalls, routers, secure access controls)
- Backup and recovery mechanisms

**Detailed Diagrams Illustrating System Components And Their Interaction**

The overall architecture of the Patient Management System (PMS) developed for Dr. Eduardo V. Roquero Memorial Hospital. The diagram presents how users, system components, and the database interact to support efficient hospital operations. At the top layer, the users or actors consist of the Admin, Front desk, Nurse, and Patient. These authorized individuals gain access to the system via a web browser, enabling them to utilize the PMS at any time within the hospital network. Each role is assigned specific access rights according to their duties, thereby ensuring data security and appropriate system utilization. The middle layer represents the Web-Based Patient Management System, developed using PHP with Object-Oriented Programming (OOP). This layer serves as the core of the system and contains the main functional modules:

- **Patient Registration Module** – captures and stores patient demographic information digitally.
- **Patient List Management** – provides a centralized list of all registered and admitted patients.
- **Electronic Medical Records (EMR)** – manages patient medical history, diagnoses, prescriptions, and treatment records.
- **Appointment Scheduling Module** – organizes and manages patient-doctor appointments.
- **Bed and Linen Tracking Module** – monitors bed availability and linen readiness in real time.
- **Reports and Dashboard** – presents summarized data such as patient counts, appointments, and bed availability for management decision-making.

The bottom layer is the Database Server, which securely houses all system data within a centralized database. It comprises tables for patient details, medical records, appointment schedules, bed and linen availability, and user accounts. This centralized data storage removes the necessity for paper records and facilitates quick access, real-time updates, and consistent data across all departments.

This architecture addresses the hospital's existing problems such as manual record-keeping, delayed data retrieval, incomplete records, and inefficient bed management. By digitizing and centralizing patient information, the system improves workflow efficiency, data accuracy, and the quality of patient care.

**Explanation Of Component Responsibilities And Relationships**

The Patient Management System (PMS) is composed of several interconnected components, each responsible for a specific hospital function. These components work together to ensure accurate data handling, efficient workflows, and improved patient care. The system follows a centralized and modular architecture, allowing seamless interaction between components while maintaining data integrity and security.

**1. Authentication and User Management Component**

**Responsibilities:**

- Manages user login and authentication.
- Enforces role-based access control (Admin, Receptionist, Nurse, Patient).
- Restricts access to sensitive patient data based on user roles.

- Ensures system security and accountability.

**Relationships:**

- Acts as the entry point to the system.
- Controls access to all other PMS components.
- Works closely with the Database Component to validate user credentials and permissions.

**2. Patient Registration Component**

**Responsibilities:**

- Records new patient demographic information such as name, address, contact details, and emergency contacts.
- Ensures required fields are completed to prevent incomplete records.
- Creates a unique patient profile for each individual.

**Relationships:**

- Sends patient data to the Patient List Component.
- Serves as the foundation for EMR, Appointment Scheduling, and Bed Management.
- Stores all patient information in the centralized database.

**3. Patient List Management Component**

**Responsibilities:**

- Displays all registered and admitted patients in a centralized list.
- Allows authorized users to search, view, and update patient information.
- Acts as the main access point for selecting patient records.

**Relationships:**

- Retrieves data from the Patient Registration Component.
- Provides patient records to the EMR, Appointment Scheduling, and Billing components.

**4. Electronic Medical Records (EMR) Component**

**Responsibilities:**

- Stores and manages patient medical history, diagnoses, prescriptions, and treatment plans.
- Allows nurses to update clinical information in real time.
- Ensures continuity of care by maintaining complete medical records.

**Relationships:**

- Receives patient identity data from the Patient List Component.
- Stores all clinical data securely in the Database Component.
- Restricted by Authentication and User Management for security.

**5. Appointment Scheduling Component**

**Responsibilities:**

- Manages patient appointments with doctors.
- Displays doctor availability and scheduled consultations.
- Prevents scheduling conflicts and double bookings.

**Relationships:**

- Uses patient data from the Patient List Component.
- Updates appointment data in the Database Component.

**6. Bed and Linen Management Component**

**Responsibilities:**

- Tracks hospital bed availability (occupied, vacant, under maintenance).
- Request Linen for new patients.
- Supports faster patient admission and room assignment.

**Relationships:**

- Receives patient admission details from Patient Registration and EMR.
- Updates real-time bed status for hospital staff.
- Stores bed and linen data in the Database Component.
- Helps reduce delays caused by manual bed checking.

**7. Reporting and Dashboard Component**

**Responsibilities**:

- Generates real-time summaries such as patient count, appointments, bed availability, and admissions.
- Provides hospital management with operational insights.
- Supports data-driven decision-making.

**Relationships:**

- Collects data from all PMS components.
- Retrieves aggregated data from the Database Component.
- Accessible mainly by Admin users through role-based access

**2. Information Systems Integration**

**Description Of How The Class's Information Systems Were Integrated**

The Patient Management System (PMS) is a system with several modules and connected to other systems of the hospital through RESTful APIs to ensure a smooth flow of data and real-time communication. The Patient Registration API receives the creation, retrieval, and update of the demographic information of the patients, as it forms the basis of the system. Patient Management API handles admissions, discharges, and update patient information, whereas the Appointment Scheduling API allows booking, rescheduling, and cancellations of appointments by communicating with Doctor and Nurse System API to check real-time doctor availability. EMR API is used to store the diagnoses, treatment plans, and notes of clinical nature, and it is linked directly to the Pharmacy API to transfer prescription data and the Laboratory API to request and obtain test results. For Inpatients, the Bed and Linen Management API follows the bed assignment and availability to provide adequate room assignment and resource monitoring. Moreover, the Billing API captures a history of billing, creates invoices and monitors payments and can be combined with services like laboratory tests and pharmacy histories to make sure that the charges are calculated accurately. Any APIs are also secured by the authentication procedure like role-based access control, which means that only authorized users like administrators, doctors, nurses and patients have access to the appropriate data. This integrated API structure helps the system to attain effective operations in the hospital, high accuracy of the data, and efficient communication among the departments.

**Diagrams Showing The Integration Points And Data Flows**

The Patient Management System (PMS) integrates its core modules and how data flows between internal components and external hospital systems. The diagrams emphasize data sharing, centralized storage, and system interoperability, consistent with the system architecture described in the uploaded document.

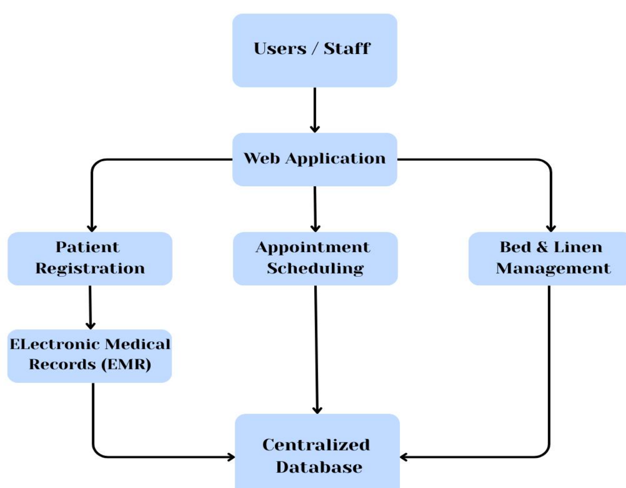


Figure 28 Illustration depicts the internal coordination of PMS components

This illustration depicts the internal coordination of PMS components. Every functional module Patient Registration, Appointment Scheduling, EMR, and Bed & Line Management interconnects via the web application and utilizes a shared centralized database. This guarantees that after patient data is input, it is instantly accessible throughout all modules without any duplication or lag.

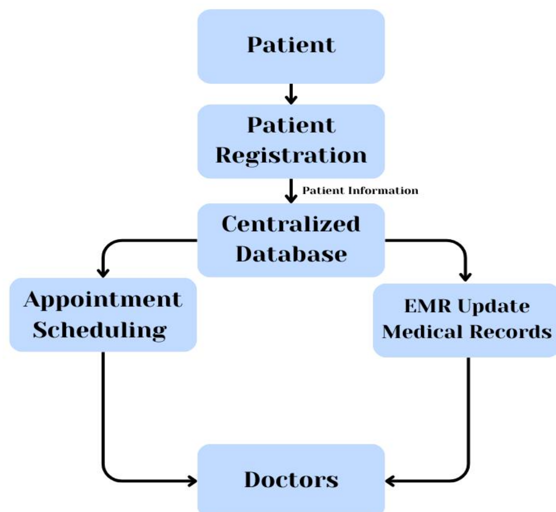


Figure 29 Illustrates how patient data moves

This data flow diagram illustrates how patient data moves through the system. Patient information is first captured during registration and stored in the database. The same data is reused for appointment scheduling and EMR updates, enabling healthcare professionals to access updated information in real time.

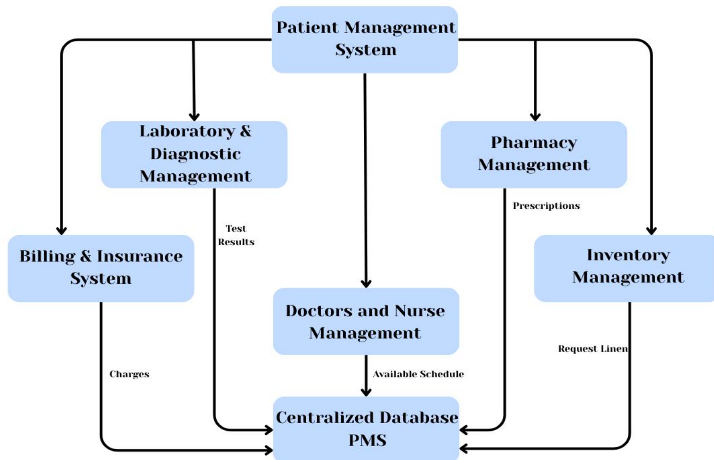


Figure 30 Integration points between PMS and other system

This diagram highlights integration points between PMS and other hospital systems. Laboratory results, pharmacy prescriptions, and billing information are exchanged through secure interfaces. The PMS acts as the central hub, ensuring synchronized data across departments and eliminating manual record transfers.

**Data Transformation And Mapping Processes**

In the Patient Management System (PMS), data transformation and mapping procedures outline the conversion of raw data gathered from different hospital sources into a uniform, organized, and functional format throughout the interconnected system modules. These processes guarantee consistency, accuracy, interoperability, and real-time accessibility of data between the PMS elements and other hospital information systems.

**1. Data Sources in the PMS**

The PMS receives data from multiple internal and external sources, including:

- Patient Registration Forms
- Appointment Scheduling Module
- Electronic Medical Records (EMR)
- Bed and Linen Management Module
- Laboratory and Diagnostic Systems
- Billing and Insurance Systems
- Pharmacy System
- Doctor and Nurse System

Each data source may use different data formats, naming conventions, or structures. Data transformation ensures that all incoming data follows a common system schema before being stored or shared.

**2. Data Transformation Process**

Data transformation is the process of converting raw input data into a standardized format suitable for processing and storage in the PMS database.

**2.1 Input Data Normalization**

- Patient details such as names, addresses, contact numbers, and dates are standardized.

- Date formats (e.g., appointment dates, admission dates) are converted into a uniform system format.
- Mandatory fields are validated to prevent incomplete records.

**2.2 Data Validation and Cleansing**

- Duplicate patient records are detected using unique identifiers (Patient ID).
- Invalid or missing values are flagged for correction.
- Data type validation ensures numeric, text, and date values are stored correctly.

This process supports the PMS objective of minimizing errors caused by rushed or manual data entry

**3. Data Mapping Process**

Data mapping defines how data elements from one module or external system correspond to fields in the PMS database.

### 3.1 Internal Module-to-Database Mapping

| PMS Module | Source Data Field | Database Field |
|---|---|---|
| Patient Registration | Full Name | patient_name |
| Patient Registration | Date of Birth | patient_dob |
| Appointment Scheduling | Appointment Date | appointment date |
| EMR Module | Diagnosis | medical diagnosis |
| Bed Management | Bed Status | bed status |

Table 28 Internal Module-to-Database Mapping

This mapping ensures all modules use consistent field definitions, allowing seamless data sharing across the system.

## 4. Data Transformation in System Integration

When integrating PMS with other hospital systems, data transformation ensures compatibility:

- Laboratory test results are converted into PMS-supported medical record formats
- Pharmacy prescriptions are mapped to patient treatment records
- Billing data is transformed into standardized charge and payment structures
- Doctors schedules are mapped to appointment scheduling for the availability of the doctors.

These transformations allow external systems to interact with the PMS through secure APIs while maintaining a single source of truth.

**5. Security and Access Control in Data Mapping**

During data transformation and mapping:

- Role-based access control determines who can view or modify mapped data
- Audit logs track data changes across modules

This ensures compliance with healthcare data protection requirements and safeguards patient confidentiality.

**6. Importance of Data Transformation and Mapping**

The implementation of data transformation and mapping processes in the PMS:

- Ensures data accuracy and consistency
- Eliminates redundancy and manual errors
- Enables seamless integration between hospital departments
- Supports real-time decision-making and patient care

These processes align with the enterprise integration and data architecture strategies defined in the uploaded Patient Management System documentation

## 3. Application Design and Development

### Detailed Information About The Software Modules And Components

The Patient Management System (PMS) is a web-based application created with PHP utilizing Object-Oriented Programming (OOP). Its purpose is to digitize and centralize hospital operations concerning patient records, appointments, medical history, and bed management. The system features a modular design, enabling each component to execute a particular function while maintaining integration with the overall system via a centralized database.

1. **Authentication and User Management**

   This controls access to the system by validating user credentials and enforcing role-based permissions.

**Functions**

- User login and logout
- Role assignment (Admin, Front Desk, Nurse, Patient)
- Access control for sensitive data

**Purpose**

This module ensures that only authorized hospital staff can access the system and that each user can only perform tasks relevant to their role.

**2. Patient Registration Module**

**Description**

This module is responsible for collecting and storing patient demographic information during their first visit.

**Functions**

- Add new patient records
- Capture personal and contact information
- Store emergency contact details
- Validate required fields to prevent incomplete records

**Purpose**

Replaces manual index card recording and ensures complete, accurate, and digital patient records.

**3. Patient List Management Module**

**Description**

This module provides a centralized directory of all registered and admitted patients.

**Functions**

- Display all patient records
- Search and filter patients
- Update patient status (admitted, discharged)

**Purpose**

Allows healthcare staff to quickly retrieve patient information without searching through physical files.

**4. Electronic Medical Records (EMR)**

**Description**

Stores and manages the complete medical history of each patient.

**Functions**

- Maintain chronological medical history
- Restrict access to authorized users only

**Purpose**

Improves clinical decision-making by providing real-time access to accurate medical data.

**5. Appointment Scheduling Module**

**Description**

This module manages patient appointments and doctor availability.

**Functions**

- Create, update, and cancel appointments
- View doctor schedules
- Prevent scheduling conflicts

**Purpose**

Reduces waiting time and ensures efficient coordination between patients and doctors.

**6. Bed and Linen Management Module**

**Description**

This module tracks bed availability and linen readiness in real time.

**Functions**

- Monitor bed status (vacant, occupied, maintenance)
- Assign beds to admitted patients
- Update bed readiness automatically
- AI predicts the length of stay

**Purpose**

Eliminates manual bed checking and improves admission efficiency, especially during peak hours.

**7. Reporting and Dashboard Module**

**Description**

This module provides summarized and real-time operational data for hospital management.

**Functions**

- Generate patient statistics
- Display appointment counts

- Show bed availability rates
- Show overall overview and analytics

**Purpose**

Supports data-driven decision-making and hospital performance monitoring.

**8. Database Management Component**

**Description**

This component serves as the centralized repository for all PMS data.

**Functions**

- Data Stored
- Patient profiles
- Medical records
- Appointments
- Bed and linen status
- User accounts and roles

**Purpose**

Ensures data integrity, consistency, and fast retrieval, eliminating paper-based storage.

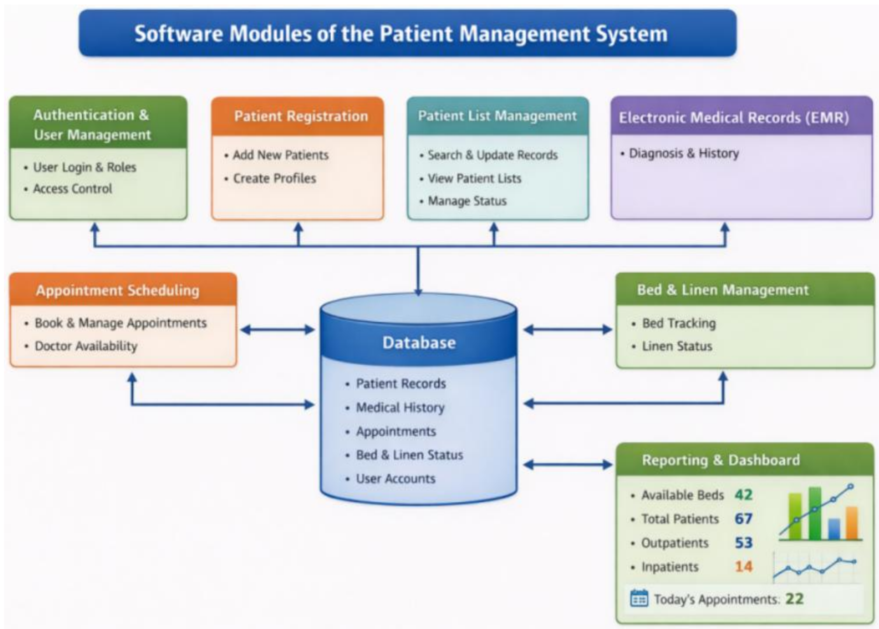


Figure 31 Software Modules and Components

**Code Snippets And Code Structure**

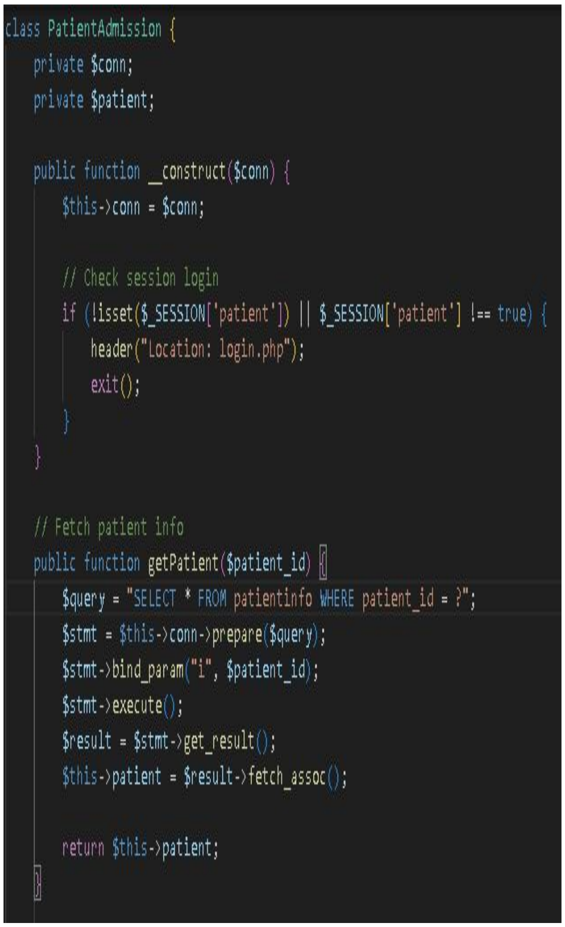


Figure 32 Code Snippet

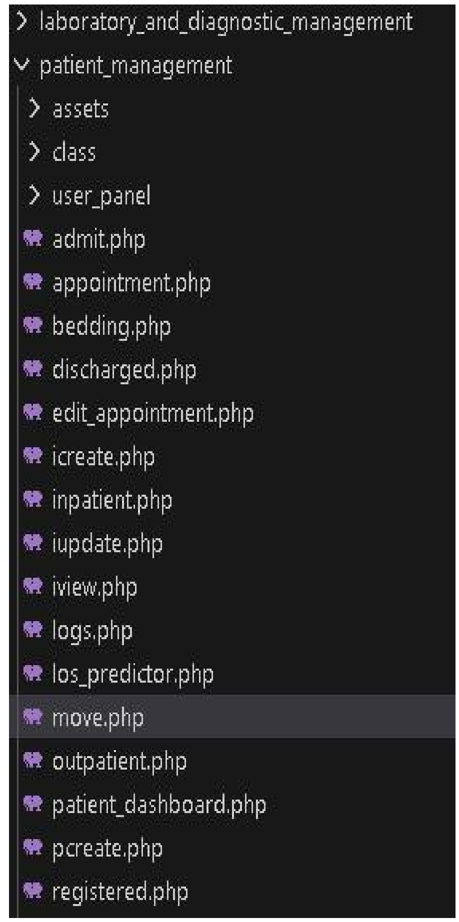


Figure 33 Code Structure

### Algorithms And Data Structures Used

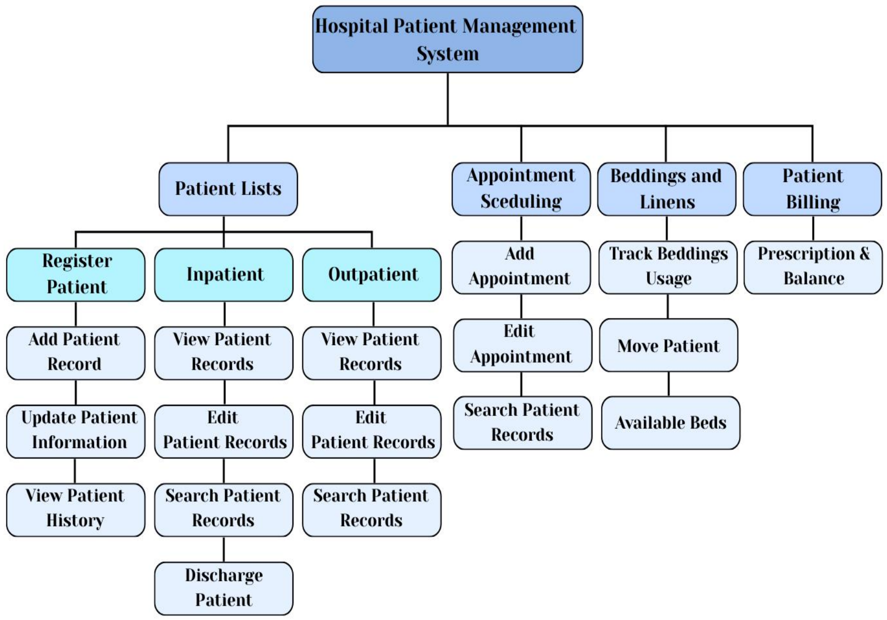


Figure 34 Algorithms and Data Structures Used

### Libraries And Frameworks Utilized

| FRAMEWORK / LIBRARY | DESCRIPTION | PURPOSE IN THE PMS (TOGAF MICRO SERVICES CONTEXT) | BENEFITS |
|---|---|---|---|
| jQuery | A JavaScript library that simplifies client-side scripting and asynchronou | jQuery assists frontend components in communicating with backend services by handling asynchronous requests and dynamic | Improves system responsiveness, reduces page |

| | | | |
|---|---|---|---|
| | s interactions. | data updates, supporting the integration layer between the presentation and application services defined in TOGAF. | reloads, and enhances user interaction**.** |
| Bootstrap | A front-end framework providing reusable UI components and responsive layouts. | As part of the presentation layer, Bootstrap ensures consistent user interface design across different system services while supporting the modular structure of the application architecture. | Speeds up development, ensures design consistency, and improves accessibility on different devices. |
| PHP | A server-side scripting language used for | In alignment with the TOGAF application architecture and microservices approach, PHP is used to build | Supports secure data processing, enables modular |

| | | | |
|---|---|---|---|
| | backend development and system processing. | independent backend services such as user authentication, patient record handling, appointment scheduling, and billing processing. Each service communicates through APIs to ensure modularity and scalability. | system design, and allows easy system expansion |
| HTML | The standard markup language used to structure web pages and application content. | HTML is used to define the structure of system pages such as login screens, patient registration forms, dashboards, and medical record views. | Provides clear content structure and ensures compatibility across web browsers. |

| | | | |
|---|---|---|---|
| CSS | A style sheet language used to control visual layout and appearance. | CSS is applied to enhance the visual presentation of the Patient Management System, ensuring readability, consistency, and a professional healthcare-oriented design | Improves user experience and maintains a clean and organized interface. |
| Font Awesome | An icon library that provides scalable vector icons | Font Awesome is used to visually represent system functions such as navigation menus, actions, and notifications, improving ease of use for medical staff and patients | Enhances usability, supports intuitive navigation, and improves visual clarity. |

Table 29 Libraries And Frameworks Utilized

# 4. Database Schema and Data Management

## Database Schema Design

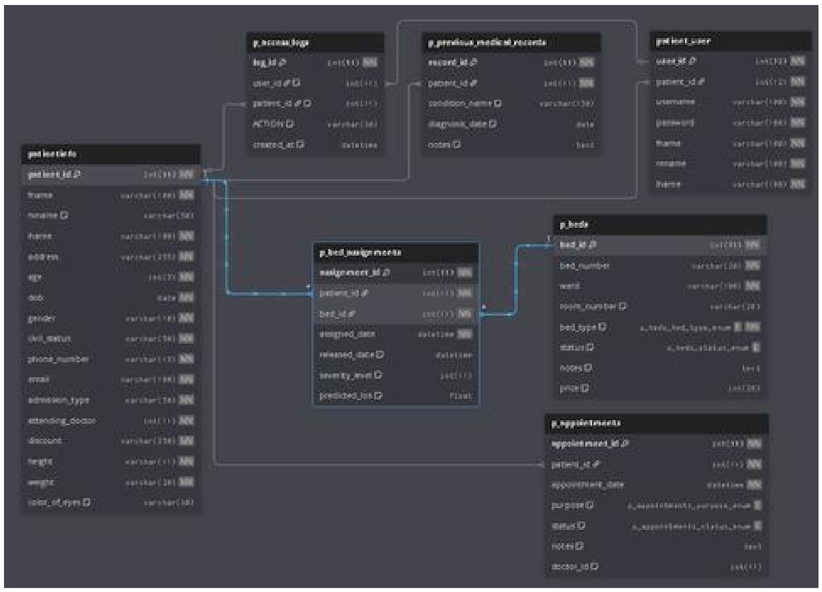


Figure 35 Database Schema Design

## Entity-Relationship Diagrams (ERDs)

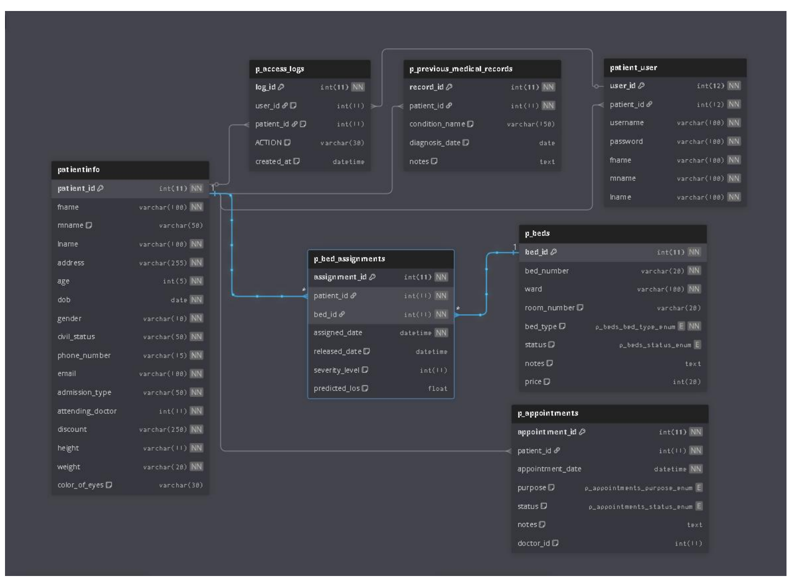


Figure 36 Entity- Relationship Diagrams (ERDs)

**Data modeling and normalization**

Data modeling for a patient management system is carried out through the creation of a systematic representation of healthcare data in a way that ensures accurate storage, retrieval, and management of patient information. It defines key entities like Patients, Appointments, Medical Records, Beds, and Billing along with their attributes and relationships. The well-designed data model will provide consistency of data, ensure system integration across different modules-Registration, EMR, Scheduling, and Billing-and offer easy reporting and decision-making.

Data normalization is the process of organizing the PMS database in such a way that data redundancy can be minimized, and data integrity is maintained. The application of the rules of normalization 1NF, 2NF, and 3NF will store the repeated details of a patient once in one table and refer them from other related tables. Normalization ensures accuracy in patient records, efficiency in data maintenance, and reliable system performance to meet secure and scalable healthcare operations.

**CRUD (Create, Read, Update, Delete) operations**

**Create**

In this function allows the authorize users (doctors, nurse and other healthcare staff) to register a new patient to the system. It is performed during the initial visit of a patient or as they being hospitalized. This also ensure that all patients are properly registered and can be tracked within the hospital system.

**Process**

- Collecting patient demographic information such as full name, date of birth, age, gender, and civil status.
- Generating a unique Patient ID to avoid duplication
- Saving patient data securely into the system database.

**Read**

Read allows authorized users to view the records of patients without modifying the records. This ensures that information is correct and individuals access information they require and provide healthcare staff quick access to accurate patient information for treatment and decision-making.

**Process**

- Viewing patient profile
- Searching patient records using Patient ID, and name
- Accessing medical history, diagnoses, or treatment records

**Update**

Update function allows authorized users to modify or update patient information when changes occur and keep patient information accurate and up to date throughout their hospital stay.

**Process**

- Updating patient information
- Updating patient admission status

**Delete**

Delete allows authorized users to remove or archived patient records from active use. It manage outdated or duplicate records.

**Process**

- Restricting deletion privileges to authorized administrators only
- Maintaining audit logs to track deleted or archived records

## 5. Network Configuration

### Network Topology And Configuration Details

#### Network Topology and Connectivity

As a web-based platform, it specifically outlines how the system integrates across different hospital units to ensure a seamless flow of data:

- **Centralized Integration:** The PMS serves as a digital backbone, connecting various departments such as Laboratory (for test results), Pharmacy (for prescriptions), Inventory (for request linens), Billing (for billing history), and Doctors (for doctors schedules).
- **API-Driven Communication**: To avoid manual data entry and ensure real-time synchronization, the application uses APIs to share patient data automatically between these integrated systems.

**System Configuration and Architecture**

The system is built using a Microservices Architecture, which breaks the application into independent, modular services.

- **Modular Services:** Key modules like Patient Registration, Appointment Scheduling, and Beddings operate independently. This configuration allows the hospital to update or fix one service without taking the entire system offline.
- **Scalability:** The configuration is designed to be scalable, meaning it can handle increasing volumes of patient data and additional staff users as the hospital grows.

- **Development Stack:** The system is developed using PHP with Object-Oriented Programming (OOP) to ensure a structured, maintainable, and reusable codebase.
- **Artificial Intelligence (AI) Integration:** A specific configuration within the bed-linen management is designed to improve bed efficiency by predicting patient length of stay and supporting proactive bed allocation and discharge planning

**Security and Governance Configuration**

To protect sensitive patient information within the network, several security protocols are implemented:

- Role-Based Access Control (RBAC): Access is restricted to authorized personnel only, ensuring that staff members can only view or update data relevant to their specific roles.
- Data Protection: The system employs firewalls, encryption, and audit trails to secure data during transmission and storage.
- CI/CD Pipeline: The technical architecture includes Continuous Integration and Continuous Deployment (CI/CD) pipelines to automate testing and deployment, ensuring the system remains reliable with minimal downtime during updates.

**Security Measures**

The Patient Management System incorporates various security protocols to safeguard the confidentiality, integrity, and accessibility of patient information. These encompass network security through firewalls, data encryption during transmission and storage, role-based access management, secure authentication methods, database safeguarding techniques, routine encrypted backups, and monitoring of user activities. These precautions guarantee that only permitted individuals can reach sensitive medical records while averting data loss, unauthorized entry, and system exploitation, thus fostering secure and dependable hospital functions.

**1. Network Security (Firewalls & Secure Access)**

To protect the PMS from unauthorized external access, network-level security controls are implemented.

**Firewall Protection**

- A firewall is deployed between the PMS server and external networks to filter incoming and outgoing traffic.
- Only essential ports (e.x, HTTPS for web access, database ports restricted to internal use) are allowed.

**2. Data Encryption (Protecting Patient Information)**

Because the PMS handles sensitive patient data (EMR, diagnoses, appointments), encryption is critical.

**Encryption in Transit**

- All data exchanged between users (front desk, nurses, and admins)
- Prevents data interception during login, patient registration, and record updates.
- Patient records stored in the database (personal details, medical history, lab results) are encrypted.

**3. Authentication and Authorization (Role-Based Access Control)**

The PMS applies strict access control based on user roles.

**User Authentication**

- Every user must log in using a unique username and password.
- Passwords are hashed and not stored in plain text.

**Role-Based Access Control (RBAC)**

- Access is granted based on role.

**4. Backup and Recovery Security**

To prevent data loss from system failure or human error:

**Regular Automated Backups**

- Daily backups of patient records, appointments, and bedding data.

**Secure Backup Storage**

- Backup files are encrypted and stored separately from the main server.

**Recovery Procedures**

- Data can be restored quickly in case of system failure or accidental deletion.

**5. Compliance and Privacy Protection**

The PMS follows healthcare data protection principles.

**Patient Data Confidentiality**

- Only authorized users can view or modify patient records.

**Regulatory Awareness**

- System design aligns with healthcare data privacy principles (e.g., confidentiality, integrity, availability).

**Protocols And Communication Methods Used**

The Patient Management System uses a variety of standardized protocols and communication tools to ensure secure, reliable, and efficient system functioning for all its functional modules. Secure web connectivity is supported by HTTP and

HTTPS, with HTTPS supporting encrypted communication to safeguard confidential patient and medical information from unauthorized access. The system is designed using RESTful APIs, which enable smooth and organized data transfer between the most important modules like patient registration, appointment scheduling, electronic medical records (EMR), beddings and linens, and patient billing. These APIs promote system interoperability, scalability, and easy integration with external healthcare applications while ensuring data consistency throughout the system.

Moreover, the system uses TCP/IP as the fundamental networking protocol to ensure stable and trustworthy communication between servers, client computers, and networked medical devices. This ensures that data packets are transmitted correctly and in the right order, reducing errors and losses. Database connectivity protocols are also used to facilitate real-time access to centralized and distributed databases, allowing reliable information exchange between departments.

**Load Balancing And Failover Mechanisms**

Load balancing and failover systems are crucial in ensuring the reliability and efficiency of the Patient Management System. Load balancing involves the distribution of system requests to a number of distributed servers, thus preventing any server from being overwhelmed with work. The system ensures optimal performance and faster response times by efficiently distributing workloads. This is particularly important in a healthcare setting where many users may be accessing patient information, making appointments, or updating medical information at the same time.

Failover systems also provide additional reliability to the Patient Management System by automatically switching operations to backup servers in the event of hardware or software failure of the primary server. The system ensures seamless continuity of critical healthcare services without requiring manual intervention. In emergency situations where uninterrupted access to patient information is critical, failover safeguards ensure that there is no loss of information or services. Load balancing and failover systems provide a comprehensive infrastructure that ensures high availability, the protection of confidential medical information, and the dependability of healthcare services.

## 6. Deployment and Infrastructure

### Deployment Strategies

The Patient Management System may be deployed using, cloud-based, or hybrid deployment strategies depending on hospital requirements. Cloud-based deployment provides scalability and accessibility, while hybrid deployment combines both approaches to ensure security, reliability, and future growth. These deployment strategies support the system's centralized architecture, secure data handling, and real-time access to patient records as defined in the proposed PMS design.

#### 1. Cloud-Based Deployment

**Description**

In cloud deployment, the PMS is hosted on a cloud service provider's infrastructure, allowing access through the internet using secure credentials.

**How it works in the PMS:**

- The PMS application and database are hosted on cloud servers.
- Authorized hospital staff access the system via secure web browsers.
- Cloud services handle scalability, backups, and uptime.

**Advantages**

- Scalable storage for long-term patient records
- Automatic backups and disaster recovery
- Accessible anytime for authorized staff

**Limitations**

- Dependence on stable internet connectivity
- Ongoing subscription costs
- Requires strict security controls

**2. Hybrid Deployment**

**Description**

Hybrid deployment combines on-premises and cloud infrastructure, offering flexibility, security, and scalability.

**How it works in the PMS:**

- Critical patient data (EMR, diagnoses, treatment records) is stored on local servers.
- Non-critical data (reports, analytics, backups) is stored in the cloud.
- Secure APIs synchronize data between local and cloud systems.

**Advantages**

- Balances data security and scalability
- Ensures continuity even if internet access is disrupted

- Reduces storage pressure on local servers

**Limitations**

- More complex setup and management

**3. Deployment Process**

**Staging Environment**

- PMS is first deployed in a testing environment.
- Nurses, front desk receptionist and admin test features (registration, appointment scheduling, beds).

**Production Deployment**

- Final PMS version is deployed to live servers.
- Hospital staff begin real operations.

**Training & Transition**

- Staff are trained to use the new PMS.
- Manual systems are gradually phased out.

**Maintenance & Updates**

- Regular updates via CI/CD practices.
- Security patches and feature enhancements applied.

**Hardware Specifications And Server Configurations**

| COMPONENENT | MINIMUM SPECIFICATION | RECOMMENDED SPECIFICATION |
|---|---|---|
| Processor | Dual – core Processor (e.g., Intel Core i3 or equivalent) | Dual – core Processor (e.g., Intel Core i5 or equivalent) |
| RAM | 8 GB | 16 GB |
| Storage | 256 GB | 512 GB SSD |
| Display Monitor | 15 – inch screen, HD Resolution | 24 – inch screen, HD Resolution |
| Operating System | Windows 10 | Windows 11 |
| Web Browser | Latest version of Chrome, Microsoft Edge | Latest version of Chrome, Microsoft Edge |
| Network Interface | LAN/Wi-Fi | Stable LAN connection |

Table 30 Hardware Specifications And Server Configurations

**Configuration Management**

**Version Control Using GitHub:** For effective collaboration, GitHub serves as the main platform for managing source code. It enables developers to track changes, review updates, and work together on features such as patient records, scheduling, and bedding without causing conflicts.

**Local Server Setup with XAMPP:** During development, a local server environment is created using XAMPP. This setup includes Apache for web hosting, MySQL for database management, and PHP for backend logic, allowing safe testing of system functions before deployment.

**Automation and Development Tools in VS Code:** As the primary coding environment, Visual Studio Code supports automation through extensions like GitHub integration. These tools simplify tasks such as code synchronization and version updates, ensuring smoother system integration and teamwork.

**Modular Application Structure:** By designing the application in a modular format using PHP and JavaScript, system components can be developed and updated independently. This structure makes it easier to enhance features such as login, patient registration, and medical records while supporting future scalability.

**Project Documentation:** Throughout the development process, detailed documentation is prepared to record system design, workflows, and technical decisions. This documentation helps ensure that the Patient Management System can be easily understood, maintained, and improved over time.

**Scalability And Resource Optimization**

The Patient Management System is built to be scalable and resource-efficient with a modular architecture, optimized database structure, automated processes, and a hybrid deployment approach. The PMS guarantees high performance, lower operational costs, and ongoing availability by enabling independent scaling of system modules and utilizing cloud resources for non-essential workloads. These methods are consistent with enterprise architecture principles and address the hospital's increasing needs for data, users, and services.

**Modular system design**

The Patient Management System is structured using a modular design approach implemented through PHP and

JavaScript, where each core function operates as an independent module. These modules include Patient Registration, Appointment Scheduling, Electronic Medical Records (EMR), Bed Management, and Billing History in patient portal. This modular structure allows developers to update, maintain, or enhance individual components without affecting the entire system.

**Load testing and performance evaluation**

Load testing is performed to assess the performance of the Patient Management System under different usage scenarios, including peak hospital times, increased patient admissions, and numerous concurrent users. This guarantees that essential functions like patient registration, EMR access, and appointment scheduling stay effective and dependable even in times of high demand.

**Database optimization**

The PMS utilizes a MySQL database deployed through XAMPP, which is regularly optimized to ensure fast data retrieval and reliable storage of patient records.

**Efficient front-end design**

Front-end performance is enhanced through HTML and JavaScript to guarantee a seamless and responsive experience for hospital personnel. Methods like decreasing page reloads, enhancing form validations, and cutting down on unnecessary scripts aid in accelerating system interactions. Effective front-end design allows users to swiftly register patients, access medical records, manage appointments, and verify bed availability without interruptions, even during peak system usage.

**Cloud deployment**

To support future growth, the Patient Management System may be deployed using a cloud-based or hybrid deployment strategy. Cloud deployment allows the system to dynamically scale resources such as storage and processing power as user demand increases. This is particularly useful for handling large volumes of patient data, system backups, and reporting services.

## 7. Security Measures

**Security Protocols And Measures Implemented**

A patient management system uses a variety of security measures to ensure the confidentiality of sensitive health information, such as role-based security, multi-factor authentication to ensure that only authorized personnel have the ability to access the health information of patients, data/transaction encryption to ensure the health information of patients is not disclosed to unauthorized individuals, as well as auditing to monitor activities carried out by the healthcare software. The software is also complaint with healthcare legislation such as HIPAA or GDPR, as it adheres to the principles of consent, data minimization, and privacy, in addition to network security techniques such as the use of firewalls, IDS, as well as secure APIs.

**Authentication And Authorization Mechanisms**

The authentication and authorization of the Patient Management System are the security processes used to control access to the system that respond to the case of ensuring appropriate access to patient data and the appropriate use of system resources.

**Authentication** - verifies the identity of the user like nurses, admin, receptionist and patient before they can log in to the system. This is normally achieved by use of safe login identities; only registered and approved users can log in to the system.

- User Login Credentials – Users login their unique username and password to access the system.
- Multi-Factor authentication – Require users to provide two or more verification such as password, one-time password code, or biometric before logging in.

- Account Lockout – Automatically blocking an account after several failed login attempts, which require admin to unlock.
- Login Logs – Which records the track who logged in.

**Authorization** – after a user is authenticated, the system will check their assigned role and permissions to determine what features they can access. This ensure that users only access information  relevant to their role, protecting patient confidentiality.

- Receptionist is authorized to register patients and view patient lists but cannot modify medical records.
- Nurses are allowed to update medical records.
- Hospital Admins can access all of the modules
- Patient has their own portal where they can view their information, billing history, medical records, and bed or room assign.

**Encryption and Data Protection**

| ASPECT | DETAILS |
|---|---|
| Encrypted Data Storage | Patient information is stored in an encrypted format to protect sensitive medical and personal data. |
| Secure Data Transfer | Data sent between users and the system is protected using encrypted connections to prevent unauthorized access. |
| Controlled Data Access | Access to patient data is limited through user roles, ensuring that only authorized users can view or update information. |
| Protected Data Backups | System data is backed up regularly, and backup files are kept secure and encrypted to avoid data loss. |
| Hidden Sensitive Information | Sensitive patient details are hidden when used for testing or system development to maintain data privacy. |

Table 31 Encryption And Data Protection

**Incident Response and Mitigation Procedures**

| ASPECT | DETAILS |
|---|---|
| Incident Response Plan | The Patient Management System implements a comprehensive incident response plan that clearly defines the roles and responsibilities of system administrators, IT support staff, and hospital management. The plan outlines procedures for responding to security breaches, system failures, data inconsistencies, and unauthorized access to patient records. |
| Detection and Analysis | Monitoring tools and system logs are used to detect suspicious activities, system errors, or abnormal usage patterns within the PMS. Login attempts, database access, and transaction records are analyzed to assess the scope and impact of incidents. |
| Containment | Containment procedures are established to limit the impact of incidents and prevent further damage. These include temporarily disabling affected user accounts, isolating |

| | |
|---|---|
| | compromised system modules, restricting database access, and switching to backup services if necessary. |
| Eradication | Once the incident is contained, eradication measures are applied to eliminate the root cause of the problem. This may involve removing malicious code, fixing software bugs, applying security patches, correcting configuration errors, or updating access control policies. |
| Post-Incident Review | After resolving the incident, a post-incident review is conducted to evaluate the effectiveness of the response. Lessons learned are documented, and system policies, security controls, and monitoring mechanisms are improved accordingly. Using the Agile Scrum methodology, |

Table 32 Incident Response And Mitigation Procedures

## 8. Testing and Quality Assurance

### Detailed Test Cases And Scenarios

The test scenarios cover all critical aspects of the Patient Management System, ensuring that every feature functions as intended, data remains accurate, and security measures are properly enforced. These tests confirm that patient records are correctly created during registration, updated during consultations, and safely stored in the system. They also verify smooth integration with related modules, such as Doctor and Nurse, and Laboratory systems, ensuring seamless workflow and continuity of patient care. Furthermore, the system’s handling of unusual cases, error situations, and user access restrictions is carefully examined to guarantee reliability, data integrity, and compliance with privacy standards.

| TEST SCENARIOS | TEST CASE ID | TEST CASE DESCRIPTION | TEST STEPS | EXPECTED RESULT | STATUS |
|---|---|---|---|---|---|
| Login Verification | TC-01 | Ensure that only authorized users can access the system and are assigned the correct role-based dashboard. | 1. Open system login page 2. Enter username and password 3. Click login | User is authenticated and redirected to their proper dashboard | PASSED |
| Dashboard Accuracy | TC-02 | Validate that the dashboard displays real-time, accurate information for each user role (admin, doctor, nurse, staff). | 1. Log in as any role 2. Access dashboard 3. Review displayed data (appointments, | Dashboard data is accurate and role-specific | PASSED |

| | | | | | |
|---|---|---|---|---|---|
| | | | notifications, patient count) | | |
| Patient Record Creation | TC-03 | Verify that new patient profiles can be added with all required details. | 1. Log in as authorized staff<br>2. Open patient registration<br>3. Enter complete patient information<br>4. Save record | Patient profile is created and accessible in the system | PASSED |
| Patient Data Update | TC-04 | Ensure that existing patient information can be edited without data | 1. Open an existing patient record<br>2. Edit necessary | Updated patient details are reflected accurate | PASSED |

| | | | | | |
|---|---|---|---|---|---|
| | | inconsistency | fields<br>3. Save changes | ly across the system | |
| Appointment Booking | TC-05 | Test booking, rescheduling, and canceling appointments for patients. | 1. Access appointment module<br>2. Book, reschedule, or cancel an appointment<br>3. Save changes | Appointment records are updated correctly and synchronized | PASSED |
| Medical Record Entry | TC-06 | Validate creation, viewing, and editing of patient medical history and treatment | 1. Open patient EMR<br>2. Add or update consultation details<br>3. Save record | Medical data is accurately stored and accessible by authoriz | PASSED |

| | | | | | |
|---|---|---|---|---|---|
| | | notes. | | ed personn el | |
| Bed Assign ment & Monitori ng | TC-07 | Test the process of assigning beds and linen to admitted patients and tracking availability. | 1. Admit patient 2. Assign bed and linen 3. Update status | Bed and linen are assigne d properly and availabili ty is updated | PA SS ED |
| Patient Transfe r & Dischar ge | TC-08 | Ensure bed and linen status updates correctly when patients are moved or discharged. | 1. Transfer or discharge patient 2. Update bed status | Bed and linen are freed and reflected in system | PA SS ED |
| Service Charge | TC-09 | Verify that patient | 1. Open billing | Charges are | PA SS |

| | | | | | |
|---|---|---|---|---|---|
| Entry | | services are recorded accurately for billing purposes. | module 2. Add service charges 3. Save record | reflected correctly in patient billing | ED |

Table 33 Detailed Test Cases And Scenarios

**Test Data And Expected Outcomes**

**Login**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-1 | Successful Login | User provides the correct username and password. | User is granted access to the Patient Management System and directed to the main dashboard. |
| TC-2 | Failed Login Attempt | User enters an incorrect username or password. | User enters an incorrect username or password. |
| TC-3 | Login with | User tries to log | System prompts for |

| | | | |
|---|---|---|---|
| | Multi-Factor Verification | in with multi-factor authentication enabled. | additional verification and grants access only after correct validation. |

Table 34 Test data(Login)

**Patient Registration and Appointment Management**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-4 | Complete Patient Registration | A new patient is registered with all required details provided. | Patient information is saved, and a unique patient ID is generated. |
| TC-5 | Incomplete Registration | Some mandatory fields are missing or invalid during registration. | System rejects the registration, logs the error, and notifies staff for correction. |
| TC-6 | Detecting Duplicate Patients | Patient with same details is attempted to be registered again. | System identifies the duplicate entry and prevents it. |

| | | | |
|---|---|---|---|
| TC-7 | Scheduling an Appointment | Patient schedules an appointment through the system. | System confirms the appointment, prevents scheduling conflicts, and updates the calendar. |

Table 35 Test data (Patient Registration)

**Electronic Medical Records (EMR)**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-8 | Adding New Medical Records | Doctor or nurse enters patient diagnoses, prescriptions, or treatments. | System saves the records in the patient's EMR with timestamp for tracking. |
| TC-9 | Viewing Patient History | Authorized users or patients view medical history via dashboard or portal. | System displays accurate historical data while maintaining privacy controls. |

| | | | |
|---|---|---|---|
| TC-10 | Editing Medical Records | Authorized staff updates existing records. | System allows edits, logs changes for audit purposes, and preserves data integrity. |

Table 36 Test data(EMR)

**Beddings Management**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-11 | Assigning a Bed | Patient is admitted and needs a bed assigned. | System updates bed availability and assigns the patient to the selected bed. |
| TC-12 | Transferring a Patient to a Different Bed | Patient is moved to another bed. | System updates bed allocation, frees the previous bed, and records the transfer. |

Table 37 Test data (Beddings)

**Patient Portal Access**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-13 | Portal Login and Data Access | Patient logs in to view personal medical records, appointments, billing history, and bed assign. | Expected Outcome: System displays relevant patient information securely and prevents unauthorized access. |

Table 38 Test data (Patient Portal)

**Security and Error Handling**

| TEST CASE NO. | TEST DATA | ACTION | EXPECTED OUTCOME |
|---|---|---|---|
| TC-14 | Unauthorized Data Access Attempt | Unauthorized user tries to access patient records. | System blocks access and records the attempt in audit logs. |
| TC-15 | Handling Data Transmissio | Patient data fails to transmit due to network or | System logs the error and retries transmission once the |

| | | | |
|---|---|---|---|
| | n Errors | system issues. | connection is restored. |

Table 39 Test data(Security and error)

**Test Results And Validation Against Requirements**

After completing the testing of the Patient Management System, the following outcomes were confirmed:

- **Patient Registration** - tests were carried out to ensure the functionality of the system to enter new patients with their entire personal, as well as health, information. The system was able to store the valid information of the patients and identify duplicates.
- **Appointment Scheduling** - Testing included appointment bookings, with conflict detection, ensuring that no overlaps are created in appointments. All these worked as expected, with seamless processes, thus fulfilling the criterion.
- **Electronic Medical Record (EMR)** - This was evaluated for secure storage and retrieval of a patient's medical history such as lab results, medications, and treatment history. The access control features allowed only authorized personnel to alter and view confidential information. The encryption

process during storage and transfer confirmed compliance with privacy policies.

- **Beddings and Linen** – was thoroughly tested to ensure proper tracking of both beds and patient linen stock. Testing involved assigning beds to patients via ward, room, and bed numbers, tracking the issue of linen to individual patients, and making updates when discharged or when the linen is replaced. The system was able to manage live updates, including the automatic release of beds and patient linen once the patient was discharged, as well as flashing the stock alert when the stock level is low, and preventing conflicts such as assigning beds twice or assigning more patient linen than required.

### Performance Testing And Optimization Efforts

#### Performance testing

**1. Load Testing** - The system was subjected to rigorous load testing with the objective of simulating the expected level of traffic generated by users accessing the system at the same time expected in a hospital environment. This involved simulating various users accessing the system in a hospital environment, including receptionists, and nurses, at the same time accessing the system for activities related to patient registration, updates in the Electronic Medical Record, and checking the availability of beds in the system.

2. **Stress Testing** - Stress testing was done to find the breaking point of the system. During stress testing, the system was subjected to data input that exceeds the expected limit. Stress testing helped the team understand the stability of the system when it is subjected to extreme conditions.

3. **Performance testing of database queries** - Since the system uses a centralized database for records of up to 10 years, some specific tests were targeted at the Data Architecture. This included measuring response time for complex search queries, especially those of historical patient data retrieval or report generation.

**Optimization Efforts**

4. **Database Indexing & Schema Optimization -** Database indexing has been implemented on frequently searched columns like Patient Name, Patient ID, and Appointment Dates to try to curtail the challenge of quick record retrieval. Such optimization greatly reduces the time the database engine spends finding specific records amongst the many, ensuring the instantaneous

nature of the search function even when a large number of patients are archived.

5. **Implementation of Caching Mechanisms -** In order to ease the server load and reduce the time needed to load the pages, caching systems were also employed. Data that is often accessed with minimal changes, like data of the various departments of a hospital, specializations of doctors, and basic medical codes, is cached.

6. **Code Optimization through Object-Oriented Programming (OOP) -** The team has optimized the PHP code using good coding practices in Object-Oriented Programming. By unifying coded snippets in reusable classes and modules based on Microservices Architecture, code duplication has been removed.

**9. System Monitoring and Maintenance**

**Tools And Techniques For Monitoring System Health**

| TOOL/ TECHNIQUE | PURPOSE | DESCRIPTION |
|---|---|---|
| Patient Process Performance Tool | Checks system speed | Monitors how fast the system handles patient registration, admission, discharge, and record viewing to avoid delays in hospital services. |
| Patient Data Monitoring Tool | Ensures data accuracy | Checks patient records, billing details, and bed assignments to make sure information is correct and updated. |
| User Activity Logging Tool | Records system actions | Tracks staff logins and system activities to support accountability and proper system use. |
| Data Backup Monitoring Tool | Protects important data | Confirms that patient records, billing information, and hospital data are regularly backed up to prevent data loss. |
| Security | Protects | Monitors access to patient data |

| | | |
|---|---|---|
| Access Monitoring Tool | patient privacy | and detects unauthorized access to keep information secure. |
| Bed and Linen Monitoring Tool | Tracks bed usage | Monitors bed availability and linen status to support proper patient accommodation. |
| Billing Monitoring Tool (Patient Portal) | Ensures correct billing | Reviews billing transactions and payments to reduce errors and maintain accurate hospital records. |
| System Availability Monitoring Tool | Ensures system uptime | Checks if the system is always available and working properly for hospital staff. |

Table 40 Tools And Techniques For Monitoring System Health

**Logging And Error Handling Mechanisms**

The Patient Management System (PMS) utilizes a structured logging and error-handling framework that is integrated into its DevOps lifecycle and Microservices Architecture. These mechanisms are designed to move the hospital from prone-to-error manual processes to a reliable digital environment.

**1. Automated Testing and CI Pipelines**

The system uses Continuous Integration (CI) to catch errors during the development phase, preventing them from affecting live hospital operations.

- **Automated Testing:** Whenever code for modules is submitted, automated pipelines run unit, integration, and regression tests.
- **Error Prevention:** This prevents bugs or integration issues from reaching the live environment, which could otherwise compromise the accuracy of sensitive patient data.

**2. Real-Time Monitoring and Alerting**

Once the system is in production, it is under constant observation through a dedicated monitoring phase.

- **Monitoring Dashboards:** Dashboards track response times, resource utilization, and error rates across all modules.
- **Log Management:** A key activity involves regular log checking and resource adjustment to maintain system speed and reliability.

**3. Resilience through Microservices**

The system's architecture provides a level of fault isolation that serves as a structural error-handling mechanism.

- Fault Isolation: Because modules like registration and appointment scheduling function as independent microservices, an error in one module can be fixed or scaled without stopping the entire PMS.
- Independent Debugging: This modularity allows the team to maintain and debug specific components easily without impacting the overall application.

**Scheduled Maintenance And Updates**

The Patient Management System is regularly maintained to keep the system secure, stable, and easy to use. Scheduled maintenance helps prevent errors, protect patient information, and ensure that hospital operations run smoothly. These activities are done weekly, monthly, or quarterly based on system needs.

| ACTIVITY | DESCRIPTION | SCHEDULE | RESPONSIBLE TEAM | PURPOSE |
|---|---|---|---|---|
| Appointment System Review | Checks appointment schedules for conflicts, overlaps, and outdated entries | Weekly | IT Support Team | Prevent scheduling errors and service delays |

| | | | | |
|---|---|---|---|---|
| | to avoid booking problems | | | |
| Beddings and Linen Status Check | Confirms bed availability and linen assignment after patient admission, transfer, and discharge | Weekly | Ward Management Team | Ensure proper patient accommodation and bed management |
| Billing Data Verification | Examines service charges, billing entries, and payment records to ensure correctness and consistency | Monthly | Billing Department | Reduce billing errors and financial disputes |
| Patient Portal Improvement | Enhances patient access to medical history, appointments, | Quarterly | Development Team | Improve patient experience and |

| | | | | |
|---|---|---|---|---|
| | and personal health information | | | system usability |
| User Access Review | Evaluates staff roles, permissions, and access levels to protect sensitive patient data | Quarterly | System Administrator | Strengthen data security and access control |
| System Data Backup Review | Confirms that patient, billing, and hospital data are regularly backed up and stored securely | Weekly | Database Administrator | Prevent data loss and ensure data recovery |
| System Performance Check | Monitors system speed, stability, and response time during peak hospital hours | Weekly | IT Support Team | Maintain smooth and reliable system operation |

Table 41 Scheduled Maintenance And Update

**Disaster Recovery And Backup Procedures**

| STEP / CATEGORY | ACTION | FREQUENCY / TOOL | SCOPE/ DESCRIPTION |
|---|---|---|---|
| Patient Information Backup | Create secure copies of patient data | Daily / Automated backup or SQL export | Protects patient profiles, appointments, billing details, bed records, and system users from data loss. |
| System File Preservation | Save application and configuration file | Weekly / Manual backup or Git repository | Allows the system to be rebuilt if files are deleted, corrupted, or damaged |
| Problem Detection | Identify errors or service interruption | When required | Helps the team quickly notice issues that may affect patient services or hospital workflows |
| System Impact Evaluation | Review affected modules and records | When required | Determines which system features such as registration, billing, or bed management |

| | | | |
|---|---|---|---|
| | | | need recovery |
| Data Restoration | Recover data from latest valid backup | When require | Brings patient and system data back to an operational state |
| Function Validation | Check system features after recovery | After recovery | Confirms that login, patient records, appointments, billing, and bed tracking work correctly |
| Security Review | Recheck user access and permissions | Monthly | Ensures patient data remains protected and accessible only to authorized hospital staff |
| System Monitoring | Observe system stability and performanc e | Weekly | Helps prevent recurring problems and supports continuous hospital operations |

Table 42 Disaster Recovery And Backup Procedures

## 10. APIs and Integration Points

### Documentation Of Apis Used For Integration

The Patient Management System (PMS) is a system with several modules and connected to other systems of the hospital through RESTful APIs to ensure a smooth flow of data and real-time communication. The Patient Registration API receives the creation, retrieval, and update of the demographic information of the patients, as it forms the basis of the system. Patient Management API handles admissions, discharges, and update patient information, whereas the Appointment Scheduling API allows booking, rescheduling, and cancellations of appointments by communicating with Doctor and Nurse System API to check real-time doctor availability. EMR API is used to store the diagnoses, treatment plans, and notes of clinical nature, and it is linked directly to the Pharmacy API to transfer prescription data and the Laboratory API to request and obtain test results. For Inpatients, the Bed and Linen Management API follows the bed assignment and availability to provide adequate room assignment and resource monitoring.

Moreover, the Billing API captures a history of billing, creates invoices and monitors payments and can be combined with services like laboratory tests and pharmacy histories to make sure that the charges are calculated accurately. Any APIs are also secured by the authentication procedure like role-based access control, which means that only authorized users like administrators, doctors, nurses and patients have access to the appropriate data. This integrated API structure helps the system to attain effective operations in the hospital, high accuracy of the data, and efficient communication among the departments.

**Api Endpoints, Methods, And Data Formats**

The Patient Management System (PMS) uses RESTful APIs to enable integration between core modules and hospital subsystems. Each API follows standard HTTP methods such as GET (retrieve data), POST (create data), PUT (update data), and DELETE (remove data). The endpoints are structured using resource-based URLs, and data is exchanged in JSON format for consistency and interoperability across systems.

| MODULE | ENDPOINT | METHOD | PURPOSE | DATA FORMAT |
|---|---|---|---|---|
| Patient Registration | /api/patient | POST | Register new patient | patient_id, fname, lname, dob, gender, phone_number, address |
| | /api/patient/{id} | PUT | Update patient information | updated demographic fields |
| | /api/patient/{id} | DELETE | Remove patient records | patient_id |
| Patient List | /api/patient-list/admit | POST | Admit patient | patient_id, admission_date, ward, attending_doctor |
| | /api/patient-list/dischar | POST | Discharge patient | patient_id, discharge_date, status |

| | | | | |
|---|---|---|---|---|
| | ge | | | |
| Appointment Scheduling | /api/appointment | POST | Book appointment | appointment_id, patient_id, doctor_id, appointment_date, status |
| | /api/appointment | GET | View appointments | appointment details |
| Electronic Medical Records (EMR) | /api/emr/{patient_id} | GET | Retrieve medical history | record_id, patient_id, medical_records |
| | /api/emr/{record_id} | PUT | Update medical records | updated clinical_notes |
| Bed and Linen | /api/bed-linen/available | GET | Check available beds | bed_id, status, bed_number, ward, room_number, bed_type |

| | | | | |
|---|---|---|---|---|
| | /api/bed-linen/assign | POST | Assign bed | patient_id, assignment_id, bed_id, assign_date, released_date, predicted_los |
| | /api/bed-linen/release | POST | Release bed | bed_id, release_date |

Table 43 Api Endpoints, Methods and Data Formats

**How External Systems Interact With Your Project**

External systems engage with the Patient Management System (PMS) mainly via an API-based architecture and a Microservices structure. This design guarantees that the PMS can effortlessly share data with other hospital departments and external services without needing manual data input.

**Mechanisms of Interaction**

The project aligns with external systems through the following integration methods:

- **API-Driven Integration -** The PMS utilizes Application Programming Interfaces (APIs) to communicate with other hospital information systems, such as **HRIS** (staff management), CRM (patient communication), and ERP (finance).

- **Third-Party Interoperability -** APIs also facilitate data sharing with external healthcare services, including laboratory systems, insurance platforms, and telemedicine tools.
- **Modular Communication -** Because the system is built on Microservices Architecture, individual modules (like registration or billing) can interact with external components independently. This allows for specific services to be updated or integrated with new external tools without affecting the entire system.
- **Real-Time Data Exchange:** The system is designed to provide real-time automated updates.

**Integrated Modules within the Hospital Ecosystem**

The PMS serves as the digital backbone that coordinates with several other departments:

- **Laboratory:** Test results are shared directly with the PMS and stored on the EMR.
- **Pharmacy:** The PMS received prescription information from this system.

- **Billing:** automatically receives service data then reflect to PMS the current billing status.
- **Doctors and Nurse Management:** provides real-time availability to the PMS to ensure proper scheduling.
- **Inventory Management:** this integrates to PMS for requesting linen for beddings.

## 11. User Documentation

### User Manuals And Guides

This section provides an introduction to the Patient Management System, explaining its purpose and benefits. The system is designed to enhance patient care, streamline administrative processes, and efficiently manage hospital resources, including appointments, patient records, and bed allocations.

#### 1. Getting Started

This part guides users on setting up and accessing the system while highlighting its advantages for smooth hospital operations.

**Step-by-Step Setup**

**Account Creation**: Instructions for registering a new user and setting up login credentials.

**Logging In**: How to securely access the system, recover forgotten passwords, and maintain account security.

**User Roles and Permissions**: Overview of roles such as Administrator, Nurse, and Patient, with details on what each role can access or modify.

**System Requirements**: Recommended hardware and software specifications to ensure the system runs efficiently and without errors.

**2. Dashboard Navigation**

This section explains how to navigate the main dashboard and utilize its features to monitor patient management activities and hospital operations effectively.

**Key Dashboard Features:**

Displays admitted patients, outpatients, total patients, number of appointments, for a quick reference. Also shows the real-time bed availability and occupancy status. Has a features of search patient, add patient, move patient, and add appointments for a quick actions. Also have a features of Analytics and Overview that generates insight on patient volumes, growth rate, and operational trends to support decision-making.

## Instructions For System Users

### Loggin In (General)

- Upon opening the web browser, the users will see the login page that need to enter the username and the password and any additional otp code if required. Healthcare personnel and patient who use this system has a different portals.

### Patient List Module

#### Registering a New Patient

- In Patient Lists Module click “Add New Patient” for adding the patient who visit for the first time and fill all the mandatory fields such as name, gender, date of birth, age, phone number, weight and other more mandatory fields. Click save once done filling all fields.

#### Updating Patient Information

- If need to update the patient information, search the name or patient Id and click edit, update the correct fields, save the changes.

#### Searching for an Existing Patient

- Go to search and enter at least one of this; Patient Id, Name, or Phone Number then click search.

## Appointment Scheduling

### Booking a New Appointment

- Search and open the patient record, click the book appointment then select the appointment type (consultation, follow-up, procedure), and which doctor. Choose an available date and time and enter any notes or reason for the appointment type, then confirm the appointment details with the patient and click confirm.

## Bedding and Linen

### Admitting a Patient

- Before admitting the patient, required to fill up the form for diagnosis, and what room or bed, here will be based on what severity level and the AI will predict the length of stay of the patient.

### Moving a Patient

- Select where bed or room to move the patient and tell the reason for the move.

**Troubleshooting And FAQs**

| ISSUE | SOLUTION / ACTION |
|---|---|
| Can not Log in | Check username/password, check role permissions, check if the account is active, try password reset, and check server availability. |
| Slow System Performance | Clear browser cache, check network connectivity, reduce parallel active sessions, and schedule maintenance or optimize database performance. |
| Unable to access patient records | Confirm the role and permission of the user, verify network connection; check for system errors or database downtimes. |
| Data entry errors | This includes verifying that input fields are validated, the format is correct, and staff has been retrained in data entry procedures to confirm system updates have not affected forms. |
| Appointment scheduling errors | Refresh system, recheck calendar configuration, re-verify staff availability, and let synchronization with integrated systems be appropriately done. |

Table 44 Troubleshooting and FAQs

## 12. Known Issues and Troubleshooting

### List Of Known Issues And Their Status

| PROBLEM | SOLUTION |
|---|---|
| Manual Incomplete Data Entry | Implemented compulsory digital input fields and validation checks that ensure all essential information has been entered before the record can be saved. |
| Inefficient of storing patient records | Implemented a computerized system that stores patient information in a centralized and searchable system that allows authorized user to instantly retrieve, and update records. |
| Manual Bed Preparation Verification | Introduced an AI-based Digital Bed Tracking that predict length of stay, which enables the staff to remotely update the status of the beds, thus offering real-time availability information on the readiness of the beds. |
| Delayed Processing | Developed an automated real-time |

| | |
|---|---|
| of Results | notification system for doctors and staff when results are ready or when records have been updated. |

Table 45 List Of Known Issues And Their Status

**Troubleshooting steps for common problems**

Although the Patient Management System is built to function reliably, certain technical concerns may still arise during daily use. The guidelines below are intended to assist users in resolving frequent issues efficiently and maintaining continuous system operation.

| PROBLEM | SOLUTION |
|---|---|
| Authentication and Login Errors | - Ensure that login credentials are typed correctly before submitting.<br>- Reset the account password using the recovery option when necessary.<br>- Reach out to the system administrator if access remains unavailable. |
| Unavailable Patient Records | - Verify the accuracy of the patient name or identification number entered.<br>- Reload the system page and attempt the |

| | |
|---|---|
| | search once more.<br>– Confirm that the patient has been properly registered in the system. |
| Patient Information Update Failures | - Review all input fields to confirm that information is correct and complete.<br>- Fill in all mandatory fields before saving changes.<br>- Forward the issue to technical support if errors continue. |
| System Navigation Difficulties | - Try clearing browser data or switching to a supported browser version.<br>- Check network stability to avoid loading issues.<br>- Inform the administrator if navigation problems persist |
| Report Generation Errors | - Confirm that all required patient details are available in the system.<br>- Generate the report again after refreshing the page.<br>- Request help from IT personnel if the report still fails to load |

| | |
|---|---|
| System Connection or Sync Issues | - Review system connections between internal modules.<br>- Validate configuration settings and user permissions.<br>- Escalate the concern to technical support if syncing does not work properly. |

Table 46 Troubleshooting Steps For Common Problems

**Contact Information For Technical Support**

| TEAM MEMBER | EMAIL / CONTACT |
|---|---|
| Princess Anne Norbie A. Sikat | princessannsikat@gmail.com |
| Jhon Robert B. Delmigues | jhonrobertdelmigues@gmail.com |
| Evelyn B. Suarez | evelyn37467@gmail.com |
| Mario L. Balangatan | mariobalangatan2@gmail.com |
| Jonel Cristian C. Balute | jonelcristianbalute@gmail.com |

Table 47 Contact Information for Technical Support

## 13. Version Control and Source Code Repository

### Description Of Version Control System

The version control system is a foundational part of the Technology Architecture of the PMS. It supports the Microservices Architecture by allowing each module (e.g., billing, pharmacy, and medical records) to be updated, maintained, and debugged independently. This modular approach ensures that a single code change in one service can be deployed without stopping the entire system.

Git is a free, open-source, distributed version control system (DVCS) used to track changes in source code and other files during software development. It allows individuals and teams to collaborate on projects, maintain a complete history of their work, and revert to previous versions if needed.

- **Distributed Architecture**: Unlike older, centralized systems, every developer has a full copy (or repository) of the project's history on their local machine. This means developers can work offline and independently, then sync their changes later with others.
- **Branching and Merging**: A core feature of Git is the ability to create branches, which are isolated lines of development. This allows developers to experiment with new features or bug fixes without affecting the main, stable code-base. Once the changes are complete and tested, they can be easily merged back into the main branch.
- **Repositories**: A Git repository is a folder that contains all the project files and the entire history of changes. Repositories

can be local (on your computer) or remote (hosted online on platforms like GitHub, GitLab, or Bitbucket) for backup and team collaboration.

**Repository Location And Access Details**

The Patient Management System (PMS) source code and documentation is placed in a centralized version control repository such that it is quickly and securely accessed, that the version is properly tracked and that the team members are working efficiently. It is built on a cloud computing platform (like GitHub, GitLab or Bitbucket) and authorized users are able to access the project remotely with high security restrictions.

Only the approved personnel have access to the repository. The developers will have read and write access according to the role they are assigned, and project managers and stakeholders will have read-only access to see. Every access is achieved through verified user accounts, and the multi-factor authentication (MFA) is implemented to minimize the threats of unapproved access.

The repository is used with an organized approach to branches, whereby there are distinct branches of development, testing and production releases. This will ensure that live code is not accidentally modified and a controlled deployment process can be taken. It has regular commits and change logs to keep track of updates, fixes, and improvements and has transparency and accountability throughout the development life cycle. The hosting platform automatically performs backup of the repository, which helps hold against data loss and allows recovering data rather fast once the system fails or is deleted by accident.

**Branching And Merging Strategies**

The Patient Management System (PMS) utilizes DevOps and CI/CD pipelines to manage its code development, which inherently involves branching and merging strategies within a version control system.

**1. Feature-Based Development**

The system is built using a Microservices Architecture, where each module (Patient Registration, Beddings, Appointment Scheduling) is an independent service.

- **Independent Workstreams**: This modular design allows developers to work on specific features or fixes independently without impacting the entire application.
- **Isolation of Changes**: Changes or updates can be applied to a single module, which suggests a branching strategy

where each service or feature is developed in its own isolated environment before being integrated.

**2. Continuous Integration (CI) and Automated Merging**

The PMS employs a Continuous Integration (CI) pipeline as part of its technology architecture.

- **Automated Testing on Commit**: When a developer completes a task and commits (merges) their code, the CI pipeline automatically triggers unit, integration, and regression tests.
- **Quality Gates**: This process ensures that new code does not introduce bugs or break existing functionalities (like patient record retrieval) before it is permanently merged into the system.

**3. Continuous Delivery (CD)**

The integration of Continuous Delivery (CD) ensures that the merged code is ready for the live hospital environment.

- **Reliable Releases**: After the code passes all automated tests in the merging phase, it is deployed to production.
- **Minimal Downtime**: This automated workflow allows for frequent updates and bug fixes, ensuring healthcare staff always have the most stable and updated version of the system.

| STRATEGY COMPONENT | ALIGNMENT IN PMS |
|---|---|
| Branching | Decentralized; organized by independent microservices (Registration, Beddings, etc.). |
| Merging | Automated through CI pipelines that run validation tests upon code submission. |
| Release Frequency | Continuous; enabled by CD to push small, tested updates frequently. |

Table 48 Troubleshooting Steps For Common Problems

## 14. DevOps and Continuous Integration/Continuous Deployment (CI/CD)

### CI/CD Pipeline Setup And Configurations

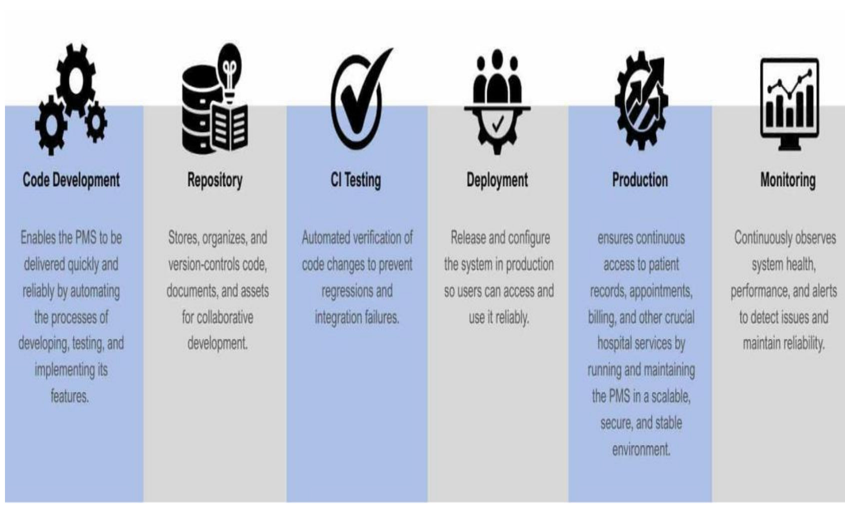


Figure 37  CI/CD Pipeline Setup And Configurations

The CI/CD (Continuous Integration/Continuous Delivery) pipeline is a core component of the project's DevOps strategy. It is designed to ensure that the web-based system is reliable, secure, and updated with minimal downtime.

**CI/CD Pipeline Setup and Workflow**

The system's pipeline follows a structured six-stage cycle to manage code updates and deployments :

1. **Code Development**

   **What happen:** Developers work with doctors, nurses, and administrative staff to create or improve parts of the Patient Management System (PMS), like patient registration, electronic medical records, scheduling appointments, and beddings.

   **Purpose:** To make sure the PMS really helps hospitals with things like speeding up patient registration, admissions, making records more accurate, and retrieving information easier.

   **Key activity:** Writing and updating code based on how things work in the hospital, feedback from staff, and changing hospital rules.

2. **Repository**

   **What happen:** A central repository (like GitHub, GitLab, or Bitbucket) stores and keeps track of all PMS source code, updates, and technical documentation.

   **Purpose:** to make it possible for the development team to collaborate effectively, keep an accurate change history, and quickly roll back earlier iterations when necessary.

   **Key activity:** Clear documentation management within the repository, including branching, merging, and committing.

3. **CI Testing**

   **What happen:** Automated pipelines test PMS modules (such as registration forms, patient records, appointment scheduling, and beddings) whenever developers submit code to the repository.

   **Purpose:** to avoid bugs or integration issues that might compromise hospital workflows or the accuracy of patient data.

   **Key activity:** setting up automated regression, integration, and unit tests for each of the PMS's essential features.

4. **Deployment**

   **What happen:** Updated PMS builds are made available to staging and production servers after passing testing, allowing hospital staff to use new features without interruption.

**Purpose:** To provide enhancements, like improved billing screens or quicker patient lookups, consistently and with the least amount of downtime possible.

**Key activity:** executing deployment scripts, setting up environments, and packaging builds for PMS servers.

5. **Production**

**What happen**: Serving nurse, administrators, and patients, the live patient management system operates in a reliable and secure setting.

**Purpose:** to ensure that pharmacy orders, consultation notes, and patient registration are always correct and available.

**Key activity:** Maintaining database performance, applying security patches, and keeping an eye on uptime even during busy hospital hours.

6. **Monitoring**

**What happen:** Monitoring dashboards, analytics, overview, and resource utilization allow for constant observation of the PMS.

**Purpose:** to keep staff and patients reliable and promptly identify problems.

**Key activity:** adjusting resources, checking logs, and setting up alerts to keep the system reliable and quick.

**Automation Scripts And Tools Used**

| CATEGORY | TOOL / SCRIPT | PURPOSE |
|---|---|---|
| Development Environment | Visual Studio Code (Extensions: Live Server, Prettier) | Automates code formatting, live preview, and efficient development workflow |
| Database Automation | MySQL / SQLyog | Automates database queries, backups, data imports, and exports for patient records |
| Local Server | XAMPP (Apache, PHP, MySQL) | Automates setup and configuration of server-side environment for testing and deployment |
| Version Control | Git & GitHub | Automates version tracking, collaboration, and system backup |

| | | |
|---|---|---|
| Front-End Automation | Bootstrap / Minified CSS & JavaScript | Automates responsive UI design and reduces manual styling |
| Form Validation | JavaScript Validation Scripts | Automates validation of patient information, appointments, and billing forms |
| Authentication Scripts | PHP Login & Session Scripts | Automates user authentication, role-based access, and session handling |
| Notification Automation | JavaScript / PHP Alert Scripts | Automates alerts for appointments, billing updates, and system notifications |
| Testing Automation | Custom PHP & JavaScript Scripts | Automates functional testing such as login, patient record management, and billing processes |

Table 49 Automation Scripts and Tools Used

**Deployment Strategies And Rollback Procedures**

To make the Patient Management System reliable and stable even in the process of upgrading, the following deployment strategies are used:

1. **Automated Deployment of CI/CD Pipeline -** In accordance with the DevOps and CI/CD process, the deployment phase is automated via a Continuous Integration/Continuous Deployment pipeline. As soon as the developers commit any changes into the repository, automated tests are triggered to test the functionality of various modules such as Patient Registration, Appointment Scheduling, and Bed Tracking. In this way, only error-free code is pushed into the live server, with minimal chances of human errors.
2. **Blue-Green deployment (Zero Downtime) -** As the hospital stays open 24X7 and system downtime impacts the hospital severely, the project implements the Blue-Green Deployment technique. In this technique, the project uses two exactly similar production environments: the Blue Environment (current live environment) and the Green Environment (new environment with the update). They update the system in the dormant environment of the Green side. They check the functionality of the system only when the system analysts

examine that the system functions completely in the new environment. They then redirect the system traffic to the new environment in a split second.

3. **Staging Environment Validation -** Before reaching the production-level server, an update is staged on a Staging Server. This staging server is essentially a replica of the production-level environment of the hospital. Such a staging allows testing of an application by the Product Owner along with certain members of the hospital to introduce functionality modifications. All potential bugs are addressed in this staging phase to not affect the production-level health service delivery by the hospital.

**Rollback Procedures**

In cases where critical bugs, instabilities, or data corruption are caused by a deployment, these following roll-back processes are invoked immediately:

1. **Migration to Reverse in the Database Structure**

   As PMS is totally dependent on its centralized database for Electronic Medical Records (EMR), it utilizes migration scripts for handling every single change that takes place in its structure. In case there is a mistake because of changes occurring in the database structure (like adding new tables or columns), the development team will run a down migration script. The script will backtrack its database structure to its former state instantaneously without eliminating or modifying the existing patient information.

**2. Code Version Control Reversion**

If a critical bug is identified within the application code (PHP) right after deployment, then the deployment team uses the version control system Git to revert changes. The team identifies the last stable Commit Hash before the failed deployment and executes a revert command. The CI/CD pipeline will re-deploy this previous stable version onto the server automatically, bringing back the system to work in a really short time.

**3. Automated Backup Restoration (Contingency Plan)**

As an additional level of protection, the whole system (application files and database) will be automatically backed up each day. As an unlikely last resort in either of the above reversals, the system will go temporarily offline. The IT team will then replace the system with the latest clean backup restored into the server. Prior to the system being brought back online, a full system check will take place for the Patient List and

Appointment modules to ensure safe patient care activities are being carried out by the hospital.

## 15. Licensing And Open Source Libraries

### Information About Licenses For Software And Libraries Used

#### Detailed Information on Licenses for Software and Libraries

The Patient Management System is built upon a specific technology stack where every software tool and library is governed by a license that dictates its usage, modification, and distribution. These licensing agreements are essential to ensure the system operates legally and effectively within the healthcare environment while supporting the management of patient records and hospital workflows.

**1. Programming Language: PHP (Hypertext Preprocessor)**

- **Usage in Project**: The core backend of the system is built using PHP with Object-Oriented Programming (OOP). This is used to handle server-side logic, database connections, and the management of patient records, appointments, and bed tracking.
- **License Type:** The PHP License (v3.01).
- **License Details:**

  This is an open-source, BSD-style license.

It allows developers to use, copy, modify, and distribute the software freely.

- **Restriction:** The name "PHP" cannot be used to endorse products derived from this software without prior written permission.

**2. Database Management: MySQL**

- **Usage in Project:** The system uses MySQL as the Relational Database Management System (RDBMS) to store patient information, medical history (EMR), and appointment schedules.
- **License Type**: GNU General Public License (GPL) v2.
- **License Details:** MySQL is open-source software.
- **Copyleft Clause:** If the researchers distribute the Hospital Management System and it includes MySQL (or incorporates it), the entire system may need to be licensed under the GPL as well, meaning the source code of the system must remain open.
- **Exception:** If the system is used strictly internally by the hospital and not distributed commercially as a standalone product, the GPL restrictions are generally more lenient regarding the sharing of the hospital's own proprietary code.

**3. Web Server Stack: XAMPP**

- **Usage in Project:** Used as the local server environment to run the PHP and MySQL components during the development and testing phases (as mentioned in the Technology Architecture).
- **License Type**: GNU GPL (for Apache) & Mixed Licenses.
- **License Details:**

- XAMPP is a compilation of free software. The core component, the Apache HTTP Server, is licensed under the Apache License 2.0.

**4. AI Integration (Beddings and Linens Module)**

- **Usage in Project:** Embedding Artificial Intelligence (AI) within the Beddings and Linens module to predict patient length of stay.
- **License Type:** Dependent on Library Used (Likely MIT or Apache 2.0).
- **License Details:**

- While the specific library isn't named in the text, common libraries for AI/ML in PHP (or integrated via Python scripts) often use the MIT License or Apache License 2.0.

- **MIT License:** This is extremely permissive, only requiring that the original copyright and license notice be included in the code.

**5. Development Tools: Visual Studio Code & GitHub**

- **Usage in Project:**

  **Visual Studio Code**: Used as the source code editor.

  **GitHub:** Used for version control and repository management.

- **License Type:**

  **VS Code**: MIT License (Microsoft). Free to use for any purpose.

  **GitHub:** Proprietary (Terms of Service). The platform is free for public repositories, and often free for students/researchers via the GitHub Student Developer Pack.

**Credits And Attributions For Open-Source Components**

| SOFTWARE / LIBRARY | PURPOSE / USAGE IN SYSTEM | LICENSE TYPE | LICENSE DESCRIPTION |
|---|---|---|---|
| PHP | Backend scripting language used to build the system's core functionalities. | PHP License v3.01 | Permits use, modification, and redistribution of PHP source code with attribution. |
| HTML5 / CSS3 / JavaScript | Used for front-end design and interactive user interfaces. | Open Web Standards (W3C License) | Freely usable standards maintained by the World Wide Web Consortium (W3C). |
| Font Awesome | Provides scalable icons and visual enhancements in the user interface. | Creative Commons Attribution 4.0 (CC BY 4.0) | Allows free use with proper credit to the original author. |
| Bootstrap | Front-end framework used | MIT License | Allows free use, modification, and |

| | | | |
|---|---|---|---|
| | for responsive web design. | | distribution with attribution. |
| MySQL / SQLyog | Database system used for storing and managing hospital data. | GNU General Public License (GPL v2) | Allows free use and modification for educational and non-commercial purposes. |
| GitHub | Platform used for version control, collaboration, and CI/CD integration. | GitHub Terms of Service | Allows public repositories for educational and non-commercial projects. |
| Postman | Used for testing APIs and verifying backend communication. | Proprietary (Free Tier License) | Free version licensed for educational and testing purposes. |
| Swagger | Used for API documentation and visualization. | Apache License 2.0 | Permits usage, modification, and distribution for open and academic |

| | | | |
|---|---|---|---|
| | | | development. |
| ML.NET | Framework used for predictive analytics and machine learning integration. | MIT License | Open-source and free to use with attribution for academic and research projects. |

Table 50 Credits and Attributes for Open Source Components

Credits and Attributions for Open-Source Components" means formally acknowledging and crediting the creators of open-source software used in a project, as required by their licenses.

**Component Tools Reference Table**

| COMPONENT TOOL | DEVELOPER / ORGANIZATION | OFFICIAL WEBSITE / SOURCE |
|---|---|---|
| PHP | The PHP Group | https://www.php.net/ |
| Bootstrap | Twitter Inc. | https://getbootstrap.com/ |
| MySQL | Oracle Corporation | https://www.mysql.com/ |
| SQLyog | Webyog / Idera Inc. | https://webyog.com/ |
| XAMPP | Apache Friends | https://www.apachefriends.org/ |

| | | |
|---|---|---|
| GitHub | GitHub, Inc. | https://github.com/ |
| Postman | Postman, Inc. | https://www.postman.com/ |
| Swagger | SmartBear Software | https://swagger.io/ |
| ML.NET | Microsoft Corporation | https://dotnet.microsoft.com/apps/machinelearning-ai/ml-dotnet |

Table 51 Components Tools Reference Table

These tools were essential in enabling secure, scalable, and efficient system development for the Patient Management System (PMS), while ensuring compliance with ethical and legal software standards.

## 16. Performance Metrics And Monitoring

### Metrics Collected And Monitored

The metrics collected and monitored for the Patient Management System (PMS), concentrate on digital transformation and optimizing hospital processes to enhance efficiency and patient services. The system shifts from a manual, paper-driven setting to an automated digital framework, gathering and tracking various essential operational sectors

**Core Metrics and Monitored Areas**

The system is designed to provide dashboards and reporting tools that track various metrics essential for hospital management:

- **Patient Flow and Volume:** The system monitors the total number of new patients, daily hospital visits, and ongoing registrations.
- **Appointment Management:** Real-time tracking of scheduled appointments, including doctor availability.
- **Bed and Facility Utilization:** Monitored metrics include real-time bed availability, patient-to-bed assignments, and the status of bed preparation.
- **Clinical Outcomes and Accuracy:** The system identifies and flags incomplete patient records such as missing medical histories or allergies ensuring critical information is captured even during emergencies.

**Alignment with Strategic Goals**

These metrics are designed to align with the hospital's operational efficiency and patient safety goals:

- **Operational Efficiency:** By automating routine tasks and tracking resource use, the PMS aims to reduce administrative burdens and operational costs.

- **Data Accuracy:** Mandatory fields and validation checks in digital forms ensure that patient records are complete and consistent, reducing the risk of medical errors.
- **User Experience:** Monitoring processing times for records and result generation allows the hospital to aim for reduced waiting times for patients.

**Tools And Dashboards Used For Performance Analysis**

1. Key Performance Indicators (KPI) Summary Cards

   Display quick statistics that helps admin quickly assess current system status.

   - Inpatient
   - Outpatient
   - Total Patient
   - Total Appointment
   - Available Beds

2. Data Visualization Charts

   Help to identify peak days, and operational patterns.

   - Bar Charts
   - Line Graphs

3. PHP (Backend Processing Tool)

   Data Retrieval - retrieves raw data from table

- Patient records
- Appointments
- Beds

Data Processing - after retrieving, performs calculation

- Total number of inpatient
- Total number of outpatient
- Total of Appointments
- Bed Occupancy rate

4. MySQL (Database Tool)

- Stores patient information, appointment, bed information, and medical records.
- Filtering and grouping data

**Actions Taken Based On Performance Data**

The Patient Management System (PMS) performance data is continuously reviewed and utilized to make system improvements and operational decisions. The metrics can be used to identify the bottlenecks in the performance of the system, the rate of errors, the level of server load, and user activity trends and anticipate the vulnerabilities before they affect patient care.

Once there are performance problems that are identified, the IT staff examines logs and trend of usage to isolate the cause. They can be activities like optimization of database queries, tuning up server resources or even system configuration to enhance speed and reliability. Future upgrades also have performance data. To illustrate, in cases where the usage of the system continues to rise, then the hardware can be scaled, or the cloud resources can be expanded in order to maintain the stability. Whenever there are trends of under performance or violation of security or compliance measures are observed, corrective action is taken immediately, such as the deployment of patches or access controls. Every activity and result is recorded and analyzed in order to ensure that the improvements are working. This feedback mechanism makes sure that PMS is dynamic, consistent, and able to deliver clinical workflow uninterrupted.

# APPENDICES B

## Appendix A: Hospital Map

V334+P84, Dr. Eduardo V. Roquero Sr. Avenue

City of San Jose del Monte, Bulacan

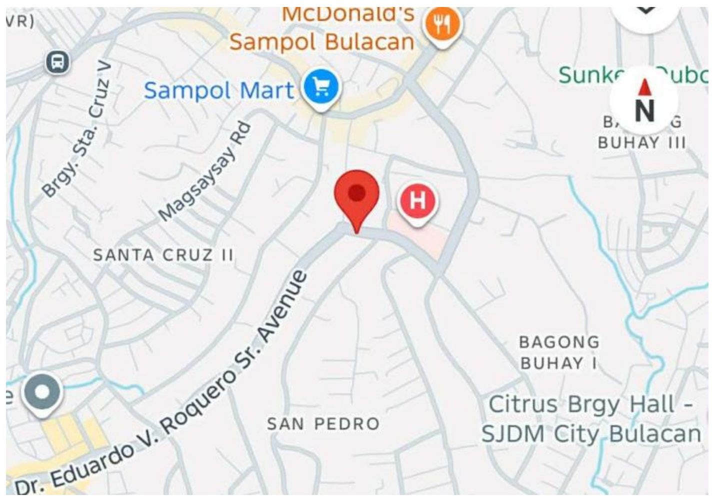


Figure 38 Hospital Map

## Appendix B: Interview

**Questionnaire**

1. How do you currently register and track patient information — such as personal details, medical history, and demographics?

2. What tools or systems (if any) do you use to manage patient records across multiple visits?

3. Are there any challenges you face in keeping patient information up to date and accurate?

4. How do you ensure that sensitive patient information is kept private and secure?

5. Can you describe how appointments are scheduled in your facility? Do you use a manual logbook, spreadsheet, software, or another method?

6. How do you handle appointment changes, like cancellations or rescheduling?

7. Are patients notified of their appointments in any way (calls, texts, etc.)? If so, how?

8. Do you ever face issues like double-booking or scheduling conflicts? How are those managed?

9. How are patient medical records currently stored and accessed — paper, electronic files, a system, or a mix?

10. How do you document important clinical information like lab results, prescriptions, or visit notes?

11. Is retrieving a patient's past medical records quick and easy? Why or why not?

12. Have you considered or tried using an Electronic Medical Record (EMR) system? If so, what do you think are the pros and cons?

13. How do you usually share patient information between departments (e.g., lab, pharmacy, billing, or radiology)?

14. When referring a patient to another facility or specialist, how is their medical information shared — printed documents, emails, digital systems, etc.?

15. Are there any challenges when transferring data between your hospital and external partners (like other hospitals, insurance providers, or government agencies)?

## Appendix C: Discussion

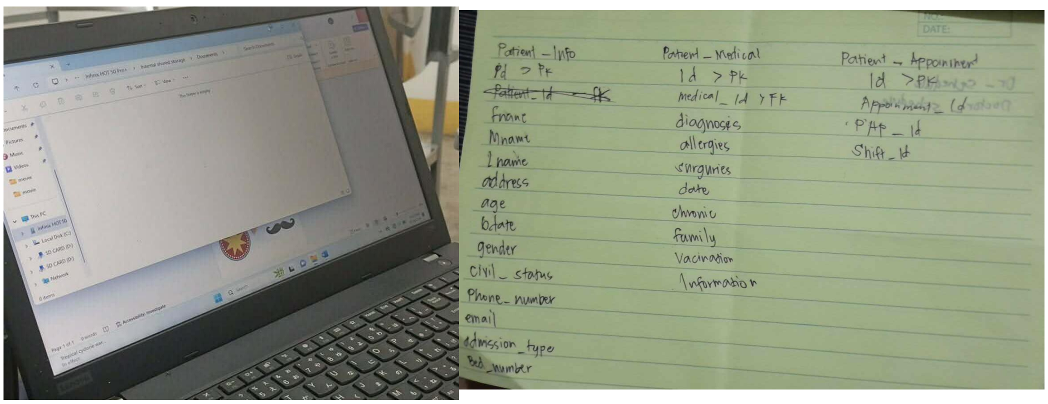


Figure 39 Discussion

## Appendix D: Documentation

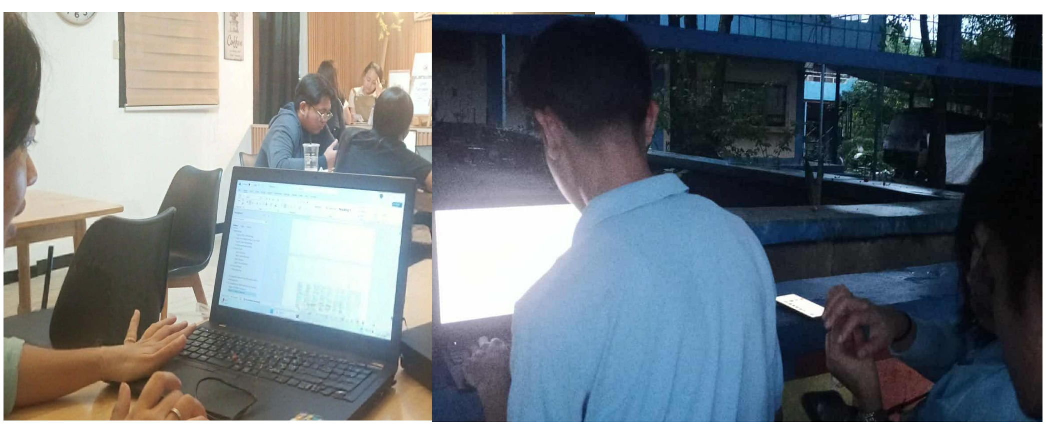

Figure 40 Documentation

## Appendix E:

## Technical Adviser's Certification

This is to certify that the Undergraduate thesis entitled

**BESTLINK COLLEGE OF THE PHILIPPINES BULACAN CAMPUS - HOSPITAL MANAGEMENT SYSTEM (PATIENT MANAGEMENT)**

Submitted by its authors

**MARIO L. BALANGATAN**

**JONEL CRISTIAN C. BALUTE**

**JHON ROBERT B. DELMIGUES**

**PRINCESS ANNE NORBIE A. SIKAT**

**EVELYN B. SUAREZ**

has been technically advised for the proper software flowcharts, system design and schematic diagram

**ENGR. JEAN LESTER M. CACHOLA**
Technical Adviser

April 8, 2026

## Appendix F:

## Certificate of English Editing

This is to certify that the Undergraduate Thesis entitled

**BESTLINK COLLEGE OF THE PHILIPPINES BULACAN CAMPUS -**

**HOSPITAL MANAGEMENT SYSTEM**

**(PATIENT MANAGEMENT)**

Submitted by its authors

**MARIO L. BALANGATAN**

**JONEL CRISTIAN C. BALUTE**

**JHON ROBERT B. DELMIGUES**

**PRINCESS ANNE NORBIE A. SIKAT**

**EVELYN B. SUAREZ**

has been edited for language, grammar, punctuation, spelling, and overall style. Neither the research contents nor the author's ideas we altered during the editing process.

**APRIL MAY A. FERNANDEZ, LPT, MAT**
Editor

April 8, 2026

## Appendix G: Photo During Defense


Figure 41 Photo During Defense

# CURRICULUM VITAE

## Researcher's Curriculum Vitae


PRINCESS ANNE NORBIE A. SIKAT

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Norzagaray, Bulacan

0921 865 3069

princessannsikat@gmail.com


### About Me

Information System student with basic experience in system development, and project development. Strong background in leading capstone projects involving hospital management systems. Eager to gain practical experiences and apply my learning in a real work environment.

### Education

**Bestlink College of the Philippines**
2022 - Present
***BSIS***

**World Citi Colleges - Quezon City**
2019-2021
***Accounting, Business and Management (ABM)***

**Manuel A. Roxas High School**
2015-2019

### Skills

- Organized
- Time Management
- Adaptability and Flexibility
- Basic Computer Literacy (MS Word, Excel, PowerPoint)
- Data Entry / Data Encoding

### Language

- English
- Filipino

# JHON ROBERT B. DELMIGUES

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Norzagaray, Bulacan

0951 108 2731

jhonrobertdelmigues@gmail.com

## About Me

To get an On-the-Job Training job where I can use my knowledge and technical abilities in Information Systems, get real-world experience in system development and data management, and help the company succeed by coming up with new ideas and being more efficient.

## Education

**Bestlink College of the Philippines**
2022 - Present
***BSIS***

**Bestlink College of the Philippines**
2022 (SHS)

**Manuel A. Roxas High School**
2020

**Rosauro Almario Elementary School**
2015-2019

## Language

- English
- Filipino

## Technical Skills

- Microsoft Office (Word, Excel, PowerPoint)
- HTML, CSS, PHP, JavaScript, MySQL
- System Analysis & Design, Database Management
- VS Code, XAMPP, Git/GitHub, PHP, JavaScript, HTML, CSS, MySQL, phpMyAdmin
- Computer hardware and software troubleshooting

## Software Skills

- Analytical thinking and problem-solving
- Communication and teamwork
- Adaptable and fast learner
- Responsible and organized

# EVELYN B. SUAREZ

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Mc Arthur 6, Wakas, Bocaue, Bulacan

0970 187 8316

evelynbobis556@gmail.com

## About Me

Currently a Bachelor of Science in Information System (BSIS) 4th year Student of Bestlink College of the Philippines, Bulacan Campus. As a student with a strong desire to learn and grow professionally. I aim to gain hands-on experience that will allow me to contribute meaningfully to projects and teams.

## Education

**Bestlink College of the Philippines**
2022 - Present
***BSIS***

**Larap National High School**
2020- 2022
***Technical Vocational***
***Livelihood- Tailoring***

## Skills

- Data Encoding
- Document Preparation
- Teamwork and collaboration

## Language

- English
- Filipino

# MARIO L. BALANGATAN

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

21 Florville St. Delta Village Extension Tandang Sora Qc.

09948105994

mariobalangatan2@gmail.com

## About Me

Reliable and detail-oriented student intern experienced in project coordination and team collaboration. Applies strong organizational skills and effective communication to achieve goals efficiently. Committed to continuous learning and professional growth.

## Education

**Bestlink College of the Philippines**

2022 - Present

***BSIS***

## Skills

- Fast learner
- Good Communication
- Teamwork and collaboration

## Language

- English
- Filipino

# JONEL CRISTIAN C. BALUTE

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Road 3 Minuyan 2 CSJDM Bulacan

09262344964

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## About Me

I am a motivated and detail-oriented Information System student with a strong interest in networking. I am eager to learn new technologies, improve my technical skills, and contribute effectively to a team. I am committed to continuous learning and professional growth in the field of Information Systems.

## Education

**Bestlink College of the Philippines**
2022 - Present
***BSIS***

**Sto. Cristo National High School**
2018 - 2022 ( Grade 8 - 12 )

**Paradise Farm National High School**
2017 - 2018 ( Grade 7 )

**Minuyan Elementary School**
2011 - 2016

## Skills

- Computer literate
- Good Communication
- Active listening
- Time Management
- Team Work
- HTML, CSS, Basic JavaScript
- Knowledgeable in Networking

## Language

- English
- Filipino

## Grammarian Curriculum Vitae

# APRIL MAY A. FERNANDEZ, LPT

**PROFILE**

[illegible]

**CONTACT**

PHONE
[illegible]

ADDRESS
[illegible]

EMAIL
[illegible]

## EDUCATION

**METRO MANILA COLLEGE**
Graduated April 2023
Master of Arts in Teaching - Major in English

**IMMACULADA CONCEPCION COLLEGES**
Graduated April 2004
Bachelor of Secondary Education - Major in English

## WORK EXPERIENCE

**TIMOTEO PAEZ INTEGRATED SCHOOL [ TEACHER III ]**
Subjects taught: Oral Communication, Reading and Writing, 21st Century Literature from the Philippines and the World, English for Academic and Professional Purposes
August 2017 – Present

**BESTLINK COLLEGE OF THE PHILIPPINES [ ASSOCIATE PROF. 1 ]**
Subjects taught: Communication Skills 1, Purposive Communication, Rizal's Life and Works, Humanities
June 2011 – July 2017

**MICROLINK INSTITUTE OF SCIENCE AND TECHNOLOGY [ INSTRUCTOR 1 ]**
Subjects taught: Communication Skills 1, Business Writing, Rizal's Life and Works, Art Appreciation
June 2010 – March 2011

**BULACAN STANDARD ACADEMY [ JHS TEACHER ]**
[illegible]
June 2007 – March 2010

**FRANCISCO HOMES MULTI-PURPOSE TRANSPORT SERVICE COOPERATIVE - FHMPTSC [ TREASURER ]**
April 2006 - April 2007

**DIVINE MAJESTY SCHOOL [ JHS TEACHER ]**
June 2005 – March 2006

**YOLA'S EDUCATIONAL SCHOOL [ ELEM. & JHS TEACHER ]**
June 2004 – March 2005

## LICENSURE

**LICENSURE EXAMINATION FOR TEACHERS**
June 2002

## AFFILIATIONS

**SENIOR HIGH SCHOOL ENGLISH TEACHERS' ASSOCIATION IN MANILA (SHSETAM)**
Regular Member since August 2017

**LEADERSHIP IN EDUCATION ACADEMY AND DEVELOPMENT (LEAD)**
Regular Member since October 2016

## RECOGNITION

**SERVICE AWARD - Speech Editor (English)**
Timoteo Paez Integrated School (SHS)
October 5, 2022

## TRAINING AND SEMINARS

**STRENGTHENING TEACHERS' COMPETENCE & COMMITMENT: A JOURNEY OF CONTINUOUS & OPTIMUM GROWTH DEVELOPMENT & SUCCESSES**
TIMOTEO PAEZ INTEGRATED SCHOOL
APRIL 5, 2024

**VIRTUAL TRAINING OF SELECTED SENIOR HIGH SCHOOL GRADE 11 TEACHERS**
DCS- MANILA QUIPPER PHILIPPINES INC.
NOVEMBER 3, 2021

**CONCEPTUALIZATION, USE AND CREATION OF SELF-MADE VIDEOS FOR ONLINE LEARNING**
VIBAL GROUP
JUNE 23, 2020

**ENHANCING ENGLISH LANGUAGE SKILLS FOR FACILITY IN WRITING RESEARCH PAPERS**
TIMOTEO PAEZ INTEGRATED SCHOOL
JUNE 23, 2020

**STRATEGIES FOR ENCOURAGING PARENTAL INVOLVEMENT IN LEARNING**
VIBAL GROUP
JUNE 18, 2020

**CREATING PARENT-FRIENDLY MODULES FOR BASIC EDUCATION**
VIBAL GROUP
JUNE 17, 2020

**WORK-TECH BALANCE AMONG WORKING PARENTS DURING HOMESCHOOL**
VIBAL GROUP
JUNE 16, 2020

**TEACHING SKILLS-BASED SUBJECTS IN ONLINE DISTANCE LEARNING**
VIBAL GROUP
JUNE 12, 2020

**CONDUCTING ASSESSMENTS ON ONLINE DISTANCE LEARNING**
VIBAL GROUP
JUNE 5, 2020

**LITERARY CONTEST OF THE SENIOR HIGH SCHOOL TEACHERS' ASSOCIATION OF MANILA**
MANUEL G. ARAULLO HIGH SCHOOL
NOVEMBER 28, 2018

**ENHANCING ENGLISH LANGUAGE SKILLS FOR FACILITY IN WRITING RESEARCH PAPERS**
TIMOTEO PAEZ INTEGRATED SCHOOL
AUGUST 10, 2018 – AUGUST 14, 2018

**SCHOOL-BASED ROLL-OUT TRAINING (SBRT) ON THE RESULT-BASED PERFORMANCE MANAGEMENT SYSTEM (RPMS) MANUAL FOR TEACHERS AND SCHOOL HEADS**
TIMOTEO PAEZ INTEGRATED SCHOOL
AUGUST 1, 2018 – AUGUST 3, 2018

**MASS TRAINING OF SENIOR HIGH SCHOOL (SHS) TEACHERS ON ACADEMIC TRACK AND TECHNICAL-VOCATIONAL ANG LIVELIHOOD TRACK ENGLISH CLUSTER**
ICON HOTEL, NORTH EDSA QUEZON CITY
JANUARY 16, 2018 – FEBRUARY 4, 2018

## PERSONAL DATA

| | |
|---|---|
| Date of Birth: | May 11, 1984 |
| Place of Birth: | Zamboanga City |
| Religion: | Roman Catholic |
| Civil Status: | Married |
| Height: | 5' 1" |
| Weight: | 60 Kilos |

## REFERENCES

**MAY F. LOPEZ**
Administrative Officer IV
Department of Education – Division of Manila
09396456865 / 09774026754

**DR. RYAN M. IGNACIO**
Dean- College of Hospitality and Business Management
Bestlink College of the Philippines
09190033667

## Technical Adviser Curriculum Vitae

**ENGR. JEAN LESTER M. CACHOLA**
Contact#: 09948003808
Email Address: jeanlestercachola28@gmail.com

**PROFILE**

A Computer Engineer, College Instructor, Research Coordinator, Research Technical Adviser and Academic Cub Adviser with experience on building Arduino Based Robotics Projects, Home Automation, IOT based Projects, Green Technology and Computer Programming. Expertise on leadership development, office management, mechanical works, and encoding.

**WORK EXPERIENCE:**

| POSITION | COMPANY | INCLUSIVE DATE |
|---|---|---|
| 1. COLLEGE INSTRUCTOR I & RESEARCH COORDINATOR | BESTLINK COLLEGE OF THE PHILIPPINES (MV Campus) | 2023 - PRESENT |
| 2. COLLEGE INSTRUCTOR I (Part Time) | BESTLINK COLLEGE OF THE PHILIPPINES (Bulacan Campus) | 2024 - PRESENT |
| 3. TREASURY OPERATIONS FILE CHECKER | UNIONBANK OF THE PHILIPPINES | 2023 - 2024 |
| 4. WEB DEVELOPER | JAE HUB WEB SERVICES | 2023 |

**EDUCATIONAL PROFILE:**

**Master of Science in Engineering major in Computer Engineering**
Rizal Technological University
School Year 2025- Ongoing

**Certificate of Professional Education (18 Teaching Units)**
Bestlink College of the Philippines
School Year 2023-2024

**College**
Bestlink College of the Philippines
Bachelor of Science in Computer Engineering
Achievement: **Achiever Awardee**
School Year 2019-2023

**Senior High School** (STEM)
School: Colegio de San Jose del Monte
Achievement: **With High Honors**
School Year: 2016-2018

**Secondary Education**
School: Colegio de San Jose del Monte
Achievement: **Rank 1**
School Year: 2012-2016

**Primary Education**
School: San Jose del Monte Central School
Achievement: **Class Valedictorian**
School Year: 2004-2012

**MEMBERSHIP/AFFILIATION:**

1. **EMBRACING THE CULTURE OF RESEARCH (ETCOR) –** Regular Membership (April 23,2025 – April 23, 2026)
2. **ASCENDENS ASIA INTERNATIONAL RESEARCHERS CLUB –** Lifetime Associate Member (April 27, 2023)

**AWARDS RECEIVED:**

- **RESEARCH EXPOSITION PANELIST –** 3rd BCPE Research Exposition, Jan 5, 2026
- **COMMITTEE CHAIRPERSON–** 3rd BCPE Research Exposition, Jan 5, 2026
- **CERTIFICATE OF RECOGNITION –** 9th SIMP – AAIRI – BCP MULTIDISCIPLINARY Research Festival (**Faculty Research**) May 6-10, 2025
- **CERTIFICATE OF PARTICIPATION –** 9th SIMP – AAIRI – BCP MULTIDISCIPLINARY Research Festival (**Faculty Research**) May 6-10, 2025
- **SEMINAR SPEAKER –** "The Internet of things: Data in Motion, Intelligence in Action" May 6, 2025
- **COMMITTEE CHAIRPERSON – 2ND BSCPE Research Expo,** March 3, 2025
- **COMMITTEE CHAIRPERSON – 2ND BLIS Research Expo,** March 3, 2025
- **RESOURCES SPEAKER – Research Seminar 2025,** August 10, 2025
- **COMMITTEE MEMBER – Research Forum 2025,** August 10, 2025
- **ACADS ADVISER, QUIZ SHOW –** December 2024
- **COMMITTEE, ENGINEERING QUIZ SHOW –** December 2024
- **SEMINAR SPEAKER –** "Smart Living: Revolutionizing Homes with Intelligent Technology" Engineering Festival, December 2024
- **Adviser – ACADS, Association of Computer Academic Driven Students,** November 2024
- **COMMITTEE – 1st BSCPE Research Expo 2024**
- **COMMITTEE – 1st BLIS Research Expo 2024**
- **FACULTY COACH QUIZ BEE –** General Education Academic Contests, February 1, 2024
- **ACHIEVER AWARDEE –** July 2023
- **TOP 1COMPUTER ENGINEERING RESEARCH PROJECT-** July 2023
- **TOP RESEARCH PAPER -** 5th SIMP-AAG-BCP Multidisciplinary Research Festival, May 23-27 2023.
- **LIFETIME ASSOCIATE MEMBER OF THE ASCENDENS ASIA INTERNATIONAL RESEARCHERS CLUB –** 27 April 2023
- **TOP 1 PRESENTER FOR COMPUTER ENGINEERING DEPARTMENT –** Research Presentation, 18 April 2023.
- **APPOINTED CLASS MAYOR –** College, 3 November 2022
- **APPOINTED CLASS MAYOR –** College, 11 July 2022
- **APPOINTED CLASS MAYOR –** College, 10 December 2021
- **BATTLE OF THE BRAINS PARTICIPANT –** CASE Department representative, 28 January 2020
- **ENGLISH DAY 2018 PARTICIPANT –** Technological Institute of the Philippines, 7 September 2018
- **TOP 1 POETRY WRITING –** Technological Institute of the Philippines, 7 September 2018
- **CEAP SCHOLAR –** 2018-2022
- **WITH HIGH HONORS –** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **BEST IN COMMUNICATION ARTS IN FILIPINO –** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **BEST IN PERFORMING ARTS (CONTEMPORARY PHILIPPINE ARTS FROM THE REGIONS) –** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **VICE MAYOR ACADEMIC EXELLENCE AWARD –** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **ST. JOSEPH CONDUCT AWARDEE –** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
  **SERVANT CONDUCTAWARD -** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **INTELLECTUAL CONDUCTAWARD -** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **APOSTOLIC CONDUCT AWARD - -** Senior High School Grade 12, Colegio de San Jose del Monte, 2018
- **WITH HIGH HONORS -** Senior High School Grade 11, Colegio de San Jose del Monte, 2017
- **EUCHARISTIC SERVIAM AWARDEE -** Senior High School Grade 11, Colegio de San Jose del Monte, 2017
- **RESPECTFUL SERVIAM AWARDEE -** Senior High School Grade 11, Colegio de San Jose del Monte, 2017
- **INTELLECTUAL SERVIAM AWARDEE -** Senior High School Grade 11, Colegio de San Jose del Monte, 2017
- **APOSTOLIC SERVIAM AWARDEE -** Senior High School Grade 11, Colegio de San Jose del Monte, 2017
- **RESEARCHERS MEET 2017 –** Catholic Education Association of the Philippines, 24 November 2017
- **1st ABR QUICK 2 GAME TOURNAMENT 2017 PARTICIPANT –** 2 December 2017
- **ISANG DAANG TULA PARA SA WIKA PARTICIPANT –** 31 August 2017
- **CONSISTENT RANK 1 STUDENT –** High School, Grade 7-10, Colegio de San Jose del Monte, 2012-2016
- **RANK 7 SHS: PAGSULAT NG SANAYSAY –** CSANPRISA 2016 Cultural Event, 16-26 September 2016
- **SERVIAM RUN PARTICIPANT –** 21 August 2016
- **1st HONORS –** Colegio de San Jose del Monte, 26 March 2015
- **MODEL STUDENT OF THE YEAR –** Colegio de San Jose del Monte, 26 March 2015
- **BEST IN SCIENCE –** Colegio de San Jose del Monte, 26 March 2015
- **PINAKAMAHUSAY SA FILIPINO –** Colegio de San Jose del Monte, 26 March 2015
- **PINAKAMAHUSAY SA HEKASI –** Colegio de San Jose del Monte, 26 March 2015
- **REGIONAL QUALIFIER SCHOOOL PRESS CONFERENCE –** Copy Reading and Headline Writing 2015

- **9th PLACE QUIZ BEE –** CSANPRISA 2015 ACADEMIC EVENT, 25 September 2015
- **CSANPRISA 2015 ACADEMIC EVENT PARTICIPANT –** 25 September 2015
- **1st HONORS –** Colegio de San Jose del Monte, 21 March 2014
- **2nd PLACE DIVISION SCHOOL PRESS CONFERECE -** Copy Reading and Headline Writing, 2014
- **BEST IN ENGLISH –** Colegio de San Jose del Monte, 21 March 2014
- **BEST IN SCIENCE –** Colegio de San Jose del Monte, 21 March 2014
- **BEST IN FILIPINO –** Colegio de San Jose del Monte, 21M arch 2014
- **BEST IN ARALING PANLIPUNAN –** Colegio de San Jose del Monte, 21 March 2014
- **1ST HONORS –** Colegio de San Jose del Monte, 21 MARCH 2013
- **BEST IN ENGLISH –** Colegio de San Jose del Monte, 21 MARCH 2013
- **BEST IN MATHEMATICS –** Colegio de San Jose del Monte, 21 MARCH 2013
- **BEST IN FILIPINO –** Colegio de San Jose del Monte, 21 MARCH 2013
- **MODEL STUDENT OF THE YEAR –** Colegio de San Jose del Monte, 21 MARCH 2013
- **VALEDICTORAN –** BATCH 2011-2012, San Jose del Monte Elementary School
- **2012 MTAP DEPED MATH CHALLENGE DIVISION LEVEL QUALIFIERS –** 10 February 2012
- **4th PLACE QUIZ BEE –** 2011 Division Level Science Fair, 2011
- **6th PLACE COPY READING AND HEADLINE WRITING –** 2011 Division Elementary School Press Conference, 10-11 November 2011
- **6th PLACE QUIZ BEE –** 2010 Division Science Fair, 23 September, 2010
- **8th PLACE DIVISION LEVEL MTAP WINNER –** Grade 1, 2007
- **BEST IN MATH –** KINDER RED A, San Jose del Monte Elementary School, 14 March 2006
- **BEST IN MATH –** KINDER NURSERY B, San Jose del Monte Elementary School, 15 March 2005

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**THESIS/PROJECT RESEARCH:**

- DEVELOPMENT OF AN INTEGRATED ROBOTIC VACUUM CLELANER AND AIR PURIFIER WITH WIFI CONNECTIVITY AND MULTI-MODE CONTROL USING ARDUINO UNO, **2025**
  - **ABSTRACT** is Published (Ascendens Asia Singapore)
- DEVELOPMENT OF ALEXA BASED SMART HOME SYSTEM WITH RFID AND LASER SECURITY USING ESP32 AND ARDUINO, **2023**
  - **ABSTRACT** is Published (AASg-BCP-JMRA Vol4 No1)
  - **JOURNAL** is Published (Ingenium, Vol 1, Issue 1, ISSN 3027-9259, pg. 26-37)
- BABY OIL AS A ROSEWOOD GUITAR FRETBOARD POLISHER, **2018**
- BREADS AND CORNFLAKES AS SUBSTITUTE FOR FISH FEEDS, **2018**
- EFFECTIVENESS OF TEC PITCHER AS A COOLING DEVICE, **2018**
- REVERB AND DELAY PEDAL AS A VOICE EFFECT MANIPULATOR, **2018**

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**COURSES HANDLED:**

Prof Ed:

1. Computer Network and Security
2. Fundamentals of Electronics Circuits
3. Engineering Drawing 2
4. Methods of Research
5. CPE Laws and Professional Practice
6. System Maintenance and Administration 2
7. Computer Architecture and Organization
8. Computer Engineering as Discipline
9. Data and Digital Communication
10. Data Structures and Algorithm
11. Digital Signal and Processing
12. CPE Elective 1
13. Emerging Technologies in CPE
14. Engineering Ethics
15. Engineering Management
16. Environmental Science and Engineering
17. Fundamentals of Mixed Signals and Sensors
18. Introduction to HDL
19. Logic Circuits and Design
20. Physics for Engineers
21. Programming Logic and Design
22. Technopreneurship

Gen Ed:

1. Purposive Communication
2. Mathematics, Science and Technology
3. Mathematics in the Modern World

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**SEMINARS AND TRAININGS:**

- IN SERVICE TRAINING FOR TEACHERS (30.95 Hours), July 16-19, 2024
- VEX IQ ROBOTICS SYSTEM, 13 JUNE 2023
- CYBERTHREAT GAME CHANGER: A NEW LOOK AT INSIDER THREATS, 26 APRIL 2023
- ARTIFICIAL INTELLIGENCE IN DIGITAL MARKETING: 7 COURSES IN 1, 25 APRIL 2023
- LEARN CSS FOR BEGINNERS, 25 APRIL 2023
- LEARN TO HOST MULTIPLE DOMAINS ON ONE VIRTUAL SERVER, 25 APRIL 2023
- DIGITAL SECURITY: CURRENT EFFORTS AND EMERGING PRACTICES IN SECURING LIBRARY SYSTEMS, 24 APRIL 2023
- PHILIPPINE NATIONAL PUBLIC KEY INFRASTRUCTURE (PNPKI) ORIENTATION, 19 APRIL 2023
- TRANSCEND: CLIMBING YOUR WAY UP IN THE INDUSTRY, 15 APRIL 2023
  "THRIVING IN A DIGITAL ECONOMY" A NEW PARADIGM FOR SCHOOLS, 15 APRIL 2023
- DEMYSTIFYING THE CISSP: KEY CONCEPTS AND STRATEGIES FOR SUCCESS IN 2023, 14 APRIL 2023
- COMPTIA NETWORK+ WEBINAR, 12 APRIL 2023
- SOFTWARE ENTREPRENEURSHIP: A GUIDE TO BUILDING AND SCALING A SUCCESSFUL TECH COMPANY, 22 MARCH 2023
- A GLIMPSE AT THE WORLD OF DATA SCIENCE AND CYBER SECURE: SECURE YOUR CYBERSPACE, SECURE YOUR DIGITAL LIFE, 18 MARCH 2023
- DISASTER PREPAREDNESS SEMINAR AND BASIC FIRST AID TRAINING – 12 JULY 2023
- MIND POWER OF ROBOTICS, 5 JUNE 2022
- NEURAL NETWORKS, 15 MAY 2022
- SYSTEM ADMINISTRATION AND MAINTENANCE, 24 APRIL 2022
  UNDERSTANDING THE BASICS OF GENDER SENSITIVITY, 19 FEBRUARY 2022
- LEADERSHIP TRAININGS FOR STUDENT LEADERS, 6 JULY 2012
- DIVISION ORIENTATION AND CONSULTATION ON THE ENHANCED K+12 BASIC EDUCATION PROGRAM, 7 FEBRUARY 2011

# Research Adviser Curriculum Vitae

**DAVID GALBAN JAURIGUE**

+639565614495 | +639684260966

Bagong Silang, North Caloocan City

aviidjaurigue07@gmail.com

**Skills:**

- Hardware
- Software
- FRONT END
- BACK END
- FRAME WORKS
- OOP LANGUAGES

**Freelance Recent Project:**

- **Inventory System (**Hausen Fabcon Inc.**)**
  **Front End:** C#
  **Back End:** PHP OOP
  2 user type system for admin and non - admin, Stock Monitoring Notification, System Testing

- **Inventory System (**Wilcon Company**)**
  **Front End:** C#
  **Back End:** LARAVEL
  Testing System function data, bug and errors

**Work Experience:**

- **Snow World Manila Corp.**
  I.T software & Hardware Maintenance
  Electrical Team Helper.
  July 17, 2014 – March 3, 2015
  (Formatting Operating system and Fixing Computer Hardware, CCTV Maintenance.)

- **Bestlink College of the Philippines**
  Senior High School ICT Instructor
  July 3, 2019 – May 1, 2020
  (Teaching Senior High School Grade 11 and Grade 12 Software and Hardware, Web Development, Project Study Baby Thesis)
  College of Computer Studies
  September 7, 2022 - Present

- **Indevfinite Company I.T Solution**
  Freelance Programmer
  December 5, 2018 – June 31, 2021
  (Front End, Back End, System Analyst Launching Game Ran Online Laptop and PC Game, testing System Features and data)

**Education:**

Bestlink College of the Philippines (Quezon City)

Bachelor of Science and Information Technology (S.Y June 5, 2015 – May 3, 2019).

Vocational TESDA Training Certificate
Caloocan City Manpower Training Center (Camarin Caloocan City)
Basic Computer Operation (BCO) with On Job Training
Computer Hardware Servicing (CHS) with On Job Training (S.Y August 2012 – July 2013)

Master's in Information Technology
Malolos Bulacan La Consolacion University Philippines
April 25 2025 - Present